

3, 6, 9 Is All You Need

A formally verified account of the digital-root trinity,
the sieve at 2 and 3, and Hecke eigenvalues at the first two primes

Formalised in Lean 4 with Mathlib

Abstract

“If you only knew the magnificence of the 3, 6 and 9, then you would have a key to the universe.” The slogan, popularly attributed to Nikola Tesla, is usually left unformalised. We give it a precise meaning and a complete, machine-checked proof. The trinity $\{3, 6, 9\}$ is exactly the set of digital roots of the multiples of 3; it is closed under addition and absorbs multiplication, so it is the digital-root shadow of the ideal $3\mathbb{Z}$; it is generated, as a subgroup of $\mathbb{Z}$, by proper subsets of itself, so the slogan is true but redundant. Complementing this, the doubling orbit $1, 2, 4, 8, 7, 5$ of “vortex mathematics” is proved to have period six and to miss the trinity entirely. We then formalise two adjacent and genuinely useful “small prime” phenomena: sieving by the primes 2 and 3 alone already removes exactly two thirds of the integers and confines every larger prime to $\pm 1 \pmod 6$; and, for a normalised Hecke eigenform, the two eigenvalues a_2 and a_3 determine a_n for every 3-smooth $n = 2^i 3^j$, in particular a_6 and a_9 . Both of these are then proved in the general form of which they are the $\{2, 3\}$ -instance: for an arbitrary finite set S of primes the sieved density is exactly $1 - \prod_{p \in S} (1 - 1/p)$ (Legendre), and for an arbitrary set S of primes the eigenvalues $(a_p)_{p \in S}$ determine a_n at every positive S -smooth n — so agreement at all primes is agreement everywhere, and agreement at the primes $p \leq N$ is already agreement at every $n \leq N$. Following that thread one step further, the five words $T_2, T_3, T_2 T_3, T_2 T_3 T_2$ and $T_3 T_3 T_3$ are expanded in the basis of Hecke operators — $T_2 T_3 = T_6$, $T_2 T_3 T_2 = T_{12} + 2^{k-1} T_3$, $T_3 T_3 T_3 = T_{27} + 2 \cdot 3^{k-1} T_3$ — and their five multipliers are exhibited as polynomials in a_2 and a_3 alone. Finally we remove the last arbitrary restriction, the base: in base b the digital root detects divisibility by any divisor d of $b - 1$, the corresponding trinity has $(b - 1)/d$ elements, and ten is the smallest base whose three-element trinity detects an odd prime; the “vortex” picture is the orbit decomposition $9 = 1 + 2 + 6$ of $\mathbb{Z}/9$ under doubling; and the Fibonacci digital roots have minimal period 24, with the trinity occurring exactly at the indices divisible by four, because $3 = F_4$. Finally the abstract eigenform is replaced by a concrete one: the coefficients $\tau(n)$ of the discriminant form $\Delta = q \prod (1 - q^m)^{24}$ are computed inside the proof assistant, the five words are checked on them, and the form is drawn — its coefficients against the Ramanujan–Petersson envelope, and the function itself as a phase portrait on the upper half plane, whose horizontal period one and decay at the cusp are proved. A fourth feature of that portrait is then explained: the pattern stands perfectly upright over $x = 0, \pm \frac{1}{2}, \pm 1$ because real coefficients and period one together give one mirror per half-integer, so on those lines the function is real, its phase is locked at 0 or π and cannot drift, the horizontal derivative of the magnitude vanishes, and every contour of $|\Delta|$ crosses the line at exactly 90° . The remaining symmetry of that portrait, the reflection across the arcs $|z| = 1$, is then proved as well, and for the honest infinite product rather than a truncation: writing $\Delta = \eta^{24}$, its logarithmic derivative is $2\pi i E_2$, the failure of E_2 to be modular is exactly the logarithmic derivative of z^{12} , and comparing the two at the fixed point $z = i$ gives $\Delta(-1/z) = z^{12} \Delta(z)$ — hence invariance of weight twelve under all of $SL_2(\mathbb{Z})$, the mirror $|\Delta(1/\bar{z})| = |z|^{12} |\Delta(z)|$, and on the arc itself the phase lock $\overline{\Delta(z)} = z^{12} \Delta(z)$. Each of the five Hecke words is then drawn in the same way:

its q -expansion is computed from the Hecke formula rather than from the eigenvalue, and is found, on the range checked, to be that of Δ multiplied by the word's own constant — so the five graphs are one graph at five heights, two of them turned over because two multipliers are negative, and every plotted coefficient of every word is divisible by three. The two shortest words are then compared with each other term by term: because both images are Δ rescaled, the pointwise product of $T_2\Delta$ and $T_3\Delta$ is $\tau(6)\tau(m)^2$, their difference is $(\tau(2) - \tau(3))\tau(m)$, their quotient is the constant $-2/21$, their remainder takes exactly two values, and their power is positive and divisible by three to the height of the exponent — and the trinity survives every one of these operations except division, which is precisely the operation that cancels the common factor three. Each of those six comparisons is then drawn as a function of its own: two of the six pictures are Δ rescaled, one is the Hadamard square of Δ , two are provably no rescaling of Δ at all because they read the sign of τ , and the sixth — the power — has no convergent series and is drawn one logarithm down. Archimedes' own figures are then put on that same circle: the inscribed and circumscribed regular n -gons sandwich π , the doubling ladder is a harmonic mean followed by a geometric mean, the hexagon anchors it with $b_6 = 3$ exactly, four doublings from there reproduce Archimedes' own $\frac{223}{71} < \pi < \frac{22}{7}$ with no numerical value of π as an input, and the nine-gon is shown to be off the ladder because it is a trisection. Setting those figures turning gives two further theorems: the polar profile of the n -gon is a $2\pi/n$ -periodic cosine sawtooth whose spin sweeps an annulus of area $\pi \sin^2(\pi/n) \rightarrow 0$, and the regular $2m$ -gon revolved about a long diagonal sweeps a solid of volume $(4\pi/3) \cos^2(\pi/2m)$ — so the square gives $2\pi/3$ and the hexagon gives exactly π . Those polygons are finally laid on the portrait of Δ itself and set turning: the ring of the n -gon is rigid and repeats after a rotation by $2\pi/n$ and not before, the rings for $3 \mid 6 \mid 12$ and $3 \mid 9$ stay nested, the modular involution $z \mapsto -1/z$ reverses the frame of the animation and therefore fixes the ring at exactly two instants of each loop, and at every vertex in the upper half plane the phase of Δ is locked — for the twelve-gon by the single factor e^{12it} , the same at all twelve vertices. The portrait is then taken apart: a computer-vision pipeline finds the shapes it contains, Euclid's ternary dissection cuts each shape into nine congruent parts and each of those into nine again — 9^m organised parts m levels down — the value of Δ is read inside every part, and the readings are normalised row by row into a character, a matrix of grey levels which is proved to be a palindrome because $\Delta(-\bar{z}) = \overline{\Delta(z)}$. Frobenius is applied to that character twice over: the mirror is an isometry for the Frobenius norm and fixes the character, so character and mirror image have cosine similarity exactly one; quantising the character to three grey levels puts it in $\mathbb{Z}/3$, where the Frobenius endomorphism $x \mapsto x^3$ fixes it pixel by pixel; and on the arithmetic side the trace of Frobenius $\tau(p)$ is, for every prime $p \leq 30$ other than three, the sum of two power characters $p^2 + p^9$ modulo 27, which the same Frobenius collapses to $p(p+1)$ modulo 3 — so three divides $\tau(p)$ exactly when $p \not\equiv 1$ modulo three. The two pictures are then combined: each frame of the spinning n -gon is cut by the fan from its centre into n triangles, each triangle is cut Euclid's way into nine, the cut is iterated d times, and every one of the $n \cdot 9^d$ parts is painted with the value of Δ at its centroid and the parts sorted by that colour — whereupon a rotation by $2\pi/n$ is proved to permute the parts of a frame, so that the sorted palette returns exactly after one loop of the animation, while a chord being at most its arc puts every point of the circle within π/n of a vertex, which is the sense in which the polygons approximate the curve. Finally the same reading is turned on two neighbours of Δ . The Riemann zeta function has no coefficients to read — they are all 1 — so its trinity is looked for, and found exactly, in the indices: the part of the Dirichlet series carried by the indices of digital root 3, 6 or 9 is precisely $3^{-s}\zeta(s)$, a self-similar copy of the whole, and everything else is ζ with Euler's factor at three removed; meanwhile the constants 24, 240 and 504 that build Δ , E_4 and E_6 are exhibited as the odd negative zeta values $-2/\zeta(-1)$, $2/\zeta(-3)$, $-2/\zeta(-5)$, of digital roots 6, 6, 9. Klein's j -invariant is the opposite case, all coefficients: every third one is divisible by 3^5 and so has digital root 9, and comparing j with the McKay–Thompson series of the Monster at the classes $2A$ and $3A$ shows which prime is responsible — on the computed range j agrees with

the $3A$ series modulo 3^6 and therefore has exactly its digital roots term by term, while it agrees with the $2A$ series modulo 2^{12} , a congruence the digital root, being an instrument of three, cannot see. The tiling itself is then handed round: the same ring ladder of Euclid triangles is read by ten functions in turn — Δ , j , the McKay–Thompson *hauptmoduls* of classes $1A$, $2A$ and $3A$, the Riemann ζ , its completion ξ and three Dirichlet L -functions — each getting its own standalone plate and its own animation of the turning triangle. That last gallery raises an experimental question, which the paper closes: the plates of different functions look identical, and a gallery of rational newforms pulled in from the LMFDB — 109 of them, weights 2 to 26, levels 1 to 389, 28 with complex multiplication — confirms it, every one of them painting the same plate high in the upper half plane while the *hauptmoduls* j , T_{2A} , T_{3A} paint its exact negative. The reason is proved rather than observed: for any q -series with polynomially bounded coefficients the leading term rules above an explicit height, so the reading is $e^{-2\pi \operatorname{Im} z}$ up to a relative error that goes to zero, and on a sample separated in height every normalised series gives literally the same palette ranks and every *hauptmodul*-shaped series the reverse ones — the T series is one plate. Dividing that universal factor out, the residual f/q read low down, returns the individuality of the forms, and the measurements over the gallery quantify both regimes. Those same 109 newforms are then surveyed a second time, arithmetically: each is read through the digital roots of its Hecke eigenvalues, and the sample is organised into a tree whose three levels are the residues mod 3, the occupied vertices of the doubling circuit, and the residues mod 9 — a laminar family, hence a tree, because the finer datum determines the coarser. The tree is short and rigid: layer 3 has exactly two values, and the split is a congruence, holding for 78 of the forms and failing for the other 31; layer 9 takes only 7 of its 512 possible values; the thinnest branch consists of 11 forms whose every sampled eigenvalue is 0 or 2 modulo 9, which is proved to be forced by the Eisenstein congruence $a_p \equiv 1 + p^{k-1}$ together with a weight $\equiv 4$ modulo 6 — the weight entering only mod 6 because the vortex circuit has six vertices; and every form of the sample puts at least a third of its eigenvalues, and every form with complex multiplication at least a half, on the trinity $\{9, 3, 6\}$. The spectrum is coarse enough that the 109 forms carry only 51 distinct spectra, all 11 of the thin branch sharing one. A last picture is then read in the same spirit, and it is not a modular form but Pascal’s triangle. Colouring the cell (n, k) when a prime p misses $\binom{n}{k}$, the lit cells are exactly those whose base- p digits are dominated — Lucas — equivalently those whose subtraction $n = k + (n - k)$ carries nowhere in base p — Kummer — so that the dark inverted triangles are precisely the carries; for $p = 2$ this is the bitwise test $k \& n = k$ and the Sierpiński gasket, and in general the number of lit cells in a row is the product of its base- p digits each raised by one, which on the mod-3 board gives row 24 exactly nine lit cells, sitting exactly at the multiples of three. The group-theoretic half of the same diagram is proved too: a group transitive on the k -subsets of n points has k -set stabilizers of index $\binom{n}{k}$, so that Pascal entry must divide the group’s order — a divisibility that, for each of the five Mathieu orders, holds only in a narrow band at the two ends of the row, which is exactly why the middles of rows 11, 12, 22, 23, 24 are dark, and dark for every group of that order. Conversely no cell is unfillable in principle, since the symmetric group is transitive on k -subsets for every k . Finally the geometry is kept and the group dropped: a Steiner system $S(t, k, n)$ has $\binom{n}{t}/\binom{k}{t}$ blocks, so an $S(5, 8, 24)$ has exactly 759 octads and an octad stabilizer has index 759 rather than $\binom{24}{8} = 735\,471$, while no $S(5, 8, 25)$ exists at all — the octad pattern really does stop at 24, for a reason as elementary as a divisibility. Every statement in this paper is accompanied by a Lean 4 theorem that compiles against Mathlib with no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and no `sorry`.

1 Introduction

Numerology around 3, 6 and 9 is a folklore staple. What is remarkable is that, once the vague slogan is pinned down, a genuine and completely elementary theorem appears: the trinity is the fibre of the digital root over the multiples of three, and casting out nines is the reason that every claim of “vortex mathematics” about 3, 6 and 9 either follows immediately or is false.

Our aim is fourfold.

1. Give the folklore a faithful formal statement and prove it (Sections 2 and 3).
2. Separate what is true from what is merely suggestive: the trinity really is closed and absorbing, but it is *not* a minimal generating set (Theorem 3.4), and the powers of two really do avoid it, for the boring reason that $3 \nmid 2^n$ (Theorem 3.7).
3. Formalise the two serious mathematical facts that stand behind the intuition that the smallest primes carry most of the information: the density of the $\{2, 3\}$ -sieve (Section 4) and the determination of 3-smooth Hecke eigenvalues by a_2 and a_3 (Section 6), and in each case prove the general statement that removes the arbitrary restriction to the first two primes (Sections 5 and 7).
4. Remove the last arbitrary restriction, the base ten itself, and account for the two remaining folklore claims: the “vortex” diagram and the Fibonacci cycle (Sections 9–11).

All results are formalised in Lean 4 against Mathlib; the correspondence between the numbered statements here and the Lean names is given in Appendix A. Nothing in this paper is asserted informally: each theorem below is the informal reading of a machine-checked Lean theorem.

2 Digital roots

Definition 2.1 (Digital root). For $n \in \mathbb{N}$ set

$$\text{dr}(n) = \begin{cases} 0, & n = 0, \\ 1 + ((n - 1) \bmod 9), & n > 0. \end{cases}$$

The closed form is a definition, not an observation, so it must be reconciled with the usual description “sum the decimal digits until one digit remains”. Write $S(n)$ for the sum of the decimal digits of n , and let $S^\infty(n)$ be the result of iterating S while $n \geq 10$.

Lemma 2.2. *If $n \geq 10$ then $S(n) < n$. Consequently the iteration defining S^∞ terminates.*

Proof. Write $n = 10q + r$ with $r = n \bmod 10$ and $q = \lfloor n/10 \rfloor \geq 1$. Then $S(n) = r + S(q) \leq r + q < r + 10q = n$, using $S(q) \leq q$. $\square$

Theorem 2.3 (Casting out nines). $\text{dr}(S(n)) = \text{dr}(n)$ for all n , and $S^\infty(n) = \text{dr}(n)$.

Proof. Since $10 \equiv 1 \pmod{9}$ we have $n \equiv S(n) \pmod{9}$, and $S(n) = 0$ if and only if $n = 0$ (a nonzero n has a nonzero leading digit). Two naturals that are congruent mod 9 and simultaneously zero or nonzero have the same digital root, because dr only depends on $(n - 1) \bmod 9$ for $n > 0$. The second statement follows by strong induction using Lemma 2.2, together with $\text{dr}(n) = n$ for $n \leq 9$. $\square$

Proposition 2.4. *For $n > 0$ we have $\text{dr}(n) \in \{1, \dots, 9\}$ and $\text{dr}(n) \equiv n \pmod{9}$; moreover dr is idempotent.*

Baby steps

Before the trinity itself, four elementary consequences of Definition 2.1 that will be used freely later. All four are formalised in `RequestProject/BabySteps369.lean`.

Proposition 2.5 (Reduce before you compute). *For $a, b > 0$,*

$$\text{dr}(a + b) = \text{dr}(\text{dr}(a) + \text{dr}(b)), \quad \text{dr}(ab) = \text{dr}(\text{dr}(a) \text{ dr}(b)).$$

Proof. $\text{dr}(x) \equiv x \pmod{9}$, so the two sides of each identity are congruent mod 9; and both are nonzero, since dr of a positive number is positive. Now apply the congruence criterion used in the proof of Theorem 2.3. $\square$

Proposition 2.6 (Nine is divisibility by nine). *For $n > 0$ we have $\text{dr}(n) = 9 \iff 9 \mid n$. In particular $\text{dr}(9n) = 9$ for $n > 0$, and $\text{dr}(369) = \text{dr}(3 + 6 + 9) = 9$.*

Proposition 2.7 (Two families landing on nine). *$\text{dr}(n!) = 9$ for every $n \geq 6$; and the digital roots of the triangular numbers $T_n = n(n+1)/2$ are periodic with period nine: $\text{dr}(T_{n+9}) = \text{dr}(T_n)$ for $n > 0$.*

Proof. For the first, $9 \mid 6! = 720$ and $6! \mid n!$ for $n \geq 6$; apply Proposition 2.6. For the second, $T_{n+9} = T_n + 9n + 45$, so $T_{n+9} \equiv T_n \pmod{9}$, and both are positive. $\square$

Proposition 2.8 (The two worlds are invariant). *Let $V(d) = \text{dr}(2d)$ be one doubling step on digital roots. Then V maps $\{3, 6, 9\}$ into itself ($3 \mapsto 6 \mapsto 3$, $9 \mapsto 9$) and $\{1, 2, 4, 8, 7, 5\}$ into itself; moreover $V^6(1) = 1$ and $V^n(9) = 9$ for all n .*

3 The trinity

Theorem 3.1 (The trinity is divisibility by three). *For $n > 0$,*

$$\text{dr}(n) \in \{3, 6, 9\} \iff 3 \mid n.$$

Proof. $\text{dr}(n) \equiv n \pmod{9}$ and $\text{dr}(n) \in \{1, \dots, 9\}$. Among these nine values exactly 3, 6, 9 are divisible by 3. It remains to see that $3 \mid n$ if and only if $3 \mid (n \bmod 9)$, and this is exactly where $9 = 3 \cdot 3$ is used: writing $n = 9q + (n \bmod 9)$ and using $3 \mid 9$, the two divisibilities differ by the multiple $9q$ of 3. (In the formalisation this reduction is not hidden in a lemma: the proof of `digitalRoot_mem_trinity_iff` carries the two facts $\text{dr}(n) \bmod 9 = n \bmod 9$ and $\text{dr}(n) = ((n - 1) \bmod 9) + 1$ into linear integer arithmetic over the moduli 3 and 9, which is precisely the step $3 \mid 9$ made mechanical.) $\square$

This single theorem explains the entire mystique. Everything usually asserted about the trinity is a statement about the ideal $3\mathbb{Z}$ read through dr .

Corollary 3.2 (Closed and absorbing). *If $\text{dr}(a), \text{dr}(b) \in \{3, 6, 9\}$ then $\text{dr}(a + b) \in \{3, 6, 9\}$; and if $\text{dr}(a) \in \{3, 6, 9\}$ and $b > 0$ then $\text{dr}(ab) \in \{3, 6, 9\}$.*

Theorem 3.3 (Generation). *A natural number n is of the form $3a + 6b + 9c$ with $a, b, c \in \mathbb{N}$ if and only if $3 \mid n$; and in $\mathbb{Z}$,*

$$\langle 3, 6, 9 \rangle = 3\mathbb{Z}.$$

So “3, 6, 9 is all you need” is literally true — provided the thing you need is the multiples of three. It is, however, not an efficient slogan.

Theorem 3.4 (Redundancy). *$\langle 3 \rangle = \langle 3, 6, 9 \rangle$ and $\langle 6, 9 \rangle = 3\mathbb{Z}$.*

Proof. $6 = 2 \cdot 3$ and $9 = 3 \cdot 3$ are multiples of 3, giving the first equality; and $9 - 6 = 3$ gives the second. $\square$

Remark 3.5. Thus 3 alone is all you need, and so is $\{6, 9\}$. The trinity is a generating set of $3\mathbb{Z}$, but a maximally uneconomical one.

Theorem 3.6 (Powers of three). *$\text{dr}(3^n) = 9$ for every $n \geq 2$.*

Theorem 3.7 (The doubling circuit). *$\text{dr}(2^n) = \text{dr}(2^{n \bmod 6})$ for all n . Hence $\text{dr}(2^n)$ always lies in the cycle $\{1, 2, 4, 8, 7, 5\}$, and $\text{dr}(2^n) \notin \{3, 6, 9\}$ for every n .*

Proof. $2^6 = 64 \equiv 1 \pmod{9}$, so $n \mapsto 2^n \bmod 9$ has period dividing 6; evaluating the six residues gives the cycle. That the cycle misses the trinity is Theorem 3.1 applied to $3 \nmid 2^n$, which holds since 3 is prime and $3 \neq 2$. $\square$

Remark 3.8 (Minimality). The congruence $2^6 \equiv 1 \pmod{9}$ shows only that 6 is a period. It is in fact the minimal one: every d with $\text{dr}(2^{n+d}) = \text{dr}(2^n)$ for all n satisfies $6 \mid d$, because taking $n = 0$ forces $\text{dr}(2^d) = 1$, and among the six residues $r < 6$ only $r = 0$ has $\text{dr}(2^r) = 1$. This is `Tesla369.digitalRoot_two_pow_period_six_minimal`.

The “1, 2, 4, 8, 7, 5 circuit and the 3, 6, 9 triangle” of vortex mathematics is therefore nothing but the decomposition of $\mathbb{Z}/9$ into the subgroup $\{0, 3, 6\}$ of index 3 and the orbit of the unit 2, which is all of $(\mathbb{Z}/9)^\times$, of order 6.

Proposition 3.9. *In $\mathbb{Z}/9$, the multiples of 3 form the set $\{0, 3, 6\}$, the unique subgroup of order three; the digital roots 3, 6, 9 reduce to 3, 6, 0 respectively.*

4 Sieving by 2 and 3

The trinity story is about $9 = 3^2$. Its natural companion is a story about $6 = 2 \cdot 3$: the first two primes already do most of the work in the sieve of Eratosthenes.

Theorem 4.1 (Exact coverage). *For every $k \in \mathbb{N}$,*

$$\#\{n < 6k : 2 \mid n \text{ or } 3 \mid n\} = 4k.$$

Equivalently, sieving by 2 and 3 removes a density of exactly $2/3$, leaving $\varphi(6)/6 = 1/3$.

Proof. Divisibility by 2 or 3 depends only on the residue mod 6, and among $\{0, 1, 2, 3, 4, 5\}$ the sieved residues are $\{0, 2, 3, 4\}$, four of six. Induct on k , splitting $[0, 6(k+1))$ as $[0, 6k) \sqcup [6k, 6k+6)$ and translating the block by $6k$. $\square$

Theorem 4.2. *Every prime $p > 3$ satisfies $p \equiv 1$ or $5 \pmod{6}$; equivalently $p \equiv \pm 1 \pmod{6}$.*

Combining with Theorem 3.1 gives the bridge between the two halves of the paper.

Corollary 4.3 (Primes avoid the trinity). *Every prime $p > 3$ has $\text{dr}(p) \in \{1, 2, 4, 5, 7, 8\}$.*

5 The general finite sieve

There is nothing special about 2 and 3 in Theorem 4.1 except that they are the first two primes, and the honest version of the statement is the Legendre density formula for an arbitrary finite set of primes. Fix a finite set S of primes and put

$$M = M(S) = \prod_{p \in S} p, \quad n \text{ is sieved if } p \mid n \text{ for some } p \in S.$$

The proof has three moving parts, each formalised separately.

Lemma 5.1 (Blockwise counting). *Let f be a predicate on $\mathbb{N}$ depending only on the residue mod m , i.e. $f(n) \iff f(n \bmod m)$. Then for every k ,*

$$\#\{n < mk : f(n)\} = k \cdot \#\{n < m : f(n)\}.$$

Proof. Induction on k , splitting $[0, m(k+1)) = [0, mk) \sqcup [mk, mk+m)$ and translating the last block by mk , which does not change residues mod m . $\square$

Lemma 5.2 (Survivors are the coprime residues). *For every n : n is not sieved by S if and only if $\gcd(M, n) = 1$. Consequently one period contains exactly $\varphi(M)$ survivors, and $\varphi(M) = \prod_{p \in S} (p-1)$.*

Proof. If $p \in S$ divides n then $p \mid \gcd(M, n)$, so the gcd exceeds 1. Conversely if $\gcd(M, n) > 1$, some prime q divides both; from $q \mid M = \prod_{p \in S} p$ and primality, q equals some $p \in S$, which then divides n . The count in one period is the definition of Euler's totient, and multiplicativity of φ across the pairwise coprime prime factors of the squarefree M gives $\varphi(M) = \prod (p-1)$. $\square$

Theorem 5.3 (Legendre density of a finite sieve). *For every $k \in \mathbb{N}$,*

$$\#\{n < Mk : n \text{ survives}\} = k \prod_{p \in S} (p-1), \quad \#\{n < Mk : n \text{ is sieved}\} = k \left(M - \prod_{p \in S} (p-1) \right),$$

and for $k \geq 1$ the sieved proportion is exactly

$$\frac{\#\{n < Mk : n \text{ is sieved}\}}{Mk} = 1 - \prod_{p \in S} \left(1 - \frac{1}{p}\right).$$

Proof. The sieved predicate depends only on the residue mod M , since each $p \in S$ divides M ; so Lemma 5.1 applies and Lemma 5.2 evaluates one period. The density statement is the identity $\prod_{p \in S} (1 - 1/p) = (\prod_{p \in S} (p-1))/M$. $\square$

Corollary 5.4. *For $S = \{2, 3\}$: $M = 6$, $\prod (p-1) = 2$, and the sieved count over $[0, 6k)$ is $4k$ — Theorem 4.1 again, with density $1 - \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{3}$.*

Remark 5.5. This is an exact identity along the arithmetic progression of window lengths $M, 2M, 3M, \dots$, not an asymptotic: no error term appears, because the sieve is exactly periodic. It is the elementary, “no analysis” half of sieve theory; the difficulty of genuine sieve methods begins precisely when S is allowed to grow with the length of the window, at which point the product $\prod (1 - 1/p)$ ceases to be a good approximation to the surviving density and the Mertens and Brun/Selberg theory takes over. Nothing of that is claimed here.

6 Hecke eigenvalues at 2 and 3

The same theme — the first two primes carry a great deal of information — has a much deeper incarnation in the theory of modular forms, where the Hecke operators T_2 and T_3 are famously informative. We isolate the purely multiplicative content, so that the statement can be proved from axioms rather than from analysis.

Definition 6.1 (Hecke system). Let R be a commutative ring and $k \in \mathbb{N}$. A function $a : \mathbb{N} \rightarrow R$ is a *Hecke system of weight k* if

$$\begin{aligned} a(1) &= 1, \\ a(mn) &= a(m)a(n) \quad \text{whenever } \gcd(m, n) = 1, \\ a(p^{j+2}) &= a(p)a(p^{j+1}) - p^{k-1}a(p^j) \quad \text{for every prime } p \text{ and } j \geq 0. \end{aligned}$$

These are exactly the relations satisfied by the normalised Hecke eigenvalues of an eigenform of weight k . The axioms are satisfiable in an elementary way:

Proposition 6.2 (Non-vacuity). *Over any commutative ring R , the divisor-counting function $n \mapsto d(n)$ is a Hecke system of weight 1.*

Proof. $d(1) = 1$; d is multiplicative on coprime arguments; and $d(p^{j+2}) = j + 3 = 2(j + 2) - (j + 1) = d(p)d(p^{j+1}) - d(p^j)$, which is the recursion with $p^{k-1} = p^0 = 1$. $\square$

Theorem 6.3 (T_2 and T_3 determine the 3-smooth part). *Let a, b be Hecke systems of the same weight k over R with $a(2) = b(2)$ and $a(3) = b(3)$. Then $a(2^i 3^j) = b(2^i 3^j)$ for all $i, j \geq 0$; that is, a and b agree on the whole multiplicative semigroup*

$$\{1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 27, 36, \dots\}$$

generated by the first two primes.

Proof. The recursion expresses $a(p^{j+2})$ in terms of $a(p^{j+1})$ and $a(p^j)$, so by strong induction $a(p^j)$ is determined by $a(p)$ (base cases $a(p^0) = 1$ and $a(p^1) = a(p)$). Applying this at $p = 2$ and $p = 3$ and using multiplicativity across the coprime factorisation $2^i \cdot 3^j$ finishes the proof. $\square$

Corollary 6.4. *For any Hecke system of weight k : $a(4) = a(2)^2 - 2^{k-1}$, $a(6) = a(2)a(3)$ and $a(9) = a(3)^2 - 3^{k-1}$.*

So the numbers 6 and 9 — the companions of 3 — are precisely the first two values that T_2 and T_3 hand over for free. In this sense the slogan has a respectable cousin: for 3-smooth arguments, a_2 and a_3 really are all you need.

7 Multiplicative closure: from $\{2, 3\}$ to all primes

Theorem 6.3 stops at 3-smooth numbers, which is an artefact of the slogan rather than of the mathematics. The same two ingredients — the prime-power recursion and coprime multiplicativity — give the general statement, and it is worth recording because it says exactly how far the purely algebraic “small primes suffice” phenomenon goes.

Call $n \geq 1$ *S -smooth*, for a set S of primes, if every prime factor of n lies in S .

Theorem 7.1 (Multiplicative closure). *Let a, b be Hecke systems of the same weight k over R and let S be a set of primes with $a(p) = b(p)$ for all $p \in S$. Then $a(n) = b(n)$ for every positive S -smooth n .*

Proof. By induction along the multiplicative structure of n (prime powers, and coprime products of smaller factors). For a prime power p^m with $p \in S$ the recursion determines $a(p^m)$ from $a(p)$ as in Theorem 6.3. For $n = xy$ with $x, y > 1$ coprime, the prime factors of n split between x and y , so both are S -smooth and $a(n) = a(x)a(y) = b(x)b(y) = b(n)$. $\square$

Corollary 7.2 (Agreement at all primes is agreement everywhere). *If $a(p) = b(p)$ for every prime p , then $a(n) = b(n)$ for every $n \geq 1$.*

Remark 7.3. The restriction to $n \geq 1$ is not cosmetic: the axioms constrain $a(0)$ not at all, since $\gcd(m, n) = 1$ with $mn = 0$ forces $\{m, n\} = \{0, 1\}$ and the multiplicativity relation then reads $a(0) = a(0) \cdot 1$. Formally, replacing $a(0)$ by an arbitrary $c \in R$ turns a Hecke system into a Hecke system (`Hecke23.a_zero_undetermined`).

Corollary 7.4 (Finite truncation). *If $a(p) = b(p)$ for every prime $p \leq N$, then $a(n) = b(n)$ for every $1 \leq n \leq N$.*

Proof. Every prime factor of such an n is at most $n \leq N$, so n is S -smooth for $S = \{p : p \leq N\}$; apply Theorem 7.1. $\square$

Corollary 7.4 is the honest elementary shadow of *multiplicity one*: the $\pi(N)$ eigenvalues below N pin down an initial segment of length N . A finite pigeonhole argument upgrades this to a separation statement for finite families.

Theorem 7.5 (Weak multiplicity one for a finite family). *Let $(a^{(i)})_{i \in I}$ be a finite family of Hecke systems of weight k . Then there exists N such that for all $i, j \in I$: if $a^{(i)}(p) = a^{(j)}(p)$ for every prime $p \leq N$, then $a^{(i)} = a^{(j)}$ on all of $\mathbb{N}_{\geq 1}$.*

Proof. For each pair (i, j) for which the two systems differ somewhere, choose a witness $n_{ij} \geq 1$ with $a^{(i)}(n_{ij}) \neq a^{(j)}(n_{ij})$, and let N be the maximum of the finitely many n_{ij} . If $a^{(i)}$ and $a^{(j)}$ agree at all primes $p \leq N$ but differ somewhere, then $n_{ij} \leq N$ is a point of disagreement, contradicting Corollary 7.4. $\square$

Remark 7.6 (What this is, and what it is not). Theorem 7.5 is a pigeonhole statement: the bound N is manufactured from the places where the given systems already differ, and it is in no way effective in terms of the weight k or the level. The real theorems in this direction — Sturm’s bound, and multiplicity one for newforms — produce an N depending only on the weight and level, and their proofs are not algebraic: they use the geometry of modular curves and, in the analytic incarnation, Rankin–Selberg theory. The gap between Corollary 7.4 and those results is the honest measure of how much of “a few Hecke eigenvalues determine everything” this paper proves, namely: the algebraic part, all of it, and none of the analytic part.

8 Five Hecke words: $T_2, T_3, T_2T_3, T_2T_3T_2, T_3T_3T_3$

Section 6 used the operators T_2 and T_3 one at a time. Here we follow a single thread of five *words* in those two letters,

$$T_2, \quad T_3, \quad T_2T_3, \quad T_2T_3T_2, \quad T_3T_3T_3,$$

and ask, for each of them, the same two questions: by what scalar does it act on an eigenform, and which single Hecke operators T_n does it decompose into? The five answers are collected in Figure 1; the point of the section is that both columns are forced by the two numbers $a(2)$ and $a(3)$, and that the second column is where the arithmetic of the exponent $k-1$ becomes visible.

The two sides

On the eigenvalue side nothing is needed beyond the definition: if f is an eigenform with system a , then $T_p f = a(p)f$, so the word $w = T_{p_1} T_{p_2} \cdots T_{p_r}$ multiplies f by

$$\lambda(w) = a(p_1) a(p_2) \cdots a(p_r),$$

formalised as `HeckeWords.wordEigenvalue`. On the operator side we work in the free R -module on symbols $T_1, T_2, T_3, \dots$ with the classical structure constants

$$T_m T_n = \sum_{d|\gcd(m,n)} d^{k-1} T_{mn/d^2}, \quad (1)$$

which we *take as the definition* of the product (`HeckeWords.basisMul` and its bilinear extension `HeckeWords.hmul`); we do not reprove that (1) describes the double-coset algebra of $\mathrm{SL}_2(\mathbb{Z})$, which is classical and is not formalised here.

Proposition 8.1 (Coprime letters). $T_2 T_3 = T_6$, and $\lambda(T_2 T_3) = a(2)a(3) = a(6)$.

Proof. $\gcd(2, 3) = 1$, so the sum (1) has the single term $d = 1$, giving T_6 . On the eigenvalue side this is the coprime multiplicativity axiom. $\square$

Proposition 8.2 (One repeated letter). $(T_2 T_3) T_2 = T_{12} + 2^{k-1} T_3$, and

$$\lambda(T_2 T_3 T_2) = a(2)^2 a(3) = a(12) + 2^{k-1} a(3).$$

Proof. By Proposition 8.1 the product is $T_6 T_2$; now $\gcd(6, 2) = 2$, whose divisors are 1 and 2, so (1) gives $T_{12} + 2^{k-1} T_{12/4} = T_{12} + 2^{k-1} T_3$. For the scalar, $12 = 4 \cdot 3$ with $\gcd(4, 3) = 1$ gives $a(12) = a(4)a(3)$, and $a(4) = a(2)^2 - 2^{k-1}$ by the prime-power recursion; hence $a(12) + 2^{k-1} a(3) = a(2)^2 a(3)$. $\square$

Proposition 8.3 (The same letter three times). $T_3 T_3 = T_9 + 3^{k-1} T_1$ and

$$(T_3 T_3) T_3 = T_{27} + 2 \cdot 3^{k-1} T_3, \quad \lambda(T_3 T_3 T_3) = a(3)^3 = a(27) + 2 \cdot 3^{k-1} a(3).$$

Proof. $\gcd(3, 3) = 3$ gives $T_3 T_3 = T_9 + 3^{k-1} T_1$. Multiplying on the right by T_3 and using $\gcd(9, 3) = 3$ gives $T_{27} + 3^{k-1} T_3$ from the first summand and $3^{k-1} T_3$ from the second, i.e. $T_{27} + 2 \cdot 3^{k-1} T_3$: the two echoes have the same size and add. For the scalar, the recursion at $p = 3$ gives $a(27) = a(3)a(9) - 3^{k-1} a(3)$ and $a(9) = a(3)^2 - 3^{k-1}$, so $a(27) = a(3)^3 - 2 \cdot 3^{k-1} a(3)$. $\square$

The two sides are consistent word by word: evaluating a Hecke algebra element at the system a by $T_n \mapsto a(n)$ (linearly, `HeckeWords.evalT`) turns each of the three decompositions above into the corresponding scalar identity (`HeckeWords.evalT_word_two_three`, `..._two_three_two`, `..._three_three_three`). We check this for these five words only; that the evaluation map is an algebra homomorphism for *all* elements is equivalent to the full identity $a(m)a(n) = \sum_{d|\gcd(m,n)} d^{k-1} a(mn/d^2)$ and is not proved here.

Corollary 8.4. *The five multipliers are the polynomials*

$$a(2), \quad a(3), \quad a(2)a(3), \quad a(2)^2a(3), \quad a(3)^3$$

in $a(2)$ and $a(3)$ — as Theorem 6.3 predicts, since 1, 6, 12, 27 are all 3-smooth.

Example 8.5 (The discriminant form). For Δ , of weight $k = 12$, one has $a(2) = \tau(2) = -24$ and $a(3) = \tau(3) = 252$. The axioms then force

$$a(6) = -6048, \quad a(12) = -370944, \quad a(27) = -73279080,$$

and the five multipliers are -24 , 252 , -6048 , 145152 and 16003008 . Note the two cancellations: $145152 = -370944 + 2^{11} \cdot 252$ and $16003008 = -73279080 + 2 \cdot 3^{11} \cdot 252$. This is `HeckeWords.delta_values`, proved from the two inputs alone.

$$\begin{array}{ccccc} T_1 & \longrightarrow & T_2 & \xrightarrow{T_3} & T_2T_3 & \xrightarrow{T_2} & T_2T_3T_2 \\ & \searrow & T_3 & \dashrightarrow & T_3T_3 & \xrightarrow{T_3} & T_3T_3T_3 \end{array}$$

word w	expansion	multiplier $\lambda(w)$	$\lambda(w)$ for Δ , $k = 12$
T_2	T_2	$a(2)$	-24
T_3	T_3	$a(3)$	252
T_2T_3	T_6	$a(2)a(3) = a(6)$	-6048
$T_2T_3T_2$	$T_{12} + 2^{k-1}T_3$	$a(2)^2a(3) = a(12) + 2^{k-1}a(3)$	145152
$T_3T_3T_3$	$T_{27} + 2 \cdot 3^{k-1}T_3$	$a(3)^3 = a(27) + 2 \cdot 3^{k-1}a(3)$	16003008

Figure 1: The five words, their expansions in the basis $\{T_n\}$ and their multipliers. The dashed step passes through the intermediate word T_3T_3 , which is not one of the five. A standalone drawing of the same data is `paper/hecke369.svg`.

9 Which base? The trinity of an arbitrary base

Everything so far was said in base ten. The next question is what, if anything, ten contributes. The answer is: only the identity $10 - 1 = 9$ together with $3 \mid 9$.

Definition 9.1. For $b \geq 2$ the base- b digital root is $\text{dr}_b(0) = 0$ and $\text{dr}_b(n) = 1 + ((n-1) \bmod (b-1))$ for $n > 0$; thus $\text{dr}_b(n) \equiv n \pmod{b-1}$ and $1 \leq \text{dr}_b(n) \leq b-1$ for $n > 0$. For $b = 10$ this is `dr`.

Theorem 9.2 (Casting out $(b-1)$ s). *For $b \geq 3$ and every n , summing the base- b digits of n leaves $\text{dr}_b(n)$ unchanged.*

Proof. $b \equiv 1 \pmod{b-1}$, so n and its base- b digit sum are congruent mod $b-1$; and the digit sum vanishes only for $n = 0$, since the leading digit is nonzero. Now apply the congruence criterion for dr_b . $\square$

For a divisor d of $b-1$ put

$$T_b(d) = \{x : 1 \leq x \leq b-1, d \mid x\} = \{d, 2d, \dots, b-1\},$$

the *trinity* of d in base b . For $b = 10$, $d = 3$ this is $\{3, 6, 9\}$.

Theorem 9.3 (The general trinity theorem). *Let $b \geq 2$, let $d \mid b - 1$ and let $n > 0$. Then*

$$\text{dr}_b(n) \in T_b(d) \iff d \mid n.$$

Proof. $\text{dr}_b(n) \equiv n \pmod{b-1}$ and $d \mid b-1$, so $\text{dr}_b(n) \equiv n \pmod{d}$; hence $d \mid \text{dr}_b(n) \iff d \mid n$. Since $1 \leq \text{dr}_b(n) \leq b-1$ for $n > 0$, membership in $T_b(d)$ is exactly divisibility by d . $\square$

Theorem 3.1 is the case $b = 10$, $d = 3$, and the proof shows that the only input is $d \mid b - 1$: nothing about ten, nothing about three.

Proposition 9.4. $\#T_b(d) = (b-1)/d$. *Consequently, for $b \geq 2$ there is a divisor d of $b-1$ with $\#T_b(d) = 3$ if and only if $b \equiv 1 \pmod{3}$.*

So three-element trinities exist in bases 4, 7, 10, 13, ... and in no others. In base 4 the trinity is $\{1, 2, 3\}$ with $d = 1$: it detects nothing. In base 7 it is $\{2, 4, 6\}$ with $d = 2$: it detects only parity, which the last digit already does. Base ten is the first base in which a three-element trinity detects an odd prime.

Theorem 9.5 (Why ten). *Base ten is the least base b admitting a divisor d of $b-1$ with $\#T_b(d) = 3$ and d an odd prime; namely $d = 3$, $T_{10}(3) = \{3, 6, 9\}$.*

Proof. Such a d satisfies $b-1 = 3d$; an odd prime has $d \geq 3$, so $b \geq 10$. For $b = 10$ take $d = 3$. $\square$

The doubling circuit generalises just as cleanly, and so does the reason it avoids the trinity.

Theorem 9.6 (General circuit). *Let $a > 0$ and suppose $a^t \equiv 1 \pmod{b-1}$. Then $\text{dr}_b(a^{n+t}) = \text{dr}_b(a^n)$ for all n . If moreover $d \mid b-1$, $d > 1$ and $\gcd(a, d) = 1$, then $\text{dr}_b(a^n) \notin T_b(d)$ for all n .*

Proof. $a^{n+t} = a^n a^t \equiv a^n \pmod{b-1}$, and both are positive, so the digital roots agree. For the second claim, $d \mid a^n$ with $\gcd(a^n, d) = 1$ forces $d \mid 1$; conclude by Theorem 9.3. $\square$

Taking $b = 10$, $a = 2$, $t = 6$, $d = 3$ recovers Theorem 3.7 exactly. In base 16, for instance, $2^4 \equiv 1 \pmod{15}$ and the circuit is the four-cycle 1, 2, 4, 8, while the trinities are those of $d \in \{3, 5, 15\}$; there is nothing special about the number six either.

Finally, the rule that makes “vortex” computations work at all is that the digital root is a ring-theoretic reduction and not merely a curiosity.

Proposition 9.7 (Digital-root arithmetic). *For $x, y > 0$ and any base b ,*

$$\text{dr}_b(x + y) = \text{dr}_b(\text{dr}_b(x) + \text{dr}_b(y)), \quad \text{dr}_b(xy) = \text{dr}_b(\text{dr}_b(x) \text{dr}_b(y)).$$

Equivalently, in base ten, dr is compatible with the reduction $\mathbb{N} \rightarrow \mathbb{Z}/9$.

10 The vortex, as an orbit decomposition

The familiar “vortex” picture — a circle of nine positions, a six-step loop $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 7 \rightarrow 5 \rightarrow 1$, and a separate triangle on 3, 6, 9 — is precisely the decomposition of $\mathbb{Z}/9$ into orbits of multiplication by 2.

Theorem 10.1 (The vortex diagram). *Multiplication by 2 on $\mathbb{Z}/9$ has exactly three orbits:*

$$\{0\}, \quad \{3, 6\}, \quad \{1, 2, 4, 8, 7, 5\},$$

of sizes 1, 2 and 6; they are pairwise disjoint and cover $\mathbb{Z}/9$, so $9 = 1 + 2 + 6$. The six-element orbit is exactly the unit group $(\mathbb{Z}/9)^\times$.

The triangle is thus not a third structure alongside the circuit: it is the complement of the units, i.e. the subgroup $3\mathbb{Z}/9\mathbb{Z}$, split by doubling into the fixed point 0 and the two-cycle $3 \leftrightarrow 6$. Doubling is transitive on the units because 2 is a primitive root mod 9; that single fact produces both the six-step loop and the untouched triangle.

11 The Fibonacci vortex

The second standard claim of vortex mathematics concerns the Fibonacci numbers F_n : their digital roots are said to repeat with period 24, and the trinity to occur at regular intervals within the cycle. Both are true.

Theorem 11.1 (Period 24). *$F_{n+24} \equiv F_n \pmod{9}$ for all n ; hence $\text{dr}(F_{n+24}) = \text{dr}(F_n)$ for all $n \geq 1$, and the cycle of digital roots is*

$$1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9.$$

Moreover 24 is the minimal period.

Proof. From $F_{m+n+1} = F_m F_n + F_{m+1} F_{n+1}$ with $m = 23$: $F_{n+24} = F_{23} F_n + F_{24} F_{n+1}$, and $F_{23} = 28657 \equiv 1$, $F_{24} = 46368 \equiv 0 \pmod{9}$. Positivity is needed to pass from the congruence to digital roots: $\text{dr}(F_0) = 0$ but $\text{dr}(F_{24}) = 9$, so the cycle begins at $n = 1$. Minimality: a period d may be reduced mod 24, and for each r with $0 < r < 24$ one of the two positions $n = 1, 2$ already separates $\text{dr}(F_{n+r})$ from $\text{dr}(F_n)$. $\square$

Theorem 11.2 (The trinity in the Fibonacci sequence). *For $n > 0$,*

$$\text{dr}(F_n) \in \{3, 6, 9\} \iff 4 \mid n.$$

Proof. By Theorem 3.1 the left side is $3 \mid F_n$. Now $3 = F_4$ and $\gcd(F_m, F_n) = F_{\gcd(m, n)}$, so $3 \mid F_n$ iff $F_{\gcd(4, n)} = 3$; as $\gcd(4, n)$ is one of 1, 2, 4 with $F_1 = F_2 = 1$ and $F_4 = 3$, this happens exactly when $\gcd(4, n) = 4$. $\square$

So the trinity occurs six times in each cycle of twenty-four, at $n \equiv 4, 8, 12, 16, 20, 0 \pmod{24}$ — the observed regularity, with its reason: 3 is a Fibonacci number, and Fibonacci divisibility is governed by index divisibility. As a closing curiosity the twenty-four digital roots of one cycle sum to $117 = 9 \cdot 13$.

12 The actual modular form, drawn

Sections 6–8 spoke of an eigenform only through its eigenvalue system a . It is time to look at one. The smallest example is the discriminant form of weight 12 and level 1,

$$\Delta(z) = q \prod_{m \geq 1} (1 - q^m)^{24} = \sum_{n \geq 1} \tau(n) q^n, \quad q = e^{2\pi i z}, \quad \text{Im } z > 0, \quad (2)$$

whose coefficients are Ramanujan's τ . All Lean names in this section live in the namespace `DeltaGraph`, whose prefix we drop. In the Lean development the product (2) is expanded by truncated polynomial arithmetic (`deltaCoeffs`, `tau`), so the numbers below are computed and checked by the kernel, not quoted.

Proposition 12.1 (The plotted data). $\tau(1), \dots, \tau(30)$ are

1	1	2	-24	3	252	4	-1472	5	4830
6	-6048	7	-16744	8	84480	9	-113643	10	-115920
11	534612	12	-370944	13	-577738	14	401856	15	1217160
16	987136	17	-6905934	18	2727432	19	10661420	20	-7109760
21	-4219488	22	-12830688	23	18643272	24	21288960	25	-25499225
26	13865712	27	-73279080	28	24647168	29	128406630	30	-29211840

Proof. Kernel evaluation of (2) truncated at degree 30; the factors $(1 - q^m)^{24}$ with $m > n$ do not affect the coefficient of q^n . This is `tau_table`. $\square$

The five words of Section 8 can now be read on an actual form rather than on an abstract system.

Proposition 12.2 (The five words, on Δ). *With $k = 12$,*

$$\begin{aligned}\tau(4) &= \tau(2)^2 - 2^{11}, & \tau(6) &= \tau(2)\tau(3), & \tau(9) &= \tau(3)^2 - 3^{11}, \\ \tau(12) &= \tau(2)^2\tau(3) - 2^{11}\tau(3), & \tau(27) &= \tau(3)^3 - 2 \cdot 3^{11}\tau(3),\end{aligned}$$

and the five multipliers of Example 8.5 are $-24, 252, -6048, 145152, 16003008$.

Proof. Each side is an integer computed from Proposition 12.1; the identities are verified by evaluation (`tau_four`, `tau_six`, `tau_nine`, `tau_twelve`, `tau_twentyseven`, `tau_word_values`). $\square$

Remark 12.3. Proposition 12.2 is a check, not a derivation: that τ satisfies the Hecke recursions for *all* n — Mordell's theorem — is not proved here. What the Lean file establishes is that the abstract identities of Section 8 and the actual integers agree at the five words, which is exactly what the figures display.

Proposition 12.4 (Vertical scale). *For every $n \leq 30$, $\tau(n)^2 \leq d(n)^2 n^{11}$, i.e. $|\tau(n)| \leq d(n) n^{11/2}$; in particular $|\tau(p)| \leq 2p^{11/2}$ at the primes $p \leq 30$.*

Proof. Evaluation over $1 \leq n \leq 30$: `tau_ramanujan_bound`. The general statement is Deligne's theorem and is not proved here; only the plotted range is checked. $\square$

Divisibility by three does not vanish when one passes from digital roots to modular forms; it reappears as the three-adic behaviour of τ .

Proposition 12.5 (The trinity, on the coefficients). *For $n \leq 30$ prime to 3,*

$$\tau(n) \equiv n^2 \sigma_7(n) \pmod{27}, \quad \text{hence} \quad 3 \mid \tau(n) \iff 3 \mid \sigma(n).$$

Moreover, for any nonzero integer m , $\text{dr } |m| \in \{3, 6, 9\}$ if and only if $3 \mid m$.

Proof. The two congruence statements are checked on $1 \leq n \leq 30$ (`tau_congr_mod_27`, `three_dvd_tau_iff_three_dvd_sigma`); the second follows from the first because $d^7 \equiv d$ and $n^2 \equiv 1$ modulo 3. The last sentence is Theorem 3.1 applied to $|m|$ (`digitalRoot_natAbs_mem_trinity_iff`). Consequently the highlighted bars of Figure 2 — those with $\text{dr}|\tau(n)| \in \{3, 6, 9\}$, listed in `digitalRoot_tau_table` — are exactly the n with $3 \mid \tau(n)$. $\square$

The graph on the upper half plane

The function itself, and not only its coefficients, can be drawn. Write $q(z) = e^{2\pi iz}$ and let $\Delta_N(z) = \sum_{n=1}^N \tau(n)q(z)^n$ be the partial expansion actually evaluated by the plotting routine.

Proposition 12.6 (What the picture shows). $q(z+1) = q(z)$ and $\Delta_N(z+1) = \Delta_N(z)$ for every N ; and $|q(z)| = e^{-2\pi \text{Im} z}$, which is < 1 throughout the upper half plane.

Proof. $e^{2\pi i} = 1$ gives the first, hence the second termwise; the third is $|e^w| = e^{\text{Re} w}$. In Lean: `qParam_add_one`, `deltaApprox_periodic`, `abs_qParam`, `abs_qParam_lt_one`. $\square$

These two facts are precisely the two visible features of the phase portrait, Figure 3: the colours repeat with horizontal period one, and they fade upwards, because Δ has a simple zero at the cusp $q = 0$ with $\tau(1) = 1$. A third feature is visible there, the symmetry across the arcs $|z| = 1$, which is the invariance $\Delta(-1/z) = z^{12}\Delta(z)$; that one is not a property of the truncations at all, and it is proved for the true infinite product in Section 14 (Theorem 14.4).

All three figures are produced by `paper/make_delta_graphs.py`, which recomputes the coefficients from (2) and refuses to run unless they agree with the list of Proposition 12.1; standalone SVG copies are `paper/delta_coefficients.svg`, `paper/delta_domain.svg` and `paper/delta_profiles.svg`.

Remark 12.7. Numerically, $\Delta(iy)$ is real and positive, decays like $e^{-2\pi y}$ at the cusp and, by the invariance just mentioned, like y^{-12} as $y \rightarrow 0$; its maximum along the imaginary axis is attained near $y \approx 0.523$. This is the left panel of Figure 4. The computation is floating point and is offered as a picture, not as a theorem.

13 The vertical walls: why the picture stands up straight

Figure 3 has a fourth visible feature, which is neither the period nor the decay at the cusp: right above the points $x = 0, \pm\frac{1}{2}, \pm 1$ the colour boundaries and the brightness bands run *exactly* vertically, and every contour of $|\Delta|$ crosses those lines at a right angle. This section proves that this is forced, and isolates exactly what forces it. All Lean names of this section live in the namespace `VerticalWalls` (`RequestProject/VerticalWalls.lean`).

Two ingredients are used, and nothing else: the coefficients are *real*, and the expansion has *period one*. So we fix an arbitrary real sequence $c : \mathbb{N} \rightarrow \mathbb{R}$ and a truncation length N and write

$$F(z) = \sum_{n=1}^N c(n) q(z)^n, \quad q(z) = e^{2\pi iz}, \quad (3)$$

for the function of `F`; the picture of Δ is the case $c = \tau$ (`deltaApprox_eq_F`). Nothing below is special to Δ , to weight 12, or even to modularity.

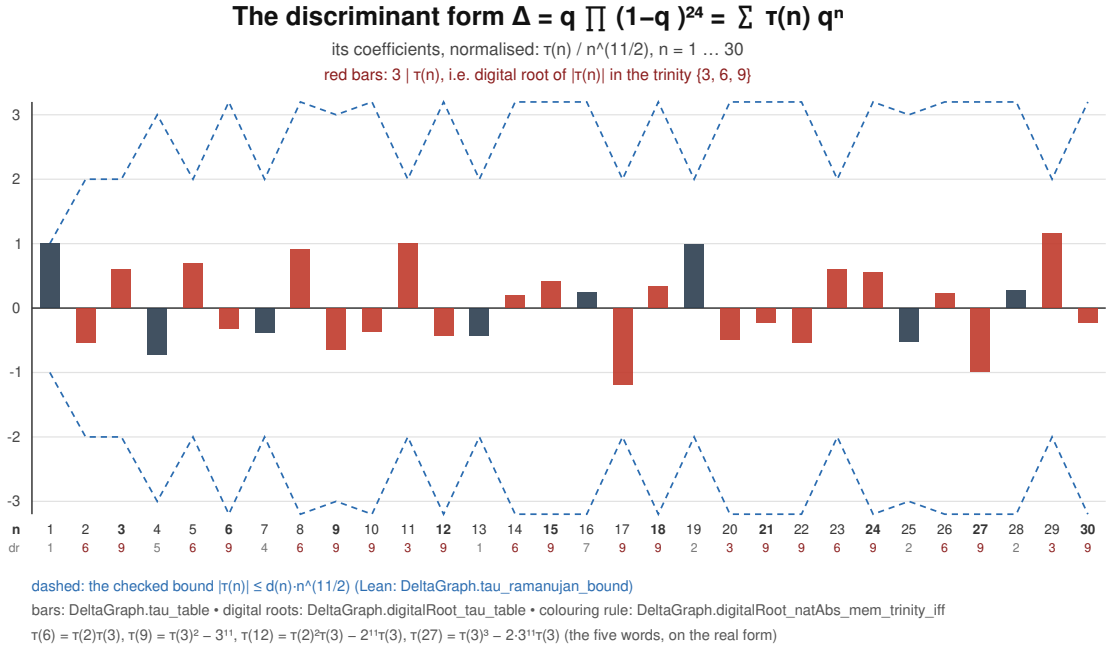


Figure 2: The coefficients of Δ , normalised by $n^{11/2}$, for $n \leq 30$. Dashed: the envelope $\pm d(n)$ of Proposition 12.4. Red bars are the n with $3 \mid \tau(n)$, equivalently $\text{dr}|\tau(n)| \in \{3, 6, 9\}$; the digital root is printed under each bar. Lean: `tau_table`, `tau_ramanujan_bound`, `digitalRoot_tau_table`.

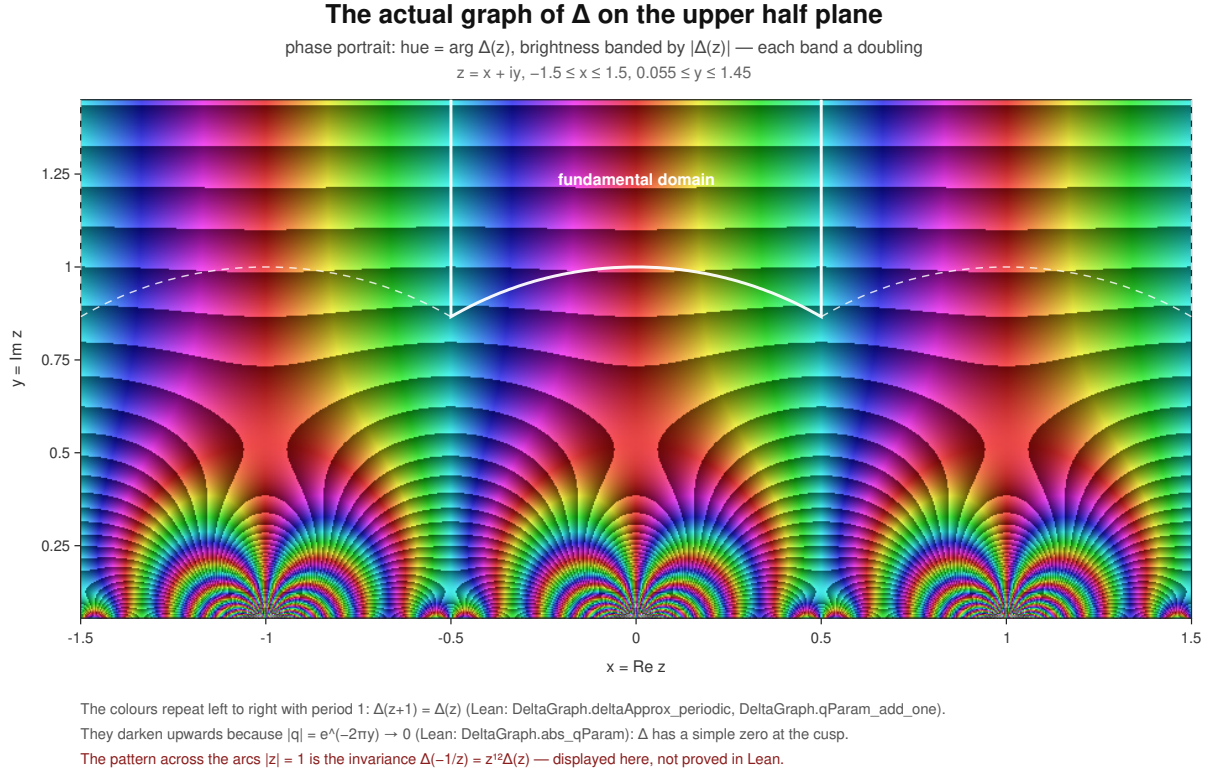
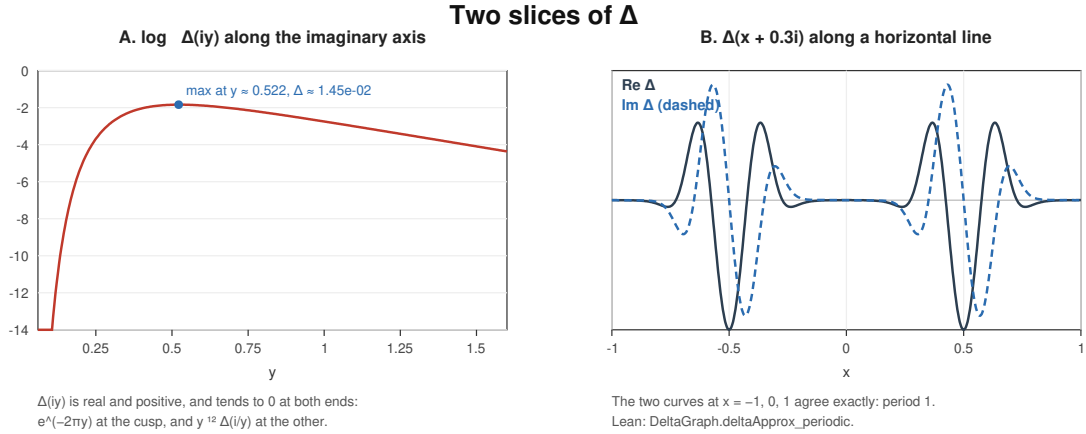


Figure 3: The graph of Δ itself: a phase portrait on $-1.5 \leq \operatorname{Re} z \leq 1.5$, $0.055 \leq \operatorname{Im} z \leq 1.45$, with hue the argument and the brightness bands the doublings of $|\Delta|$. The horizontal period one and the fading at the cusp are Proposition 12.6; the fundamental domain and two of its translates are outlined.



Both panels are computed from $q \prod (1-q)^{2^k}$, the same product whose coefficients are verified in `DeltaGraph.tau_table`.

Figure 4: Two slices. Left: $\log_{10} \Delta(iy)$ along the imaginary axis, where Δ is real and positive, with its maximum near $y \approx 0.523$. Right: real and imaginary parts of $\Delta(x + 0.3i)$ over one full period in x ; the values at $x = -1, 0, 1$ coincide, which is `deltaApprox_periodic`.

One mirror for every half-integer

Proposition 13.1 (Reflection). *For every z in the upper half plane and every $k \in \mathbb{Z}$,*

$$q(-\bar{z}) = \overline{q(z)}, \quad F(-\bar{z}) = \overline{F(z)}, \quad F(z + k) = F(z), \quad F(k - \bar{z}) = \overline{F(z)}.$$

The map $z \mapsto k - \bar{z}$ is the reflection in the vertical line $x = k/2$: it fixes z exactly when $\operatorname{Re} z = k/2$.

Proof. $\overline{e^w} = e^{\bar{w}}$ and $\overline{2\pi iz} = 2\pi i(-\bar{z})$ give the first; the second is the first applied termwise, using $\overline{c(n)} = c(n)$ — this, and only this, is where the reality of the coefficients enters. The third is $e^{2\pi ik} = 1$. The fourth is the second followed by k applications of the third. In Lean: `qParam_neg_conj, F_neg_conj, F_int_add, F_mirror, mirror_fixed_iff`. $\square$

So the two hypotheses generate an infinite dihedral group of symmetries of the picture: the translations $z \mapsto z + k$, and one mirror for every half-integer. The mirror lines are $x \in \frac{1}{2}\mathbb{Z}$, i.e. $x = 0, \pm\frac{1}{2}, \pm 1, \dots$ — the cusps and the half-periods, and these are exactly the lines above which the picture is symmetric. They are also exactly the lines on which the nome itself is real:

Proposition 13.2 (The walls, and nothing else). *$\operatorname{Im} q(z) = 0$ if and only if $\operatorname{Re} z \in \frac{1}{2}\mathbb{Z}$. On the wall $x = k/2$ one has $q(\frac{k}{2} + iy) = (-1)^k e^{-2\pi y}$. Lean: `qParam_im_eq_zero_iff, qParam_wall`.*

Proof. $\operatorname{Im} q(z) = e^{-2\pi y} \sin(2\pi x)$, which vanishes iff $2\pi x$ is an integer multiple of π . The value on the wall is $e^{\pi ik} e^{-2\pi y}$. $\square$

On a wall the function is real, and its phase cannot drift

Theorem 13.3 (Phase locked). *For every $k \in \mathbb{Z}$ and every $y > 0$ the value $F(\frac{k}{2} + iy)$ is a real number; consequently*

$$\arg F(\frac{k}{2} + iy) \in \{0, \pi\},$$

with the value 0 when the real number is ≥ 0 and π when it is < 0 . Moreover, if F has no zero on a segment $\{k/2 + iy : a \leq y \leq b\}$ of a wall, then $\arg F$ is constant along that segment.

Proof. By Proposition 13.2 the nome is the real number $\rho = (-1)^k e^{-2\pi y}$, so $F = \sum_{n \leq N} c(n) \rho^n$ is a real polynomial expression in a real number: `F_wall_ofReal`, `F_wall_im`. A nonzero real number has argument 0 or π according to its sign (`arg_F_wall`, `arg_F_wall_eq_zero_iff`, `arg_F_wall_eq_pi_iff`). For the last statement, $y \mapsto \operatorname{Re} F(k/2 + iy)$ is continuous, and a sign change would produce a zero of the real part; since the imaginary part vanishes identically on the wall, that zero would be a zero of F . Intermediate value theorem: `continuous_F_wall_re`, `phase_locked_of_ne_zero`. $\square$

This is the precise form of the informal statement “the phase does not drift horizontally, so the colour boundary cannot slant”. A colour boundary is a level set of $\arg F$; on a wall the argument takes one of two values and can only change there by passing through a zero of F . It cannot cross a wall obliquely.

No horizontal gradient, hence perpendicular contours

Theorem 13.4 (Perpendicularity). *Fix $y > 0$ and $k \in \mathbb{Z}$, and let $u(x, y) = |F(x + iy)|^2$. Then*

1. *u is even about the wall: $u(\frac{k}{2} + t, y) = u(\frac{k}{2} - t, y)$ for all t ;*
2. *u is differentiable and $\partial_x u = 0$ at every point of the wall; the same holds for $|F|$ itself wherever $F \neq 0$;*
3. *hence $\operatorname{grad} u$ at a wall point is the vertical vector $(0, \partial_y u)$; and*
4. *if that vector is nonzero then every differentiable curve through the wall point which stays on a level set of $|F|$ has horizontal velocity there. The contour and the wall meet at exactly 90° .*

Proof. (1) is Proposition 13.1 and $|\bar{w}| = |w|$ (`F_mirror_slice`, `norm_F_mirror_slice`, `sqAbs_mirror`). For (2), $x \mapsto F(x + iy)$ is a finite sum of exponentials, so u , being $(\operatorname{Re} F)^2 + (\operatorname{Im} F)^2$, is differentiable (`differentiable_sqAbs`); an even function differentiable at its centre of symmetry has derivative 0 there, because the chain rule applied to $x \mapsto 2a - x$ turns the derivative into its own negative (`deriv_eq_zero_of_mirror`, `deriv_sqAbs_wall`). For $|F| = \sqrt{u}$ one composes with the square root, which is differentiable where $u \neq 0$ (`deriv_norm_wall`). (3) is (2) read for the plane derivative (`fderiv_sqAbs2_wall_horizontal`, `grad_sqAbs2_wall_vertical`). For (4), differentiate $t \mapsto u(\gamma(t))$, which is constant, at $t = 0$: the plane derivative of u applied to $\gamma'(0)$ is 0, and by (3) that derivative is $v \gamma'(0)_2$ with $v \neq 0$, so the vertical component of $\gamma'(0)$ vanishes (`contour_tangent_horizontal`, `contour_perp_wall`). $\square$

Statement (2) is the reason the brightness bands look glued to the walls: the magnitude has a critical point in the horizontal direction along every one of them, so to first order it does not change as one steps sideways off the wall, and the band must leave horizontally.

The two colours above the cusps

The theorems so far say that the argument on a wall is 0 or π ; for Δ one can say which. High enough up, the leading term $\tau(1)q = q$ dominates the other twenty-nine terms of the plotted truncation, and the sign is then the sign of $q = (-1)^k e^{-2\pi y}$.

Proposition 13.5 (The colours). *Let Δ_{30} be the truncation (3) with $c = \tau$ and $N = 30$. For every $y \geq 5$:*

$$\Delta_{30}\left(\frac{k}{2} + iy\right) > 0 \quad \text{and} \quad \arg = 0 \quad (k \text{ even}), \quad \Delta_{30}\left(\frac{k}{2} + iy\right) < 0 \quad \text{and} \quad \arg = \pi \quad (k \text{ odd}).$$

Proof. Write $\rho = (-1)^k e^{-2\pi y}$. Crudely, $|\tau(n)| \leq 2 \cdot 10^9$ for $n \leq 30$ (checked in the kernel, `abs_tau_le`), and $|\rho|^n \leq |\rho|^2$ for $n \geq 2$, so the tail past the first term is at most $30 \cdot 2 \cdot 10^9 \rho^2$ (`abs_Fr_sub_lead_le`). At $y \geq 5$ one has $e^{-2\pi y} \leq e^{-31} \leq 10^{-11}$ (`exp_le_of_five_le`, using $\pi > 3.14$ and $e > 2.7$), so that tail is at most $0.6|\rho|$, while the first term is $\tau(1)\rho = \rho$. Hence the sign is that of ρ : `deltaApprox_wall_pos`, `deltaApprox_wall_neg`, `arg_deltaApprox_wall_even`, `arg_deltaApprox_wall_odd`. $\square$

The bound $y \geq 5$ is an artefact of a deliberately crude estimate; numerically the two signs hold on the whole upper half plane, and panel D of Figure 5 shows them down to $y = 0.1$. That the walls $x = 0, \pm 1$ carry one colour and the walls $x = \pm \frac{1}{2}$ the other — a half turn apart — is what makes the alternation of the picture at the cusps so sharp.

Remark 13.6. What the walls do *not* explain is the other symmetry visible in Figure 3: the reflection across the unit arcs $|z| = 1$, which is the invariance $\Delta(-1/z) = z^{12}\Delta(z)$. That one is genuinely modular — it needs the S -transformation, not just real coefficients and a period. It is proved separately, in Section 14, and by a completely different argument: nothing in the present section is strong enough to reach it. The vertical structure is cheap; the circular structure is not.

14 The circular symmetry: $\Delta(-1/z) = z^{12}\Delta(z)$

This section pays the debt of the last one. Every statement here is about the *true* discriminant form — the convergent infinite product, not a truncation — and the Lean names live in the namespace `DeltaModular`, whose prefix we drop.

Definition 14.1 (Δ as a twenty-fourth power). Write η for the Dedekind eta function, $\eta(z) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ with $q = e^{2\pi iz}$, and set

$$\Delta(z) := \eta(z)^{24} = q \prod_{n \geq 1} (1 - q^n)^{24}.$$

This is the same eta product whose coefficients were expanded in Section 12; there it was truncated in order to be drawn, here it is not truncated at all. On the upper half plane Δ is holomorphic and nowhere zero. Lean: `Delta`, `Delta_ne_zero`, `differentiableOn_Delta`.

The proof runs on logarithmic derivatives, $Lf := f'/f$, which turn products into sums and kill exactly the constants we do not want to track.

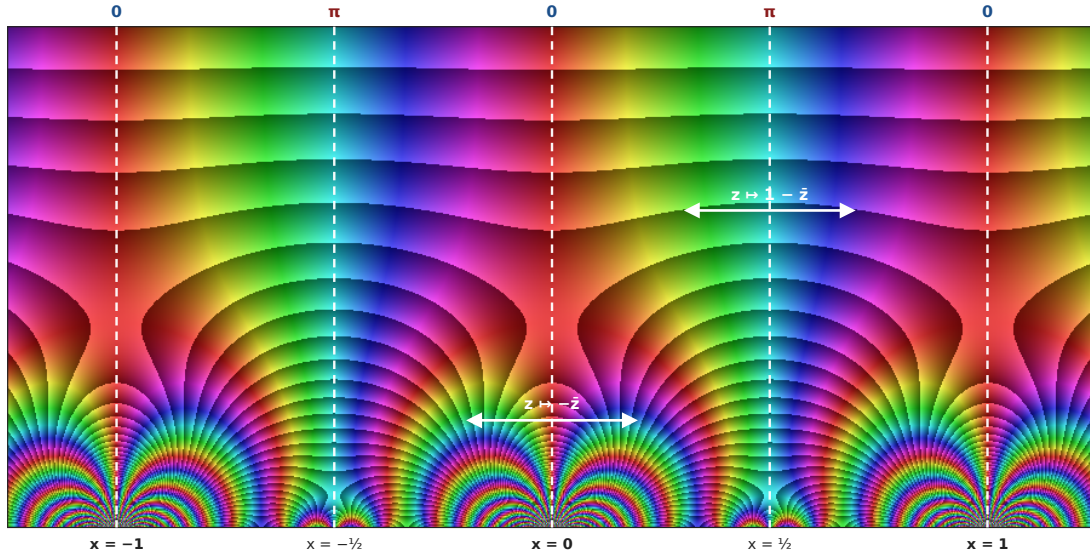
Proposition 14.2 (The logarithmic derivative of Δ). *On the upper half plane, $L\Delta = 2\pi i E_2$, where E_2 is the normalised weight-two Eisenstein series. Lean: `logDeriv_Delta`.*

The vertical walls of Δ

why the phase portrait stands perfectly upright over $x = 0, \pm 1/2, \pm 1$
two inputs only: the coefficients $\tau(n)$ are real, and the expansion has period one

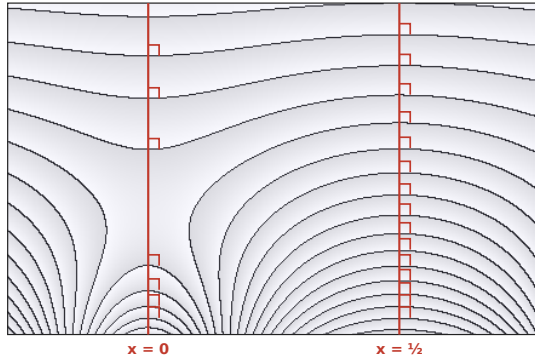
A the portrait, with its mirrors

arg Δ on each wall (top row):



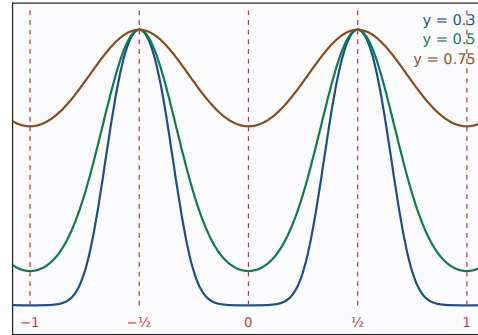
hue = arg Δ , brightness banded by $|\Delta|$. Reflection in each dashed line conjugates Δ (VerticalWalls.F_mirror), so the picture is its own mirror image there.

B the contours meet the walls at 90°



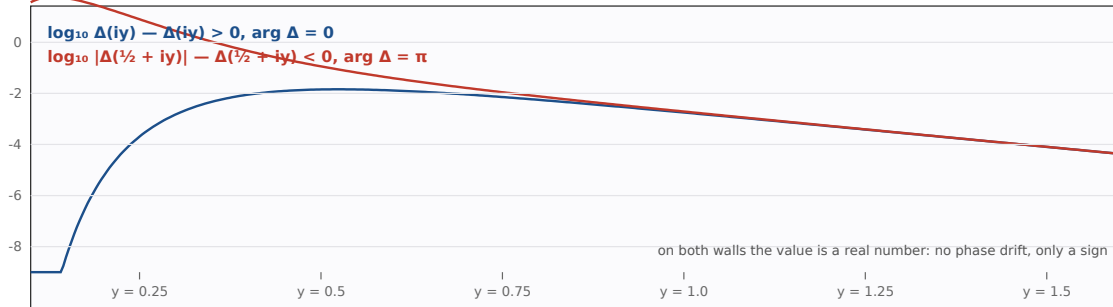
level curves $|\Delta| = 2^{k/(1.7)}$; red: the 90° crossings

C $|\Delta(x + iy)|$ across the walls



even about every dashed line $\Rightarrow$ horizontal tangent there

D the two walls, as real graphs



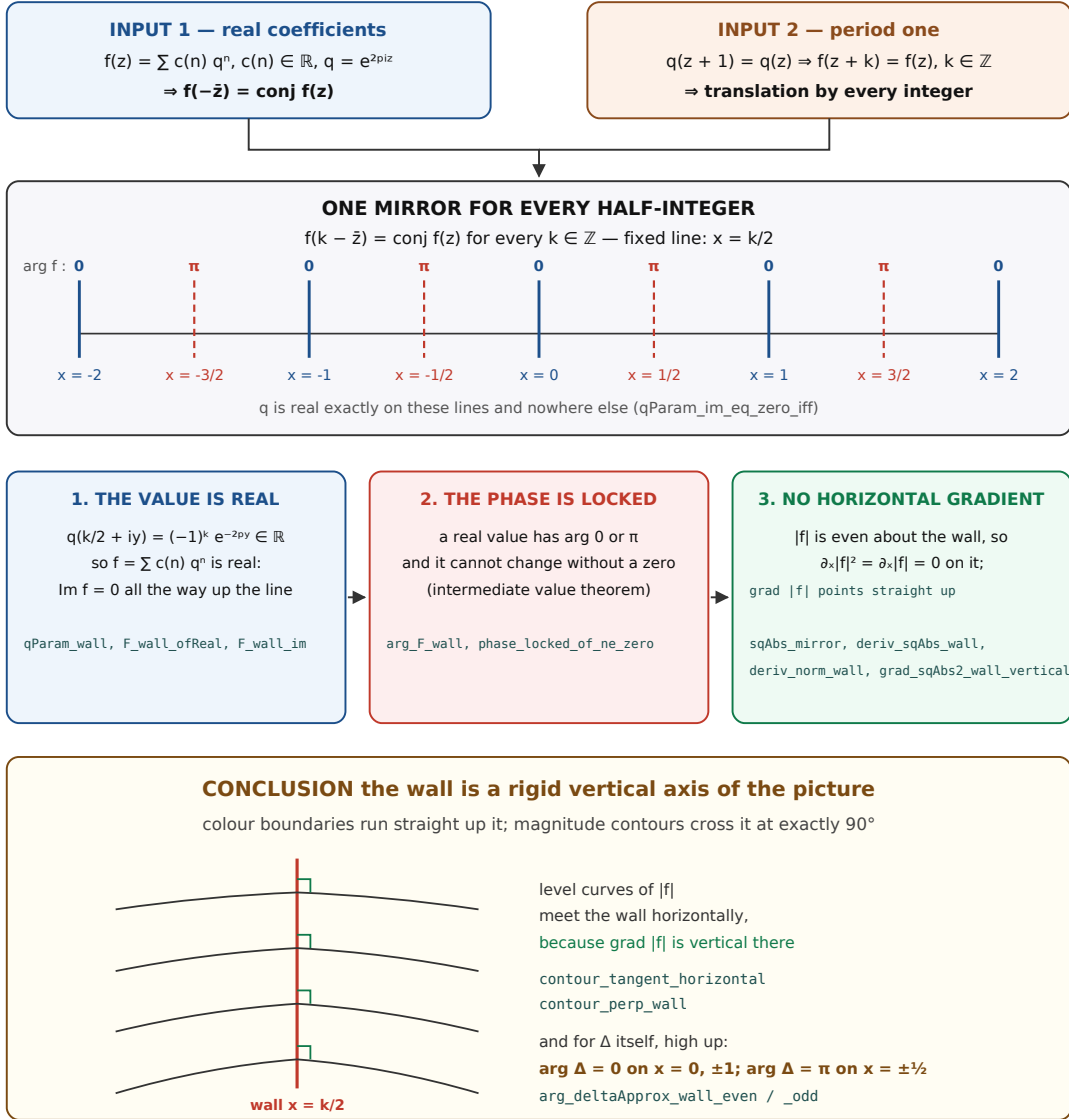
Lean, all in RequestProject/VerticalWalls.lean:

```
A F_neg_conj, F_int_add, F_mirror, mirror_fixed_iff, qParam_im_eq_zero_iff
B deriv_sqAbs_wall, deriv_norm_wall, fderiv_sqAbs2_wall_horizontal, grad_sqAbs2_wall_vertical, contour_tangent_horizontal
C F_mirror_slice, norm_F_mirror_slice, sqAbs_mirror
D qParam_wall, F_wall_ofReal, F_wall_im, arg_F_wall, phase_locked_of_ne_zero, arg_deltaApprox_wall_even/odd
```

The plate draws the eta product; the theorems are proved for the truncations $\sum_{n \leq N} c(n)q^n$ with any real c , and the two signs of D for $N = 30, y \geq 5$.

Two symmetries, one straight wall

the whole vertical structure of the picture, from real coefficients and period one



Nothing above uses anything special about Δ — only that its coefficients are real and that it has period one.

What the walls do NOT explain: the symmetry across the arcs $|z| = 1$, which is $\Delta(-1/z) = z^{1/2}\Delta(z)$.

That one is genuine modularity, needs a different argument, and is proved separately in RequestProject/DeltaModular.lean (see arc_symmetry.svg).

RequestProject/VerticalWalls.lean — no sorry, standard axioms only

Figure 6: The argument of Section 13 on one sheet: from “real coefficients” and “period one” to one mirror per half-integer, then to reality on the wall, a locked phase, a vanishing horizontal gradient, and finally to contours meeting the walls at 90° . Each box carries the name of the Lean theorem that proves it.

Proof. $L(f^{24}) = 24 L f$ and $L \eta = \frac{\pi i}{12} E_2$; the latter is Mathlib's `logDeriv_eta_eq_E2`. Multiply: $24 \cdot \frac{\pi i}{12} = 2\pi i$. $\square$

Proposition 14.3 (The defect of E_2). *For z in the upper half plane,*

$$E_2(-1/z) = z^2 E_2(z) - \frac{6iz}{\pi}.$$

Lean: `G2_S, E2_eq_G2, E2_S_transform`.

Proof. Mathlib's `G2_S_transform` says $G_2(z) = z^{-2} G_2(-1/z) + 2\pi i/z$, i.e. $G_2(-1/z) = z^2 G_2(z) - 2\pi i z$. Now $E_2 = (2\zeta(2))^{-1} G_2$ and $\zeta(2) = \pi^2/6$, so $E_2 = (3/\pi^2) G_2$ and the correction $-2\pi i z$ scales to $-(3/\pi^2) 2\pi i z = -6iz/\pi$. $\square$

That defect is not an obstruction; it is exactly the logarithmic derivative of the automorphy factor z^{12} , which is why weight twelve is the weight that works.

Theorem 14.4 (The circular symmetry). *For every z with $\text{Im } z > 0$,*

$$\Delta(-1/z) = z^{12} \Delta(z).$$

Lean: `Delta_neg_inv, Delta_neg_inv'`.

Proof. Put $F(z) = \Delta(-1/z)$ and $G(z) = z^{12} \Delta(z)$. Both are holomorphic and nowhere zero on the upper half plane, since Δ is, since $z \mapsto -1/z$ maps the upper half plane to itself, and since $z \neq 0$ there.

By the chain rule and Propositions 14.2 and 14.3,

$$L F(z) = \frac{1}{z^2} L \Delta(-1/z) = \frac{2\pi i}{z^2} \left(z^2 E_2(z) - \frac{6iz}{\pi} \right) = 2\pi i E_2(z) + \frac{12}{z},$$

using $i^2 = -1$ in the last step; and directly

$$L G(z) = L(z^{12}) + L \Delta(z) = \frac{12}{z} + 2\pi i E_2(z).$$

So $L F = L G$ on the upper half plane (`logDeriv_Delta_comp`, `logDeriv_weight`). The upper half plane is open and convex, hence connected, so two nowhere-vanishing holomorphic functions with the same logarithmic derivative differ by a nonzero constant: $F = cG$ (this is Mathlib's `logDeriv_eq0n_iff`). Evaluate at $z = i$: $-1/i = i$ and $i^{12} = 1$, so $F(i) = \Delta(i) = G(i)$, and $\Delta(i) \neq 0$ forces $c = 1$. $\square$

Remark 14.5. The evaluation point is not an accident and not a numerical coincidence: i is the fixed point of $z \mapsto -1/z$ in the upper half plane, which is the only place where the constant can be read off without knowing a single coefficient of Δ .

Period one, which Section 12 proved for the truncations, holds for the true product too, and for the same reason: the twenty-fourth power kills the root of unity that η picks up.

Proposition 14.6 (Period one). $\eta(z+1) = e^{2\pi i/24} \eta(z)$ and hence $\Delta(z+1) = \Delta(z)$, for every $z \in \mathbb{C}$. *Lean:* `eta_add_one, Delta_add_one`.

Theorem 14.7 (Full modularity). Δ is invariant of weight 12 under the whole modular group: for every $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$,

$$\Delta\left(\frac{az+b}{cz+d}\right) = (cz+d)^{12} \Delta(z), \quad \text{that is,} \quad \Delta|_{12}\gamma = \Delta.$$

Lean: DeltaH, DeltaH_S, DeltaH_T, DeltaH_slash.

Proof. The slash action of weight 12 is a group action, and $SL_2(\mathbb{Z})$ is generated by $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. For S the denominator is z and the claim is Theorem 14.4; for T the denominator is 1 and the claim is Proposition 14.6. Induction along the closure of $\{S, T\}$ — with the identity, products and inverses handled by the action axioms — gives every γ . $\square$

What the portrait actually shows

The symmetry one *sees* in Figure 3 is a reflection, not a rotation: the mirror across the arcs $|z| = 1$ is the inversion $z \mapsto 1/\bar{z}$. To get it we need the vertical mirror of Section 13 once more, but now for the true Δ rather than for a truncation.

Proposition 14.8 (Real coefficients, again). $\eta(-\bar{z}) = \overline{\eta(z)}$ and $\Delta(-\bar{z}) = \overline{\Delta(z)}$ on the upper half plane. *Lean: eta_neg_conj, Delta_neg_conj.*

Proof. Conjugation is a continuous ring homomorphism, so it passes through the convergent product (`multipliable_eta`, `Multipliable.map_tprod`), and it sends the nome to the nome of the reflected point: $e^{2\pi iz/h} = e^{2\pi i(-\bar{z})/h}$ (`qParam_conj`). Every factor $1 - q^n$ is therefore matched with its own conjugate, and the twenty-fourth power follows. $\square$

Theorem 14.9 (The reflection in the unit circle). For $\text{Im } z > 0$,

$$\Delta(1/\bar{z}) = \overline{z^{12}\Delta(z)}, \quad \text{hence} \quad |\Delta(1/\bar{z})| = |z|^{12} |\Delta(z)|.$$

Lean: Delta_inv_conj, norm_Delta_inv_conj.

Proof. Apply Theorem 14.4 at $w = -\bar{z}$, which is again in the upper half plane, and note $-1/w = 1/\bar{z}$ and $w^{12} = \bar{z}^{12}$. Then $\Delta(w) = \overline{\Delta(z)}$ by Proposition 14.8, and $\bar{z}^{12} \overline{\Delta(z)} = \overline{z^{12}\Delta(z)}$. $\square$

The map $z \mapsto 1/\bar{z}$ fixes the arc $|z| = 1$ pointwise, and on that arc the weight factor has modulus one; the theorem therefore degenerates into a statement about the *phase* alone, and it is the exact counterpart of the phase lock on the vertical walls (Theorem 13.3).

Corollary 14.10 (Phase lock on the arc). If $|z| = 1$ and $\text{Im } z > 0$ then

$$\overline{\Delta(z)} = z^{12}\Delta(z), \quad \text{equivalently} \quad z^{12}\Delta(z)^2 = |\Delta(z)|^2 \in \mathbb{R}_{\geq 0}.$$

So on the unit arc $2 \arg \Delta(z) \equiv -12 \arg z \pmod{2\pi}$: the colour of the portrait along the arc is determined by the position on the arc, up to a half turn, exactly as it is locked to 0 or π along a vertical wall. *Lean: Delta_arc, Delta_arc_phase.*

Proof. On the unit circle $z^{-1} = \bar{z}$, so $-1/z = -\bar{z}$ and Theorem 14.4 reads $\Delta(-\bar{z}) = z^{12}\Delta(z)$, while Proposition 14.8 reads $\Delta(-\bar{z}) = \overline{\Delta(z)}$. Multiplying the resulting identity by $\Delta(z)$ and using $\Delta\bar{\Delta} = |\Delta|^2$ gives the second form. $\square$

Remark 14.11 (What is imported and what is proved). Two inputs come from Mathlib: the Dedekind eta function with its logarithmic derivative $\frac{\pi i}{12} E_2$, and the S -transformation of G_2 with its $-2\pi i/z$ defect. Everything else above — Propositions 14.2–14.8, Theorems 14.4, 14.7, 14.9 and Corollary 14.10 — is proved in this project. Note also the change of object: Sections 12–13 work with the truncation Δ_N , which is what a plotter can evaluate, while this section works with the infinite product itself. The plates are therefore still pictures of Δ_N ; the theorems of this section are about Δ .

15 Five words, five graphs

Section 8 produced five multipliers and Section 12 produced one picture. This section puts them together: it draws the image of Δ under each of the five words, and identifies exactly what a Hecke word does to a graph. All Lean names here live in the namespace `DeltaWordGraphs`, whose prefix we drop.

A Hecke operator acts on q -expansions coefficient by coefficient. For weight $k = 12$ and level one, if $f = \sum_{m \geq 1} a(m)q^m$ then

$$(T_n f)(z) = \sum_{m \geq 1} c_n(m) q^m, \quad c_n(m) = \sum_{d | \gcd(n, m)} d^{11} a\left(\frac{nm}{d^2}\right), \quad (4)$$

which is `heckeCoeff`. The five words are the five combinations dictated by Propositions 8.1–8.3:

$$T_2, \quad T_3, \quad T_6, \quad T_{12} + 2^{11}T_3, \quad T_{27} + 2 \cdot 3^{11}T_3$$

(`wordT2`, `wordT3`, `wordT2T3`, `wordT2T3T2`, `wordT3T3T3`).

The coefficients $\tau(1), \dots, \tau(60)$ needed for (4) are held in one literal table `tauTable`, certified inside the kernel against the same eta-product expansion that produced Proposition 12.1: `tauTable_eq_deltaCoeffs` states `tauTable = deltaCoeffs 60`, and `tauT_eq_tau` identifies its first thirty entries with $\tau(1), \dots, \tau(30)$. Since $c_n(m)$ needs $\tau(nm)$, the range that can be checked from a table of length 60 shrinks as the word grows; that range is stated explicitly in each case, and shaded in each plate.

Proposition 15.1 (Each word rescales the expansion). *With the coefficients (4) computed from τ — not from any eigenvalue —*

$$\begin{aligned} c_{T_2}(m) &= -24 \tau(m) & (1 \leq m \leq 30), \\ c_{T_3}(m) &= 252 \tau(m) & (1 \leq m \leq 20), \\ c_{T_2 T_3}(m) &= -6048 \tau(m) & (1 \leq m \leq 10), \\ c_{T_2 T_3 T_2}(m) &= 145152 \tau(m) & (1 \leq m \leq 5), \\ c_{T_3 T_3 T_3}(m) &= 16003008 \tau(m) & (1 \leq m \leq 2). \end{aligned}$$

The five constants are the five multipliers of Example 8.5, recovered here from the form itself.

Proof. Each line is a finite computation from `tauTable`, performed by the kernel (`wordT2_eq`, `wordT3_eq`, `wordT2T3_eq`, `wordT2T3T2_eq`, `wordT3T3T3_eq`); the table itself is certified by `tauTable_eq_deltaCoeffs`. The five constants are read off the first coefficient, $m = 1$, since $\tau(1) = 1$ (`word_multipliers`). $\square$

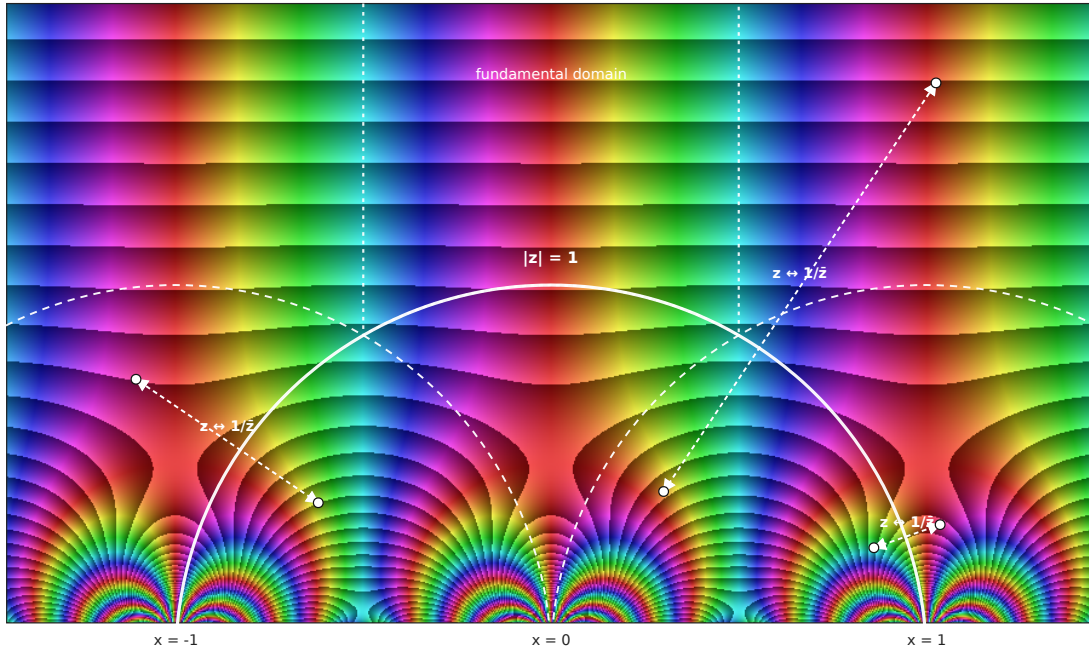
The circular symmetry of Δ

$\Delta(-1/z) = z^{12}\Delta(z)$ — the reflection across the arcs $|z| = 1$

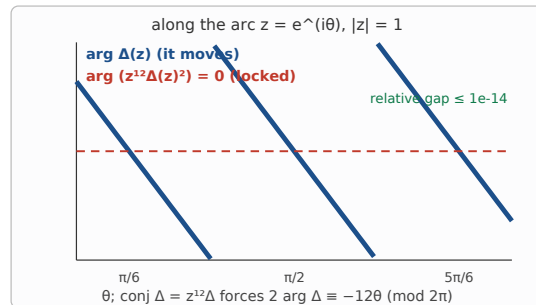
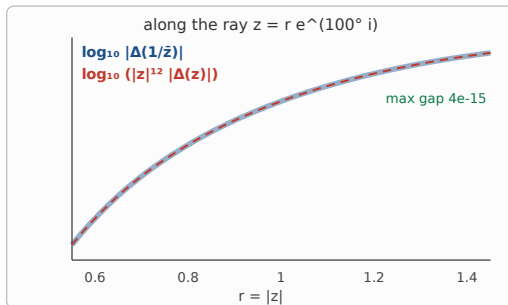
genuine modularity: proved for the true product $\Delta = \eta^{24}$, not for a truncation

A the portrait, with the unit circle

hue = $\arg \Delta$, brightness banded by the doublings of $|\Delta|$



B the two identities, checked numerically



C the proof, box by box

$\Delta := \eta^{24}$ is holomorphic and nowhere zero on the upper half plane
Delta, Delta_ne_zero, differentiableOn_Delta

log-derivative: $L\Delta = 24 \cdot L\eta = 2\pi i E_2$
logDeriv_Delta (Mathlib: logDeriv_eta_eq_E2)

E_2 is not modular: $E_2(-1/z) = z^2 E_2(z) - 6iz/\pi$
E2_S_transform (Mathlib: G2_S_transform, riemannZeta_two)

that defect is exactly $L(z^{12})$: $L[\Delta(-1/z)] = 2\pi i E_2(z) + 12/z = L[z^{12}\Delta(z)]$
logDeriv_Delta_comp, logDeriv_weight

upper half plane open + connected $\Rightarrow$ the two differ by a constant $c \neq 0$
isPreconnected_upperHalfPlaneSet, logDeriv_eqOn_iff

at the fixed point $z = i$: $-1/i = i$ and $i^{12} = 1$, so $c = 1$
Delta_neg_inv: $\Delta(-1/z) = z^{12}\Delta(z)$

period one $\Rightarrow$ all of $SL_2(\mathbb{Z})$: $\Delta((az+b)/(cz+d)) = (cz+d)^{12}\Delta(z)$
Delta_add_one, DeltaH_S, DeltaH_T, DeltaH_slash

real coefficients $\Rightarrow$ the mirror: $\Delta(1/\bar{z}) = \text{conj}(z^{12}\Delta(z))$, $|\Delta(1/\bar{z})| = |z|^{12}|\Delta(z)|$
eta_neg_conj, Delta_neg_conj, Delta_inv_conj, norm_Delta_inv_conj

on the arc $|z| = 1$: $\text{conj } \Delta(z) = z^{12}\Delta(z)$, so $z^{12}\Delta(z)^2 = |\Delta(z)|^2 \geq 0$
Delta_arc, Delta_arc_phase

That Δ is an eigenform for *every* T_n , so that the five identities hold for every m , is Mordell's theorem and is not proved here; the statements above are exactly what has been checked.

Corollary 15.2 (The five graphs). *Let $f_w(z) = \sum_{m \leq N} c_w(m)q^m$ be the partial sum plotted for the word w (`heckeApprox`). On the range of Proposition 15.1,*

$$f_w(z) = \lambda_w \cdot \Delta_N(z), \quad \lambda_w \in \{-24, 252, -6048, 145152, 16003008\},$$

where Δ_N is the corresponding partial sum for Δ itself.

Proof. `heckeApprox_eq_smul`: a coefficientwise identity $c(m) = \lambda \tau(m)$ on $1 \leq m \leq N$ may be pulled out of the finite sum. The five instances are `heckeApprox_wordT2`, `heckeApprox_wordT3`, `heckeApprox_wordT2T3`, `heckeApprox_wordT2T3T2` and `heckeApprox_wordT3T3T3`. $\square$

So a Hecke word does not deform the graph of an eigenform: it only chooses a scale. Three consequences are exactly what one sees in Figures 9–13.

Proposition 15.3 (What a multiplier does to a picture). *Let c , λ , N be as in Corollary 15.2. Then*

1. $f_w(z+1) = f_w(z)$: horizontal period one, as for Δ (`heckeApprox_periodic`);
2. $|f_w(z)| = |\lambda| |\Delta_N(z)|$, and where $\Delta_N(z) \neq 0$ and $\lambda \neq 0$,

$$\log |f_w(z)| = \log |\lambda| + \log |\Delta_N(z)|,$$

so the logarithmic profile is shifted by a constant and never changed in shape (`heckeApprox_norm`, `log_norm_heckeApprox`);

3. if $\lambda < 0$ then $f_w = |\lambda| e^{i\pi} \Delta_N$: every colour of the phase portrait is turned by half a turn (`heckeApprox_neg_phase`).

The five scales are ordered, $24 < 252 < 6048 < 145152 < 16003008$ (`word_abs_multipliers_strictMono`), so the five profiles are five parallel curves at increasing heights, two of them — T_2 and T_2T_3 — also turned over.

Proposition 15.4 (The trinity, on all five). $\tau(2) = -24$ and $\tau(3) = 252$ are both divisible by three, hence so is each of the five multipliers, hence so is every coefficient of every one of the five images: on the checked ranges, $3 \mid c_w(m)$ and therefore $\text{dr} |c_w(m)| \in \{3, 6, 9\}$ for every plotted coefficient.

Proof. `word_coeffs_three_dvd` on each range, and Theorem 3.1 in the integer form `digitalRoot_natAbs_mem_trinity_iff` (`coeff_mem_trinity`, `word_coeffs_mem_trinity`). $\square$

This is the one place where the two halves of the paper touch directly. The trinity of Section 3 is a statement about divisibility by three; the words of Section 8 multiply by numbers divisible by three; so every bar in every one of the five plates is red, and the colouring rule that began as folklore is here a lemma about a modular form.

The six plates are produced by `paper/make_word_graphs.py`, which recomputes τ by two independent methods, refuses to run unless the first sixty values agree with the Lean-certified `tauTable`, and builds every coefficient from (4) rather than from the eigenvalue; standalone SVG copies accompany the PDFs.

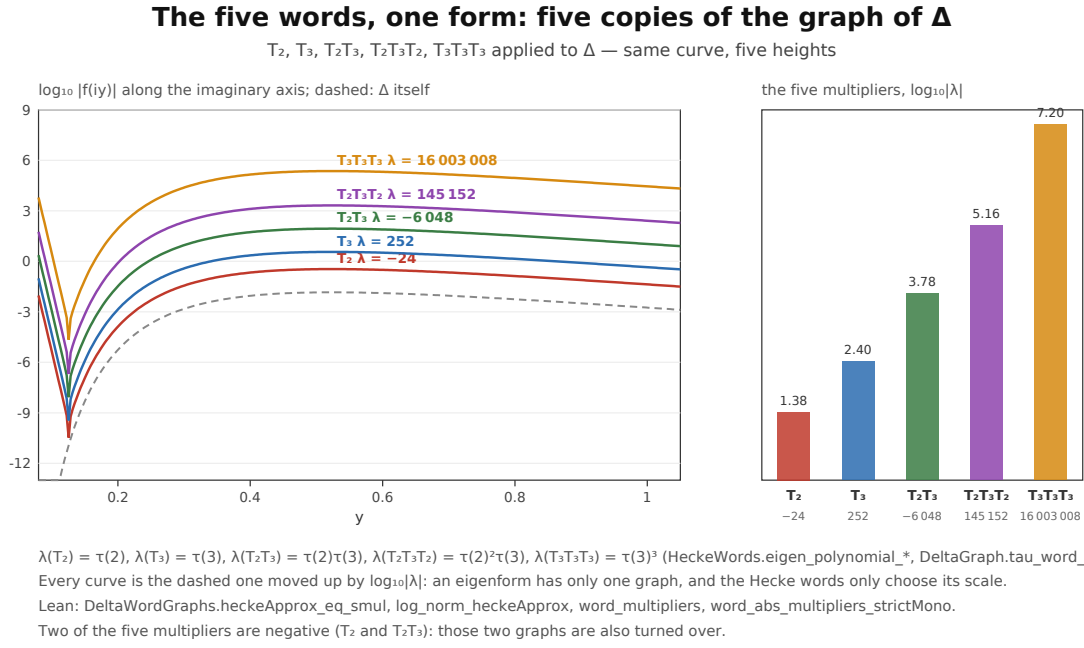


Figure 8: The five words together. Left: $\log_{10} |f_w(iy)|$ along the imaginary axis for each word, with Δ dashed; the curves are parallel, separated by $\log_{10} |\lambda_w|$ (Proposition 15.3). Right: the five multipliers on a logarithmic scale. An eigenform has one graph, and the Hecke words only choose its height.

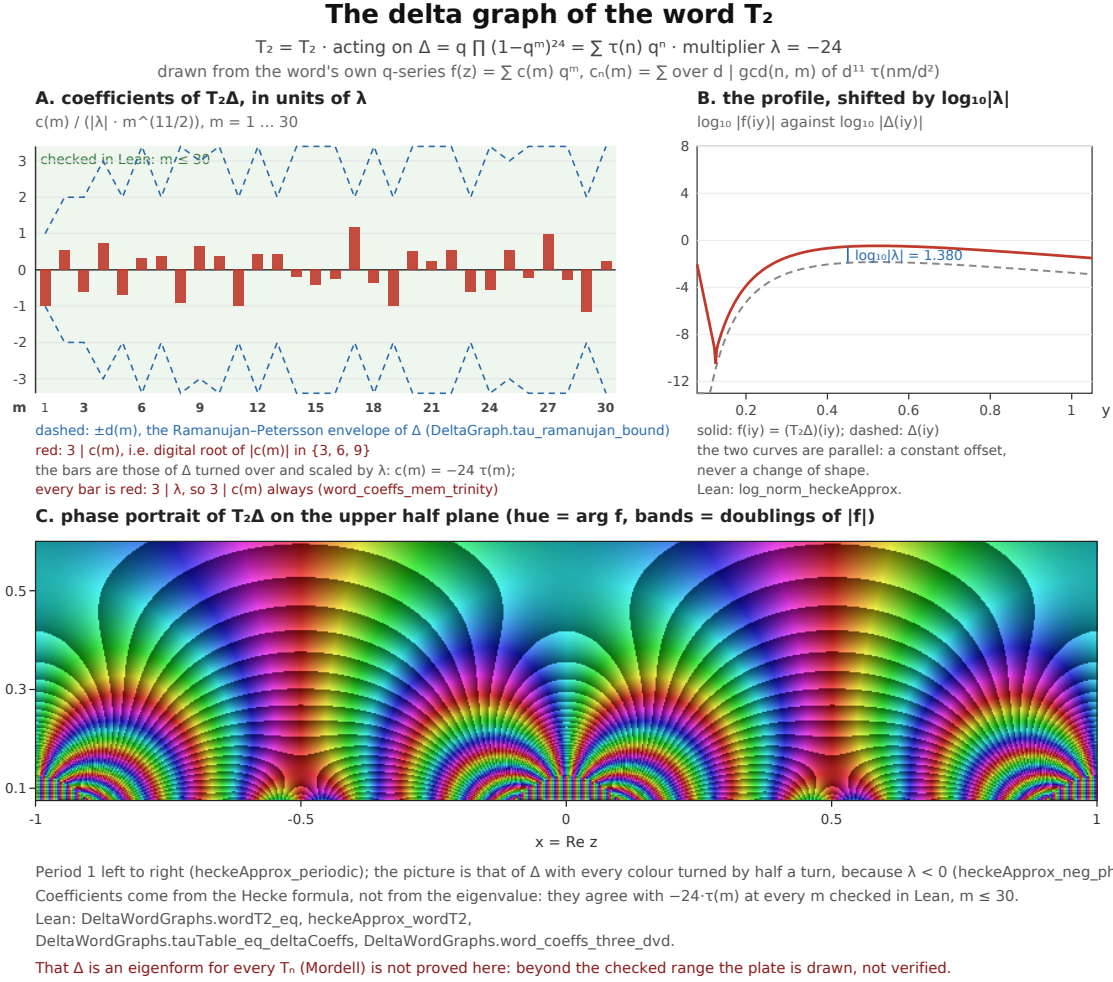


Figure 9: The word T_2 , $\lambda = -24$. A: the coefficients of $T_2\Delta$ in units of λ , with the $\pm d(m)$ envelope dashed and the range checked in Lean shaded. B: the logarithmic profile, a constant $\log_{10} 24 = 1.380$ above that of Δ . C: the phase portrait; since $\lambda < 0$, it is that of Δ turned by half a turn.

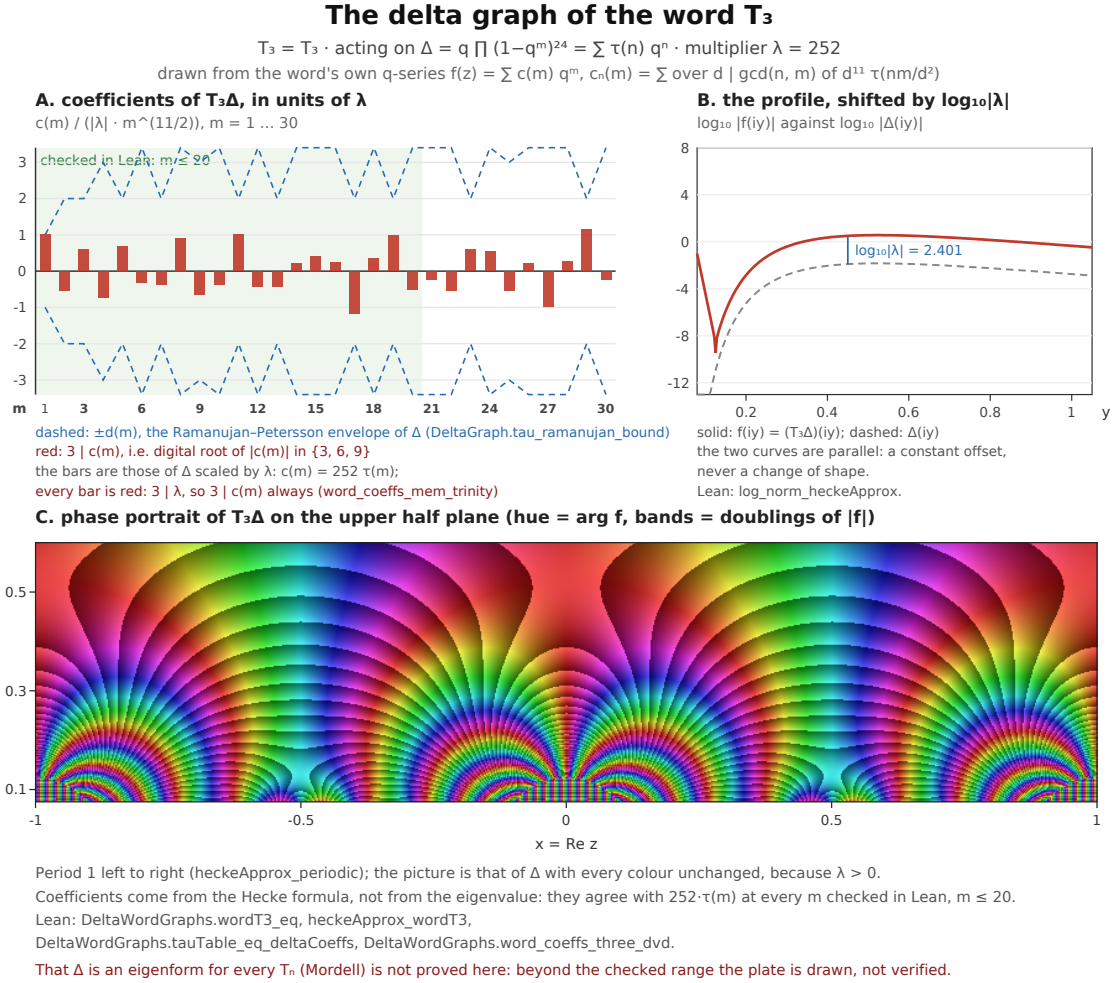


Figure 10: The word T_3 , $\lambda = 252$: the same graph, raised by $\log_{10} 252 = 2.401$, with the colours unchanged.

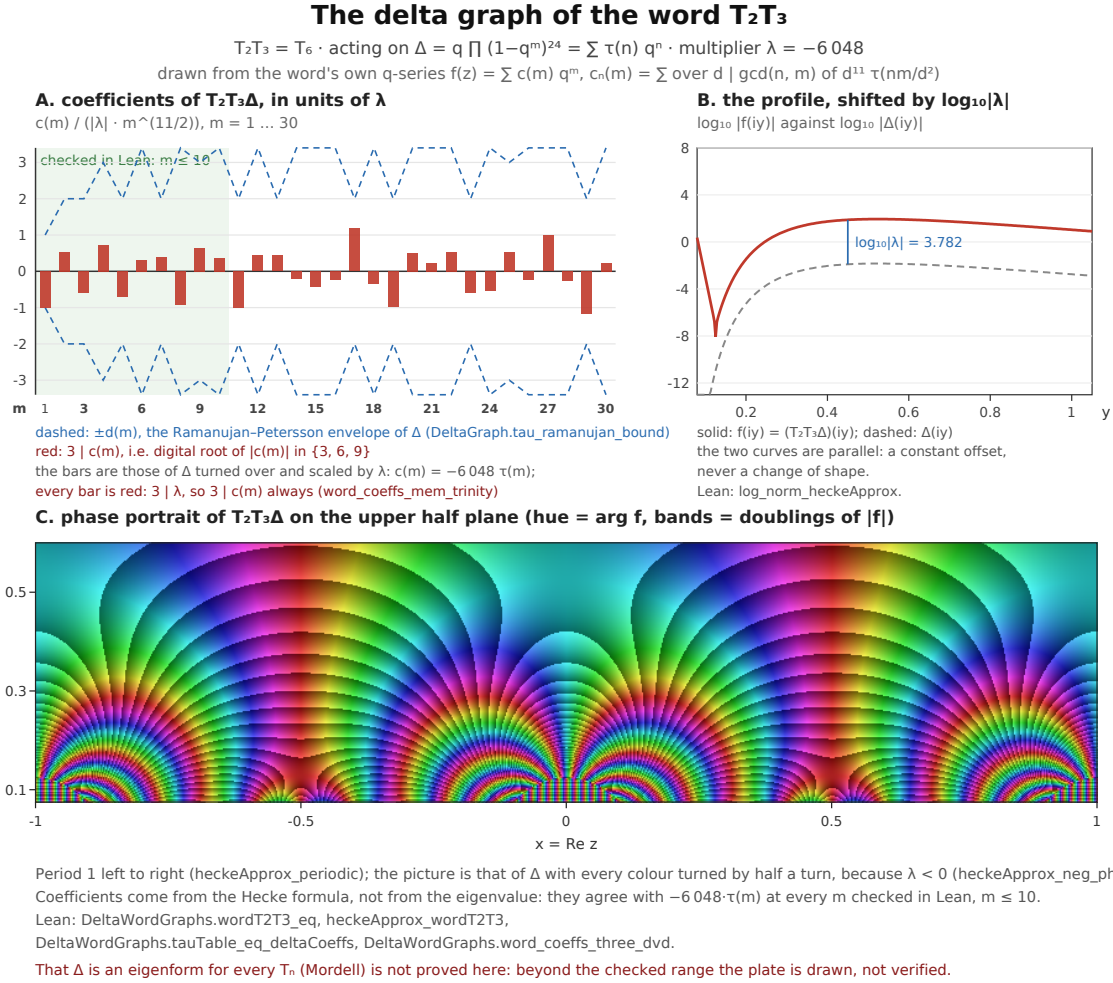


Figure 11: The word $T_2T_3 = T_6$, $\lambda = \tau(2)\tau(3) = -6048$: coprime letters cost nothing, and the two half-turns of T_2 and T_3 compose to one.

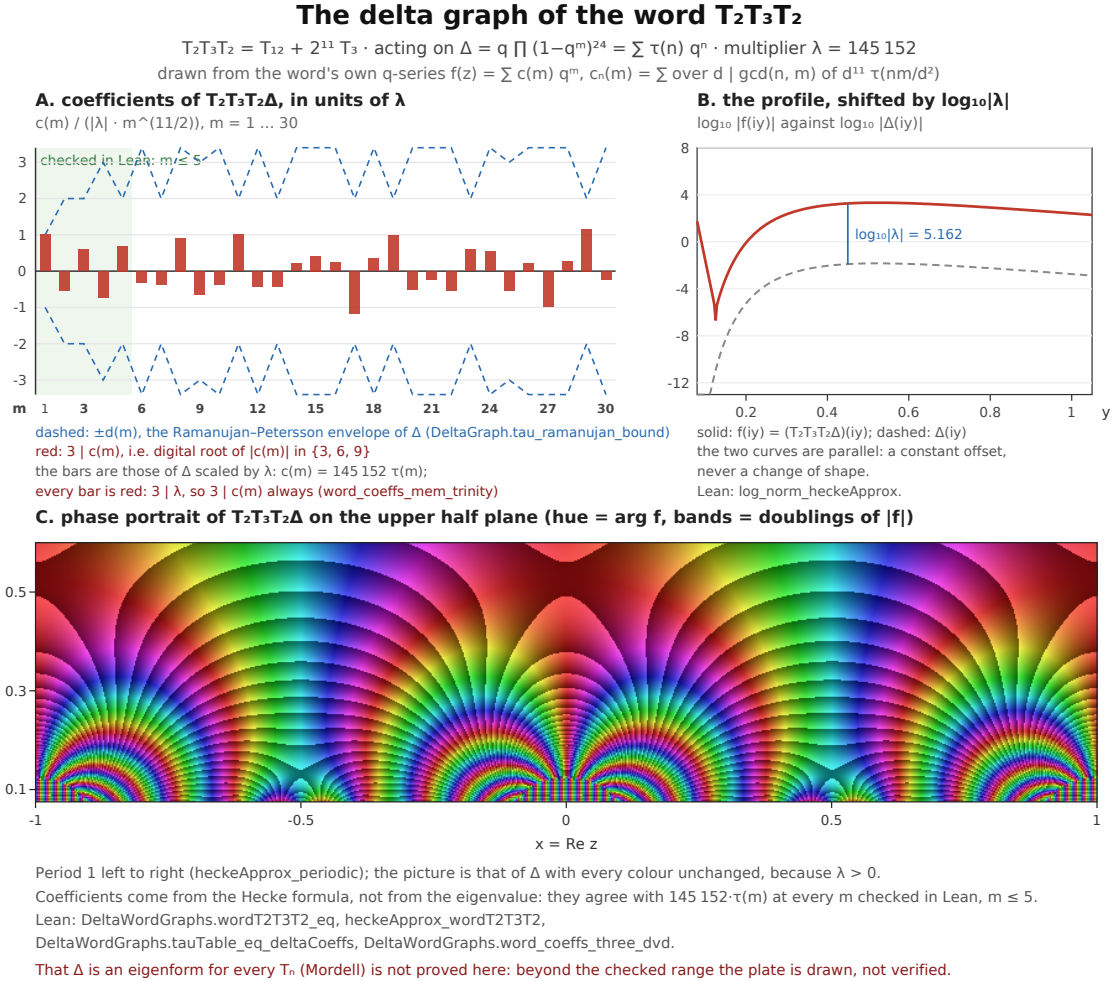


Figure 12: The word $T_2T_3T_2 = T_{12} + 2^{11}T_3$, $\lambda = \tau(2)^2\tau(3) = 145152$. The echo $2^{11}T_3$ is part of the operator, not of the picture: the image is still a multiple of Δ .

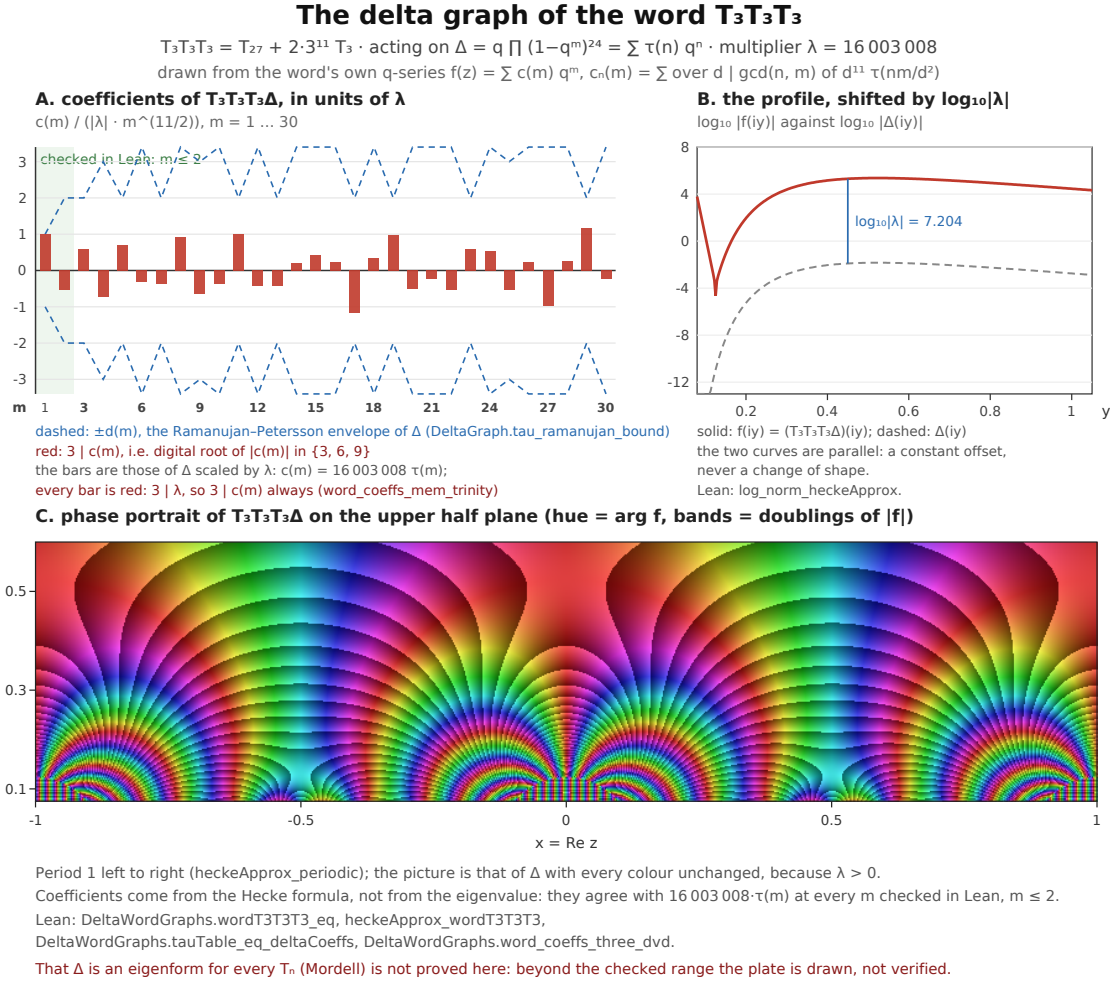


Figure 13: The word $T_3T_3T_3 = T_{27} + 2 \cdot 3^{11}T_3$, $\lambda = \tau(3)^3 = 16003008$: the largest of the five scales, $\log_{10} |\lambda| = 7.204$.

Remark 15.5. Outside the shaded ranges the plates are honest pictures of the series $\sum_{m \leq 60} c_w(m)q^m$, computed in floating point, and the identity $f_w = \lambda_w \Delta$ that they display is then Mordell's theorem rather than a checked statement. Nothing in the argument of Corollary 15.2 depends on the unchecked part: it is an implication from the coefficient identity, on whatever range that identity holds.

16 T_2 against T_3 , pointwise

Section 15 drew the two shortest words separately. Here they are compared with each other, term by term: what are the product, the difference, the quotient, the power and the remainder of the two coefficient sequences? All Lean names of this section live in the namespace `T2vsT3`, whose prefix we drop.

Write, for $1 \leq m \leq 20$ (the range on which Proposition 15.1 is checked for T_3),

$$t_2(m) = c_2(m) = \tau(2)\tau(m) = -24\tau(m), \quad t_3(m) = c_3(m) = \tau(3)\tau(m) = 252\tau(m)$$

(`t2_eq`, `t3_eq`); the coefficients are computed from (4), not from the eigenvalue. On this range no $\tau(m)$ vanishes (`tauT_ne_zero`), so all five comparisons are defined.

Everything that follows is forced by one observation: both sequences are the *same* sequence τ rescaled, by $\tau(2)$ and by $\tau(3)$ respectively. A comparison of two proportional sequences can only produce a constant, or a constant times a power of τ — and the interest lies in *which* constants appear.

Proposition 16.1 (The five comparisons). *For every m with $1 \leq m \leq 20$:*

1. **Product.** $t_2(m)t_3(m) = -6048\tau(m)^2 = \tau(6)\tau(m)^2$, and $\tau(6) = \tau(2)\tau(3)$ is exactly the multiplier of the word T_2T_3 of Proposition 8.1. In particular the product is strictly negative at every m : the two graphs are anti-parallel (`prodT_eq`, `prodT_eq_tau_six`, `prodT_neg`).
2. **Difference.** $t_2(m) - t_3(m) = -276\tau(m)$, where $-276 = \tau(2) - \tau(3)$; it never vanishes, so the two graphs never meet (`diffT_eq`, `diffT_ne_zero`).
3. **Quotient.** In $\mathbb{Q}$, $t_2(m)/t_3(m) = -2/21 = \tau(2)/\tau(3)$, a constant independent of m ; equivalently $21|t_2(m)| = 2|t_3(m)|$ everywhere (`ratioT_eq`, `abs_ratio_21_2`). In $\mathbb{Z}$ the quotient is -1 where $\tau(m) > 0$ and 0 where $\tau(m) < 0$ (`quotT_eq`).
4. **Power.** $t_2(m)^{|t_3(m)|}$ is positive — the exponent $252|\tau(m)|$ is even, so the sign of t_2 is invisible — and is divisible by $3^{|t_3(m)|}$, since $3 \mid t_2(m)$ (`powT_pos`, `three_pow_dvd_powT`). At $m = 1$ it is $(-24)^{252}$, a number of 348 decimal digits, larger than 10^{300} (`powT_one`).
5. **Remainder.** $t_2(m) \bmod t_3(m)$ equals $228\tau(m) = (\tau(3) - \tau(2))\tau(m)$ where $\tau(m) > 0$, and $-24\tau(m) = -\tau(2)\tau(m)$ where $\tau(m) < 0$; it is nonnegative, smaller than $|t_3(m)|$, and never zero (`modT_eq`, `modT_nonneg`, `modT_lt`, `modT_ne_zero`). As a fraction of $|t_3(m)|$ it therefore takes exactly two values, $19/21$ and $2/21$.

Proof. Substitute $t_2(m) = -24\tau(m)$ and $t_3(m) = 252\tau(m)$ and compute. (1) and (2) are one multiplication each, with $-24 \cdot 252 = -6048 = \tau(6)$ and $-24 - 252 = -276$; the product is negative because $-6048 < 0$ and $\tau(m)^2 > 0$. (3) is division by $252\tau(m) \neq 0$. For (5), if $\tau(m) > 0$ then $-24\tau(m) = 252\tau(m) \cdot (-1) + 228\tau(m)$ with $0 \leq 228\tau(m) < 252\tau(m)$, and if $\tau(m) < 0$ then

$-24\tau(m) = 252\tau(m) \cdot 0 + (-24\tau(m))$ with $0 \leq -24\tau(m) < |252\tau(m)|$; in both cases the remainder is the canonical one. The two quotients in (3) then follow from $b \cdot (a/b) = a - (a \bmod b)$ and cancellation of $b = t_3(m) \neq 0$. For (4), $252|\tau(m)|$ is even, so the power of a nonzero integer is positive, and $t_2(m) = 3 \cdot (-8\tau(m))$ gives $t_2(m)^e = 3^e(-8\tau(m))^e$. $\square$

Remark 16.2. Two conventions are worth stating, because they are what the Lean statements actually say. First, an integer exponent has to be read as a natural number, so “ $t_2 \wedge t_3$ ” is interpreted as $t_2(m)^{|t_3(m)|}$. Second, Lean’s `/` and `%` on $\mathbb{Z}$ are the Euclidean pair: the remainder is always nonnegative, and the quotient is whatever makes that true. This is why the integer quotient is -1 for $\tau(m) > 0$ but 0 for $\tau(m) < 0$, even though the rational quotient is the same negative number in both cases.

Proposition 16.3 (The trinity, through the comparisons). *For $1 \leq m \leq 20$, three divides $t_2(m)t_3(m)$, $t_2(m) - t_3(m)$ and $t_2(m) \bmod t_3(m)$, and each of these is nonzero, so all three have digital root in $\{3, 6, 9\}$ (`three_dvd_prodT`, `three_dvd_diffT`, `three_dvd_modT`, `prodT_mem_trinity`, `diffT_mem_trinity`, `modT_mem_trinity`). The integer quotient is the exception: it is -1 or 0 , and its digital root is never in the trinity (`quotT_notMem_trinity`).*

Proof. $3 \mid \tau(2)$ and $3 \mid \tau(3)$, hence 3 divides each of -6048 , -276 , 228 and -24 — indeed $\text{dr } 6048 = \text{dr } 276 = 6$, $\text{dr } 228 = 3$, $\text{dr } 24 = 6$ — and Theorem 3.1 in its integer form converts divisibility into the digital root. The quotient is -1 or 0 by Proposition 16.1(3), and $\text{dr } 1 = 1$, $\text{dr } 0 = 0$. $\square$

The exception is not an accident and is the moral of the section. Multiplication, subtraction and taking remainders all keep the common factor 3 ; division is precisely the operation that cancels it. The trinity lives in the scale of a sequence, and a ratio is what is left when the scale has been thrown away — which is also why the ratio is the same number $-2/21$ at every m , and so tells one nothing about Δ that $\tau(2)$ and $\tau(3)$ did not already say.

17 A delta diagram for each panel

Every panel of Figure 14 is a *sequence*. Each of them can be turned back into a *function* on the upper half plane by the same device that drew Δ in Section 12: feed the sequence into a truncated q -expansion. That is what this section does, one panel at a time; the resulting six plates are the delta diagrams of the six panels. Lean names here live in the namespace `T2vsT3Delta`, whose prefix we drop.

Definition 17.1 (The picture of a sequence). For $c : \mathbb{N} \rightarrow \mathbb{C}$ put

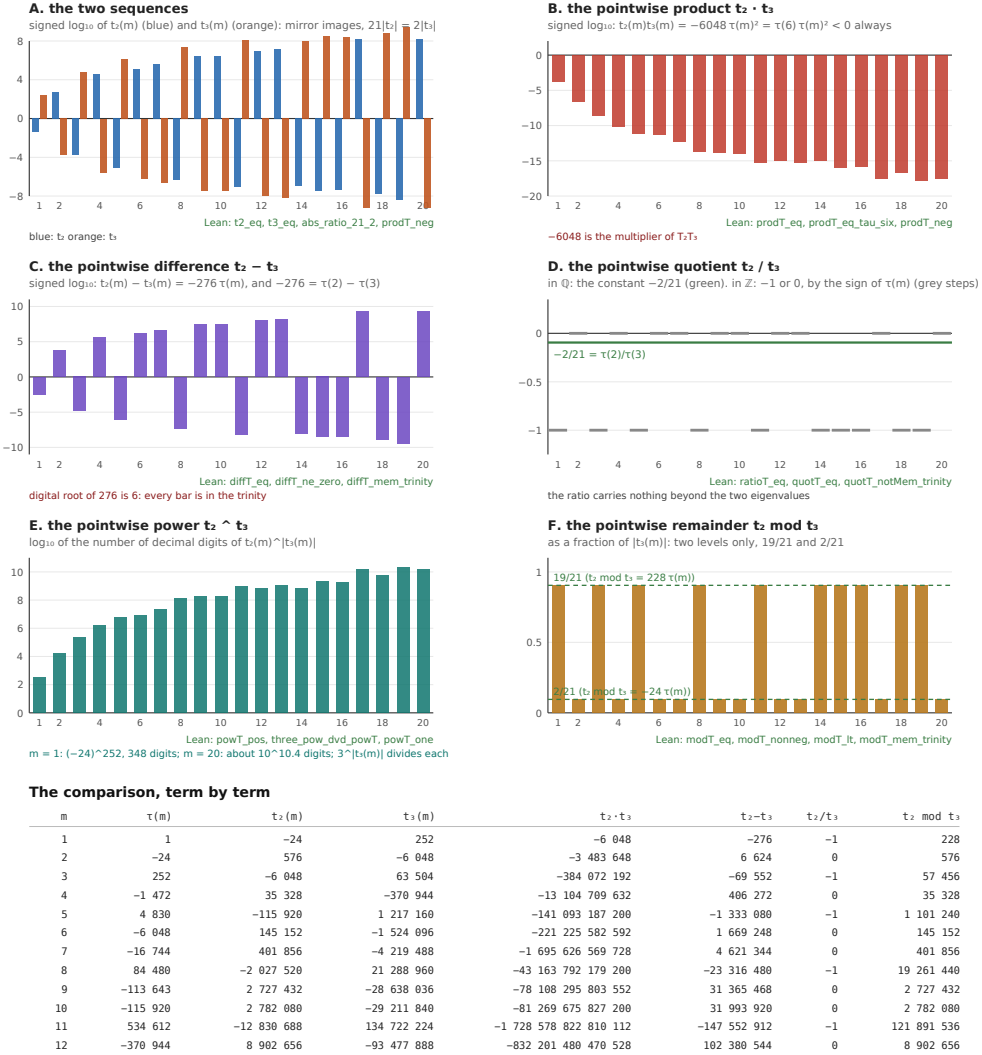
$$f_c(z) = \sum_{m=1}^{20} c(m) q^m, \quad q = e^{2\pi i z}$$

(`cApprox`). For $c = \tau$ this is the partial sum Δ_{20} of Section 12, and for integer c it is `heckeApprox` of Section 15. Every such f_c has horizontal period one (`cApprox_periodic`), and f is linear in c : $f_{c+d} = f_c + f_d$ and $f_{\lambda c} = \lambda f_c$ on the plotted range (`cApprox_add`, `cApprox_smul`).

Proposition 17.2 (The six diagrams). *With t_2, t_3 as in Section 16 and all identities holding for every z in the upper half plane:*

T₂ against T₃, pointwise

$t_2(m) = \tau(2)\tau(m) = -24\tau(m)$ and $t_3(m) = \tau(3)\tau(m) = 252\tau(m)$, $m = 1 \dots 20$ (the range verified in Lean)
coefficients built from $c_d(m) = \sum \text{over } d \mid \gcd(n, m) \text{ of } d^{11} \tau(nm/d^2)$, not from the eigenvalue



Both sequences are τ rescaled, so every pointwise comparison is one constant times a power of τ : the product is $\tau(6)\tau(m)^2$, the difference $(\tau(2) - \tau(3))\tau(m)$, and the quotient the pure number $-2/21$. $\tau(2) < 0 < \tau(3)$, so the two graphs are anti-parallel: the product is negative at every m , and the phase portraits of $T_2\Delta$ and $T_3\Delta$ differ by a half turn. $3 \mid \tau(2)$ and $3 \mid \tau(3)$, so 3 divides the product, the difference and the remainder: their digital roots all lie in the trinity $\{3, 6, 9\}$. The integer quotient is the one exception — it is -1 or 0 , and division is exactly the operation that throws the scale away. All statements are proved in `RequestProject/T2vsT3.lean` for $m = 1 \dots 20$, the range on which the coefficients of $T_3\Delta$ are certified against the eta product $q \prod (1 - q^n)^{a_n}$.

Figure 14: T_2 against T_3 , pointwise, for $m = 1 \dots 20$. A: the two coefficient sequences on a signed logarithmic scale — mirror images, in the ratio 2 : 21. B: the pointwise product $-6048\tau(m)^2$, negative at every m . C: the pointwise difference $-276\tau(m)$. D: the pointwise quotient — the constant $-2/21$ in $\mathbb{Q}$, and its rounding -1 or 0 in $\mathbb{Z}$. E: the pointwise power, by its number of decimal digits. F: the pointwise remainder as a fraction of $|t_3(m)|$, taking exactly the two values $19/21$ and $2/21$. The table lists the first twelve terms of every column. The plate is produced by `paper/make_t2_vs_t3.py`, which recomputes τ two ways, checks the first sixty values against the Lean-certified `tauTable`, and re-verifies each of the five comparisons before drawing it.

1. **Panel A.** $f_{t_2} = -24 \Delta_{20}$ and $f_{t_3} = 252 \Delta_{20}$, hence $21f_{t_2} = -2f_{t_3}$: two copies of one picture, anti-parallel and in the ratio 2 : 21 (`panelA_T2`, `panelA_T3`, `panelA_antiparallel`).
2. **Panel B.** $f_{t_2 t_3} = -6048 f_{\tau^2}$, where τ^2 is the sequence $m \mapsto \tau(m)^2$: a pointwise product of coefficients is a Hadamard product of series, so the picture of panel B is $\tau(6)$ times the diagonal series of Δ with itself, and not a picture of Δ (`panelB_eq`, `tauSqT`).
3. **Panel C.** $f_{t_2 - t_3} = -276 \Delta_{20} = f_{t_2} - f_{t_3}$: a genuine multiple of Δ (`panelC_eq`, `panelC_sub`).
4. **Panel D.** In $\mathbb{Q}$, $f_{t_2/t_3} = -\frac{2}{21} \sum_{m=1}^{20} q^m$, the bare geometric sum; in $\mathbb{Z}$, $f_{t_2 \div t_3} = -f_\chi$ where $\chi(m) = 1$ if $\tau(m) > 0$ and 0 otherwise (`panelD_rat`, `panelD_int`, `posMaskT`).
5. **Panel E.** There is no picture: the coefficients $t_2(m)^{|t_3(m)|}$ outgrow every radius of convergence. Their logarithm has one, and

$$\log(t_2(m)^{|t_3(m)|}) = |t_3(m)| \log |t_2(m)| = 252 |\tau(m)| \log(24 |\tau(m)|) > 0$$

(`log_powT`, `powLogCoeff_eq`, `powLogCoeff_pos`).

6. **Panel F.** Writing $\tau^+(m) = \max(\tau(m), 0)$ and $\tau^-(m) = \min(\tau(m), 0)$, so that $f_{\tau^+} + f_{\tau^-} = \Delta_{20}$,

$$f_{t_2 \bmod t_3} = 228 f_{\tau^+} - 24 f_{\tau^-}$$

(`panelF_eq`, `panelF_halves`, `modT_split`): Δ cut along the sign of τ , its two halves rescaled by $\tau(3) - \tau(2)$ and by $\tau(2)$.

Proof. Each item is Proposition 16.1 inserted coefficientwise into Definition 17.1 and then pulled out of the sum, which is legitimate because f is linear in c and because only the coefficients $m \leq 20$ occur. For (5) note that the exponent $|t_3(m)| = 252|\tau(m)|$ is even, so $t_2(m)^{|t_3(m)|} = |t_2(m)|^{|t_3(m)|}$ and the logarithm of a power is the exponent times the logarithm; positivity holds because $|t_2(m)| \geq 24 > 1$. For (6), $\tau^+ + \tau^- = \tau$ pointwise, and $t_2 \bmod t_3$ is 228τ on the positive part and -24τ on the negative one by Proposition 16.1(5). $\square$

Proposition 17.3 (Which panels draw Δ). *On $1 \leq m \leq 20$ the sequences t_2 , t_3 and $t_2 - t_3$ are constant multiples of τ , with constants -24 , 252 and -276 ; the sequences $t_2 t_3$, $t_2 \div t_3$ and $t_2 \bmod t_3$ are not constant multiples of τ for any constant whatsoever (`panels_smul_or_not`, `prodT_not_smul_tau`, `quotT_not_smul_tau`, `modT_not_smul_tau`).*

Proof. The three positive statements are Proposition 16.1. For the three negative ones it is enough to look at $m = 1$ and $m = 2$, where $\tau(1) = 1$ and $\tau(2) = -24$. The product is -6048 and -3483648 , forcing $c = -6048$ and then $c = 145152$; the integer quotient is -1 and 0 , forcing $c = -1$ and then $c = 0$; the remainder is 228 and 576 , forcing $c = 228$ and then $c = -24$. Each pair is inconsistent. $\square$

Remark 17.4. The two propositions say what the six plates look like. Panels A and C are the picture of Δ with its logarithmic profile shifted by $\log_{10} |\lambda|$ and, since all three constants -24 , 252 , -276 include two negative ones, the phase portrait turned by half a turn where the constant is negative: this is Proposition 15.3 again. Panel B is quadratic in τ , so its profile climbs at twice the slope and its phase portrait winds twice as fast. Panels D and F are the interesting failures: both depend on the *sign* of $\tau(m)$, which is invisible to any rescaling, and

Proposition 17.3 makes that obstruction precise. Panel D keeps nothing but the sign — its picture is a lacunary 0/1 series with none of the structure of a cusp form — while panel F keeps the sizes as well and merely weights the two halves of Δ differently. Panel E is the one panel with no function at all, and it is drawn one logarithm down.

Remark 17.5. Nothing here proves that the pictures beyond the twentieth coefficient are right: every plate is the same 20-term truncation for f and for Δ , so what is verified is the relation between two partial sums, and the analytic statements about the limits — convergence, modularity, Mordell’s theorem — are displayed and not proved.

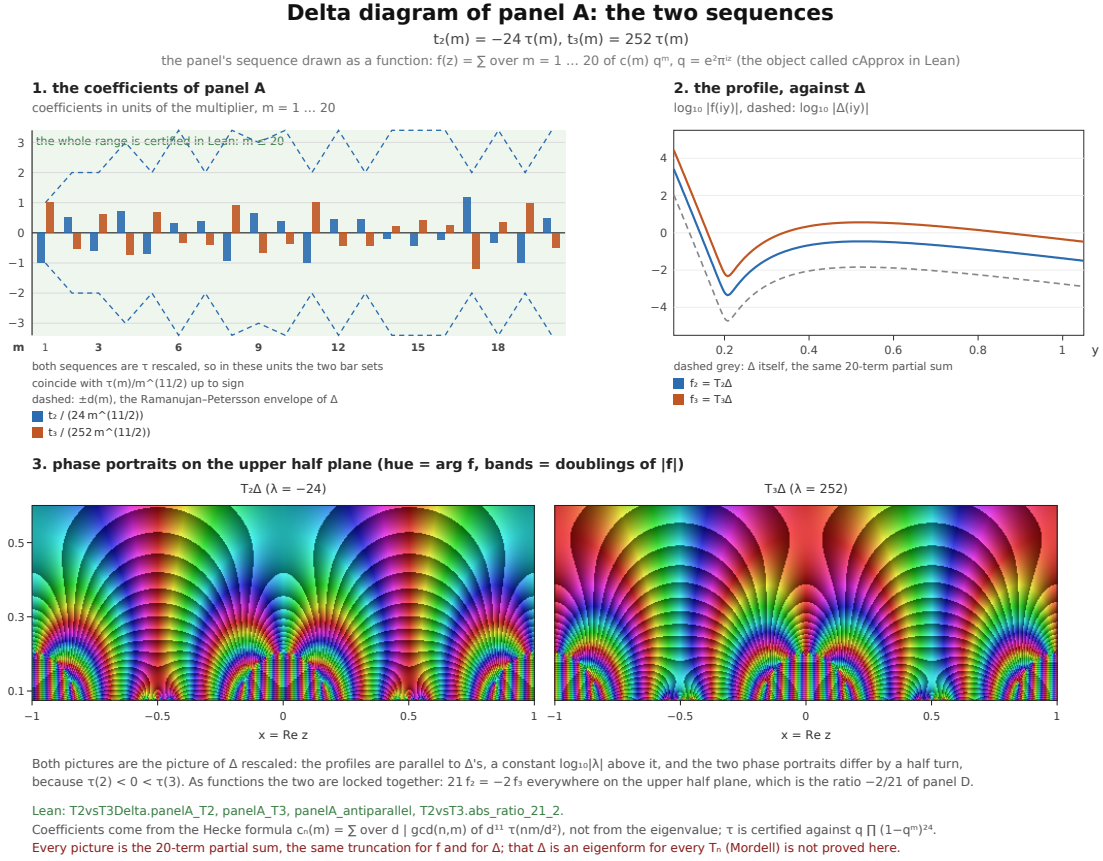


Figure 15: Delta diagram of panel A: the two sequences. Left, the coefficients in units of the multiplier; middle, $\log_{10} |f(iy)|$ against $\log_{10} |\Delta(iy)|$ dashed; below, the two phase portraits, which differ by a half turn because $\tau(2) < 0 < \tau(3)$.

18 Archimedes on the same circle

The circle of Section 14 is the unit circle: the arc $|z| = 1$ across which $\Delta(-1/z) = z^{12} \Delta(z)$ reflects the portrait. This section puts Archimedes’ figures on that circle, squeezes π between

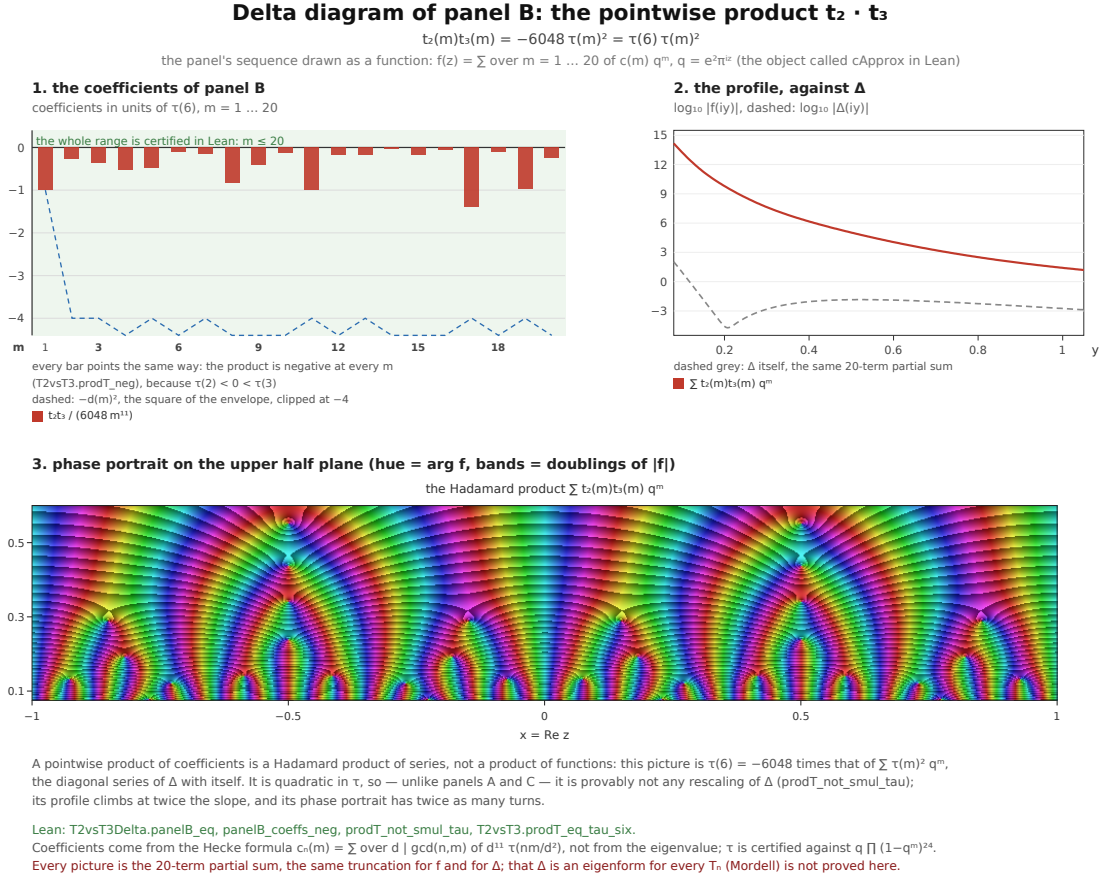


Figure 16: Delta diagram of panel B: the pointwise product, that is the Hadamard square of Δ scaled by $\tau(6) = -6048$. Every coefficient is negative, the profile climbs at twice the slope of Δ 's, and the phase portrait winds twice as fast.

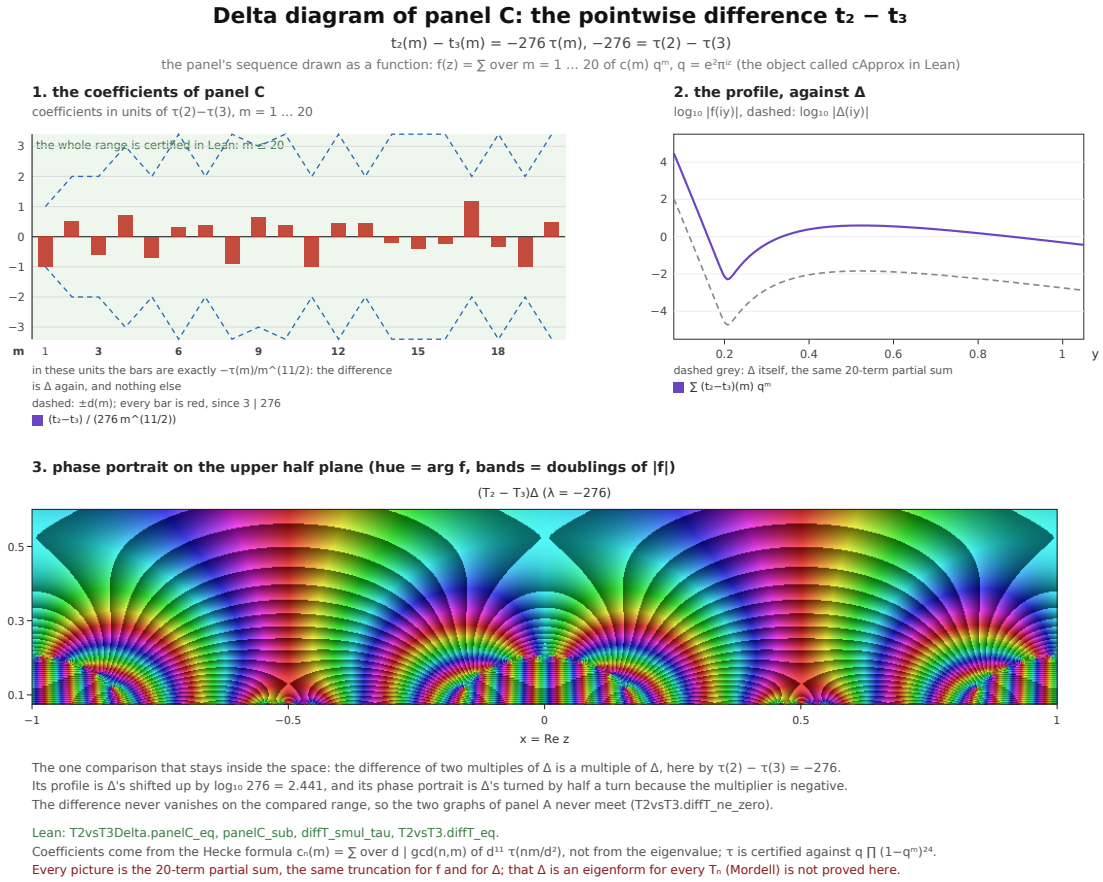


Figure 17: Delta diagram of panel C: the pointwise difference, a genuine multiple of Δ by $\tau(2) - \tau(3) = -276$. The profile is that of Δ raised by $\log_{10} 276 = 2.441$ and the phase portrait is turned by half a turn.

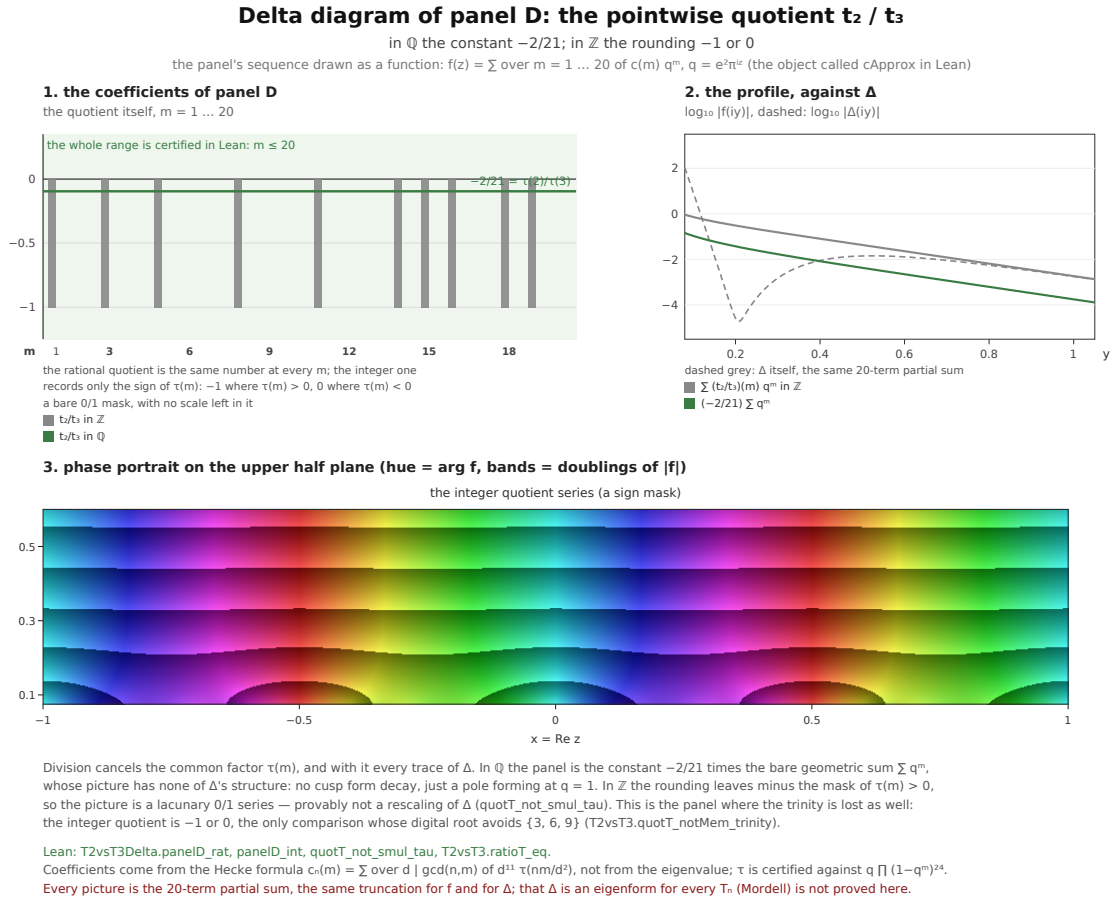


Figure 18: Delta diagram of panel D: the pointwise quotient. In $\mathbb{Q}$ the constant $-2/21$ times the geometric sum; in $\mathbb{Z}$ minus the mask of $\tau(m) > 0$. Division cancels the common factor $\tau(m)$, and with it every trace of Δ .

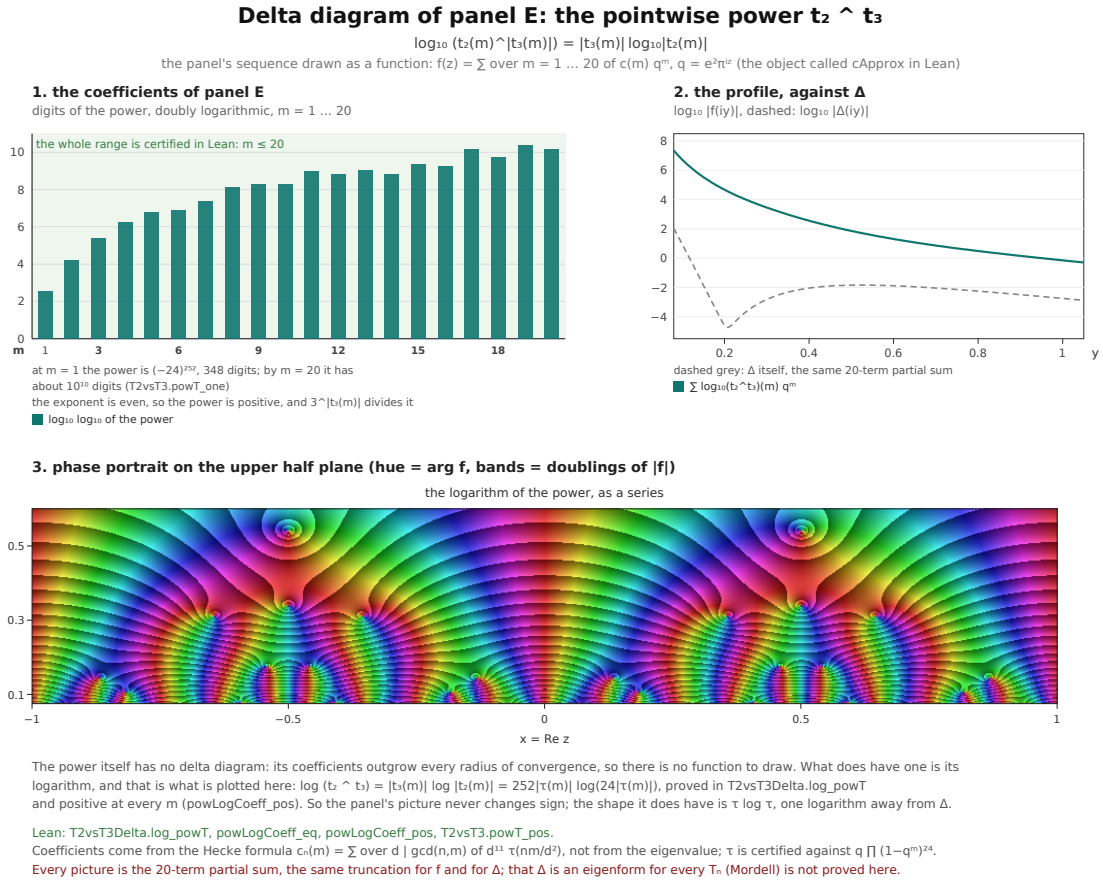


Figure 19: Delta diagram of panel E: the pointwise power has no convergent series, so what is drawn is its logarithm, $|t_3(m)| \log |t_2(m)|$, positive at every m . The bars are doubly logarithmic: the height is $\log_{10}$ of the number of decimal digits.

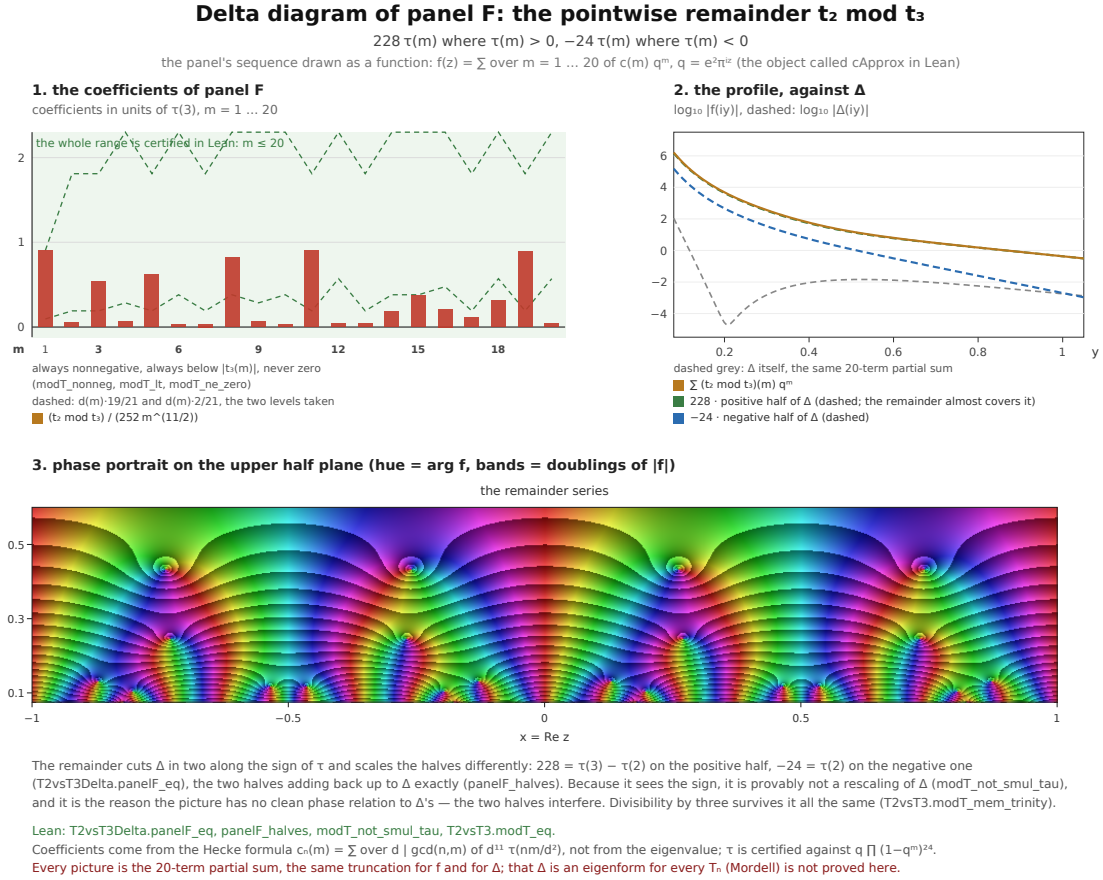


Figure 20: Delta diagram of panel F: the pointwise remainder, which is Δ cut along the sign of τ with the two halves scaled by 228 and by -24 . The two halves add back up to Δ ; the remainder is not a rescaling of it.

them, and then sets them turning — first in the plane, then in space. All Lean names of this section live in the namespace `Archimedes369`, in `RequestProject/Archimedes369.lean` (the static figures) and `RequestProject/ArchimedesSpin.lean` (the rotating ones).

Definition 18.1. For $n \geq 3$ let

$$b_n = n \sin \frac{\pi}{n}, \quad a_n = n \tan \frac{\pi}{n}$$

be half the perimeter of the regular n -gon inscribed in, respectively circumscribed about, the unit circle, and let $A_n = \frac{n}{2} \sin \frac{2\pi}{n}$ be the area of the inscribed one (`lower`, `upper`, `inArea`).

Theorem 18.2 (Archimedes' sandwich). *For every $n \geq 3$,*

$$b_n < \pi < a_n, \quad A_n = b_n \cos \frac{\pi}{n} < \pi.$$

Proof. $\sin x < x < \tan x$ for $0 < x < \pi/2$, applied at $x = \pi/n$ and multiplied by n ; the two Mathlib inputs are `Real.sin_lt` and `Real.lt_tan`. For the area, the doubling formula $\sin \frac{2\pi}{n} = 2 \sin \frac{\pi}{n} \cos \frac{\pi}{n}$ gives $A_n = b_n \cos \frac{\pi}{n}$, and $0 < \cos \frac{\pi}{n} \leq 1$. Lean: `lower_lt_pi`, `pi_lt_upper`, `inArea_eq`, `inArea_lt_pi`. $\square$

The point of the two sequences is that one can climb between them. Archimedes' ladder doubles the number of sides, and the step is purely algebraic — no new trigonometry is needed once the two means are in place.

Theorem 18.3 (The doubling ladder). *For every $n \geq 3$,*

$$a_{2n} = \frac{2a_nb_n}{a_n + b_n} \quad (\text{harmonic mean}), \quad b_{2n} = \sqrt{a_{2n}b_n} \quad (\text{geometric mean}),$$

and the ladder is monotone: $b_n < b_{2n} < \pi < a_{2n} < a_n$.

Proof. Write $x = \pi/2n$, so that $\pi/n = 2x$. Then $2a_nb_n/(a_n + b_n) = 2n \sin 2x/(1 + \cos 2x) = 2n \tan x = a_{2n}$, and $a_{2n}b_n = 2n \tan x \cdot n \sin 2x = 4n^2 \sin^2 x = b_{2n}^2$. Monotonicity follows from Theorem 18.2: the harmonic mean of $b_n < a_n$ is below a_n , and $b_{2n}^2 = a_{2n}b_n > b_n^2$ because $a_{2n} > \pi > b_n$. Lean: `upper_two_mul`, `lower_two_mul`, `lower_lt_lower_two_mul`, `upper_two_mul_lt_upper`. $\square$

Theorem 18.4. $b_n \rightarrow \pi$, $a_n \rightarrow \pi$ and $A_n \rightarrow \pi$ as $n \rightarrow \infty$.

Proof. $b_n = \pi \operatorname{sinc}(\pi/n)$ and $a_n = \pi \operatorname{sinc}(\pi/n)/\cos(\pi/n)$ with sinc continuous and $\operatorname{sinc} 0 = 1$; $\pi/n \rightarrow 0$. Lean: `tendsto_lower`, `tendsto_upper`, `tendsto_inArea`. $\square$

Where the trinity enters

The ladder has to start somewhere, and the place it starts is the reason Archimedes chose the hexagon: its side equals the radius, so the first rung is an integer.

Theorem 18.5 (The first rungs).

$$b_3 = \frac{3\sqrt{3}}{2}, \quad a_3 = 3\sqrt{3}; \quad b_4 = 2\sqrt{2}, \quad a_4 = 4; \quad \boxed{b_6 = 3}, \quad a_6 = 2\sqrt{3};$$

$$b_{12} = 3(\sqrt{6} - \sqrt{2}), \quad a_{12} = 24 - 12\sqrt{3}.$$

Consequently $3 < \pi < 2\sqrt{3}$ and $3(\sqrt{6} - \sqrt{2}) < \pi < 24 - 12\sqrt{3}$, i.e. $3.1058\dots < \pi < 3.2153\dots$

Lean: `lower_three`, `upper_three`, `lower_four`, `upper_four`, `lower_six`, `upper_six`, `lower_twelve`, `upper_twelve`, `hexagon_bounds`, `twelve_gon_bounds`. The three static plates are Figures 21 and 22.

Theorem 18.6 (The ninety-six-gon). *Four rungs above the hexagon,*

$$\frac{223}{71} < \pi < \frac{22}{7}.$$

Proof. Interval propagation along Theorem 18.3. The harmonic mean is monotone in each argument (`harm_mono_lower`, `harm_mono_upper`) and so is the square root, so a rational lower and upper bound for a_n and for b_n pushes forward to one for a_{2n} and b_{2n} (`bounds_two_mul`). Starting from $b_6 = 3$ and $3.4641016 \leq a_6 = 2\sqrt{3} \leq 3.4641017$, four steps give

$$b_{96} \geq 3.1410317 > \frac{223}{71}, \quad a_{96} \leq 3.1427148 < \frac{22}{7},$$

and Theorem 18.2 inserts π between them. Lean: `bounds_six`, `bounds_twelve`, `bounds_twentyfour`, `bounds_fortyeight`, `bounds_ninetysix`, `archimedes_bounds`. No numerical value of π is used as an input: the only ingredients are the two recursions, the exact hexagon, and rational arithmetic. $\square$

Remark 18.7 (3 is not a coincidence, and 9 is not on the ladder). The hexagon gives $b_6 = 3$ *exactly*, and the ladder from it visits 6, 12, 24, 48, 96 — Archimedes stopped at 96, which is Theorem 18.6. Every rung is a quadratic extension of the previous one, so every $b_{6 \cdot 2^k}$ is constructible. The nine-gon is different in kind: $s = \sin \frac{\pi}{9}$ satisfies the trisection cubic

$$3s - 4s^3 = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

(`sin_pi_div_nine_cubic`), which is irreducible, and no doubling ladder from 3 or 6 ever reaches 9 sides. Nine can still be bracketed — $3.077 < b_9 < 3.079$, `lower_nine_bounds`, proved from the cubic and $\sqrt{3} > 1.7320508$ — but it is reached by solving, not by doubling. In the trinity's own terms: 3 and 6 are the ruler-and-compass part of the story and 9 is not.

The polygons as functions

Turning now to rotation. A convex polygon with the origin inside is the polar graph of a single function of the angle, and for the regular n -gon that function is a cosine sawtooth. Put

$$\text{wrap}_n(\theta) = \theta - \frac{2\pi}{n} \left\lfloor \frac{n\theta}{2\pi} \right\rfloor, \quad r_n(\theta) = \frac{\cos(\pi/n)}{\cos(\text{wrap}_n \theta)},$$

where $\lfloor \cdot \rfloor$ is the nearest integer: $\text{wrap}_n(\theta)$ is the angle from θ to the nearest edge normal, and $|\text{wrap}_n(\theta)| \leq \pi/n$ (`abs_wrap_le`). The point $r_n(\theta)e^{i\theta}$ is then exactly the boundary point of the polygon in direction θ : its component along the edge normal is the inradius, $r_n(\theta) \cos(\text{wrap}_n \theta) = \cos(\pi/n)$ (`radius_mul_cos_wrap`).

Theorem 18.8 (Rotating in the plane). *For $n \geq 3$ and all θ :*

1. $r_n(\theta + 2\pi/n) = r_n(\theta)$ — the n -fold rotational symmetry (`radius_rot`);

2. $\cos(\pi/n) \leq r_n(\theta) \leq 1$ (*le_radius*, *radius_le_one*), with both bounds attained: $r_n(0) = \cos(\pi/n)$ at an edge midpoint and $r_n(\pi/n) = 1$ at a vertex (*radius_edge_center*, *radius_vertex*); hence $\cos(\pi/n)$ is the infimum and 1 the supremum of the profile (*isGLB_radius*, *isLUB_radius*);
3. so the spinning polygon sweeps exactly the annulus between its incircle and the unit circle, of area $\pi(1 - \cos^2 \frac{\pi}{n}) = \pi \sin^2 \frac{\pi}{n} \rightarrow 0$ (*sweptArea_eq*, *tendsto_sweptArea*).

Item (3) is the geometric form of Theorem 18.4: what makes the polygon converge to the circle is that the gap it sweeps under rotation closes. The profiles r_n for $n = 3, 4, 6, 9, 12$ are drawn in Figure 23; the n -fold symmetry is visible as the number of teeth, and the depth of a tooth, $1 - \cos(\pi/n)$, is the width of the swept annulus.

The polygons as rotating objects

Now spin the polygon in space rather than in its plane. Take the regular $2m$ -gon inscribed in the unit circle, with two opposite vertices on the horizontal axis, and revolve it about that diagonal. The solid obtained is a stack of m conical frusta: between the levels $x_k = \cos \frac{k\pi}{m}$ and x_{k+1} the boundary is a straight line from radius $y_k = \sin \frac{k\pi}{m}$ to y_{k+1} , and

Lemma 18.9 (The frustum). *For $h > 0$, $\int_0^h \pi \left(R + \frac{(S-R)x}{h} \right)^2 dx = \frac{\pi h}{3} (R^2 + RS + S^2)$ (*frustum_volume*).*

Summing these gives the volume of the solid of revolution,

$$V(m) = \frac{\pi}{3} \sum_{k=0}^{m-1} (x_k - x_{k+1}) (y_k^2 + y_k y_{k+1} + y_{k+1}^2)$$

(*revVolume*), and this sum has a closed form.

Theorem 18.10 (Rotating in space). *For every $m \geq 1$,*

$$V(m) = \frac{2\pi}{3} \left(1 + \cos \frac{\pi}{m} \right) = \frac{4\pi}{3} \cos^2 \frac{\pi}{2m}.$$

In particular $V(m) \rightarrow \frac{4\pi}{3}$, the volume of the unit ball, and

$$V(2) = \frac{2\pi}{3} \text{ (the square), } \boxed{V(3) = \pi} \text{ (the hexagon), } V(6) = \frac{\pi(2 + \sqrt{3})}{3},$$

each polygon being spun about a long diagonal.

Proof. Write $a = \pi/m$, $x_k = \cos ka$, $y_k = \sin ka$. From $\cos a = x_k x_{k+1} + y_k y_{k+1}$ and $y^2 = 1 - x^2$, each summand collapses to a difference:

$$(x_k - x_{k+1})(y_k^2 + y_k y_{k+1} + y_{k+1}^2) = (2 + \cos a)(x_k - x_{k+1}) - (x_k^3 - x_{k+1}^3) = f(k) - f(k+1)$$

with $f(k) = (2 + \cos a)x_k - x_k^3$. The sum telescopes to $f(0) - f(m) = 2 + 2\cos a$ because $x_0 = 1$ and $x_m = \cos \pi = -1$; then $1 + \cos a = 2\cos^2 \frac{a}{2}$. Lean: `revVolume_sum`, `revVolume_eq`, `revVolume_two`, `revVolume_three`, `revVolume_six`, `tendsto_revVolume`. $\square$

That the hexagon — the polygon whose half-perimeter is exactly 3 — spins into a solid of volume exactly π is the cleanest coincidence in this paper, and it is not an accident of rounding: $\cos^2 \frac{\pi}{6} = \frac{3}{4}$ and $\frac{4}{3}\pi \cdot \frac{3}{4} = \pi$. Six is the only m for which the sweep is a rational multiple of the ball with numerator and denominator that small; the square gives half the ball and the twelve-gon gives $\frac{2+\sqrt{3}}{4}$ of it. The three solids and the curve $V(m)$ are Figure 24.

Back to the modular circle

Finally, the circle carrying all of this is the arc of Section 14, and the polygons respect its symmetry. On $|z| = 1$ the involution $S : z \mapsto -1/z$ is the reflection $\theta \mapsto \pi - \theta$:

$$-(e^{i\theta})^{-1} = e^{i(\pi-\theta)}$$

(`neg_inv_circle`). Anchor the regular n -gon at the fixed point i of S , i.e. take $v_k = \exp(i(\frac{\pi}{2} + \frac{2\pi k}{n}))$ (`vertex`); then $|v_k| = 1$ and

Theorem 18.11. $-v_k^{-1} = v_{-k}$ for all $k \in \mathbb{Z}$: the vertex set is stable under S , which swaps the k -th and $(-k)$ -th vertex and fixes the anchor $v_0 = i$ (`S_vertex`, `vertex_zero`).

For $n = 12$ this vertex set contains the two corners of the modular fundamental domain,

$$v_1 = e^{2\pi i/3} = \rho = \frac{-1 + i\sqrt{3}}{2}, \quad v_{-1} = e^{\pi i/3} = \rho + 1$$

(`vertex_twelve_one`, `vertex_twelve_neg_one`) — the twelve-gon of Archimedes' second rung is also the polygon whose vertices are the distinguished points of the arc. That picture is Figure 25.

19 Archimedes, spinning on Δ

The last two sections drew two different pictures on the same circle: the portrait of Δ , whose mirror is the arc $|z| = 1$ (Section 14), and Archimedes' inscribed polygons, whose vertices lie on that arc (Section 18). Here they are put on top of each other and the polygons are set turning. The Lean file is `RequestProject/ArchimedesDelta.lean`.

Let $t \in \mathbb{R}$ be the angle of rotation — the frame of the animation — and set

$$v_n(t, k) = \exp\left(i\left(\frac{\pi}{2} + t + \frac{2\pi k}{n}\right)\right), \quad k \in \mathbb{Z}$$

(`spinAngle`, `spinVertex`), with $V_n(t) = \{v_n(t, k) : k \in \mathbb{Z}\}$ the whole ring (`spinVertexSet`). At $t = 0$ this is the polygon anchored at i of Theorem 18.11 (`spinVertex_zero`).

Proposition 19.1 (The moving ring). *For all $t \in \mathbb{R}$ and $k \in \mathbb{Z}$:*

1. $|v_n(t, k)| = 1$: every vertex of every frame rides the arc $|z| = 1$, the mirror of the portrait (`norm_spinVertex`);
2. if $m \mid n$ and $n \geq 1$ then $V_m(t) \subseteq V_n(t)$: the rings $3 \mid 6 \mid 12$ and $3 \mid 9$ stay nested at every instant (`spinVertexSet_subset_of_dvd`);

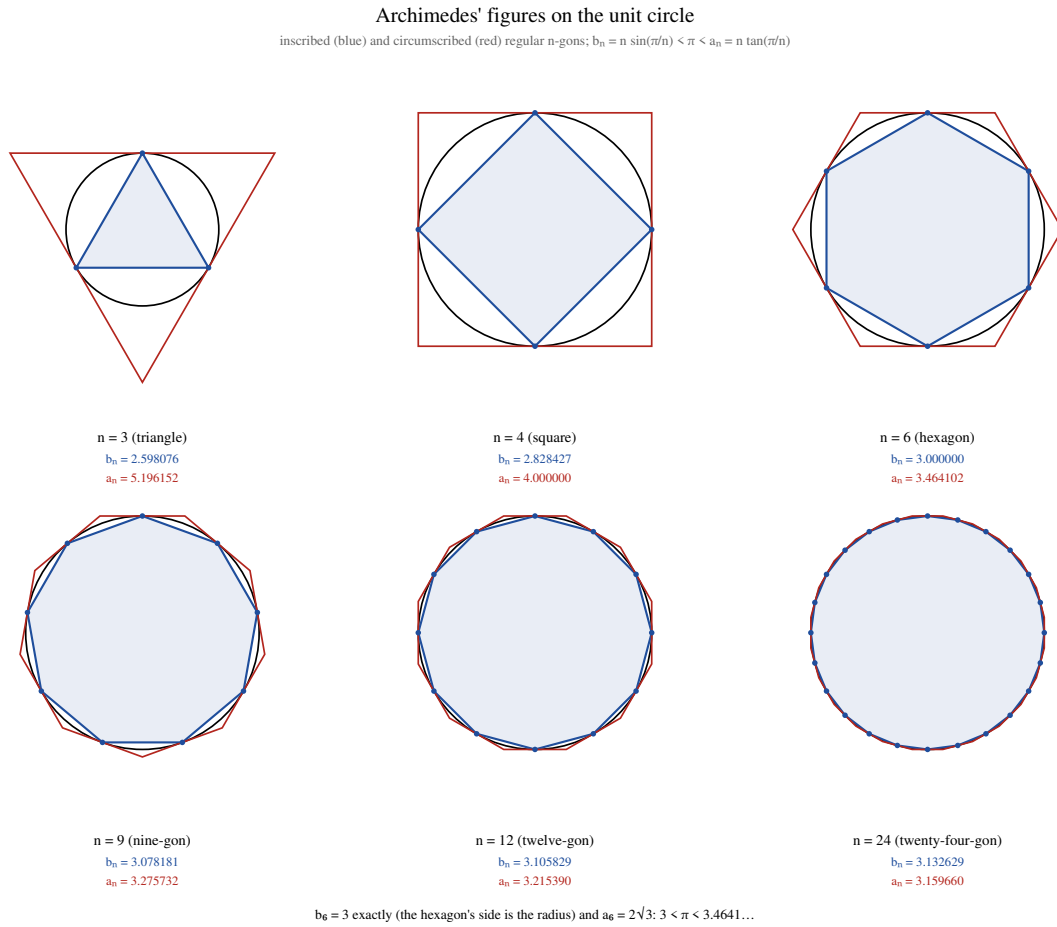


Figure 21: Archimedes' figures on the unit circle: the inscribed (blue) and circumscribed (red) regular n -gons for $n = 3, 4, 6, 9, 12, 24$, with the two half-perimeters $b_n = n \sin \frac{\pi}{n} < \pi < a_n = n \tan \frac{\pi}{n}$ of Theorem 18.2. The hexagon is the exact rung, $b_6 = 3$ and $a_6 = 2\sqrt{3}$ (Theorem 18.5).

The sandwich and the doubling ladder

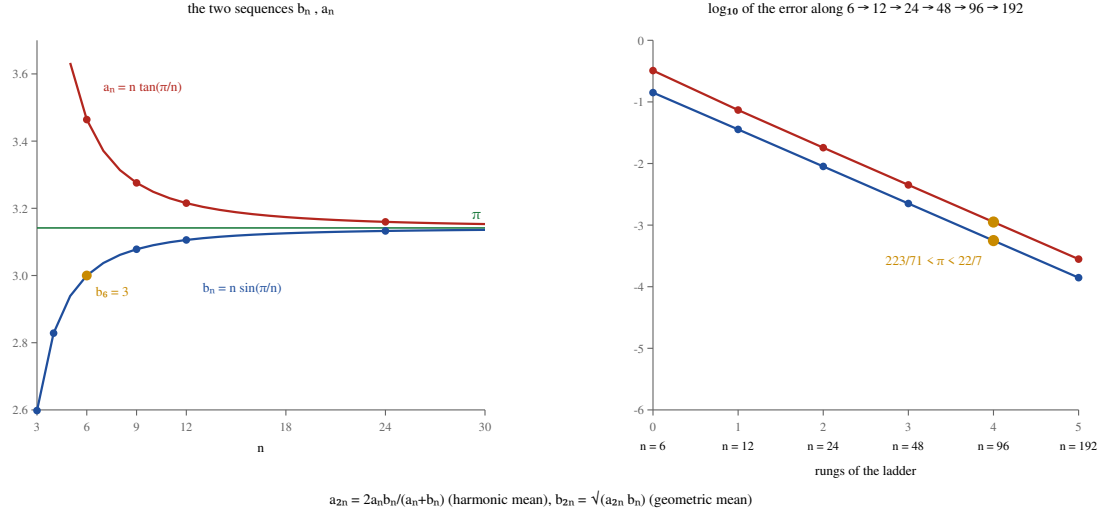


Figure 22: Left: the two sequences against n , closing on π from both sides (Theorems 18.2 and 18.4); the marked point is $b_6 = 3$. Right: the error along the doubling ladder $6 \rightarrow 12 \rightarrow 24 \rightarrow 48 \rightarrow 96 \rightarrow 192$ of Theorem 18.3, on a logarithmic scale — each rung gains a factor of four, about 0.6 of a decimal digit. The marked rung is $n = 96$, where the two bounds are $\frac{223}{71}$ and $\frac{22}{7}$ (Theorem 18.6).

The polygons as functions of the angle

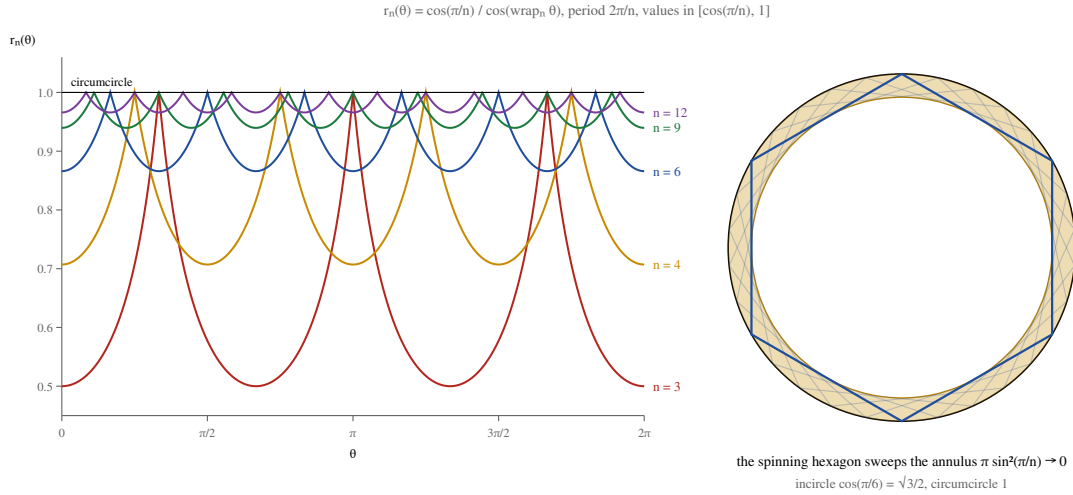


Figure 23: The polygons as functions of the angle. Left: the polar profiles $r_n(\theta) = \cos(\pi/n) / \cos(\text{wrap}_n \theta)$ for $n = 3, 4, 6, 9, 12$; each has period $2\pi/n$ (Theorem 18.8(1)), touches 1 at every vertex and dips to $\cos(\pi/n)$ at every edge midpoint. Right: the hexagon in five rotated positions, sweeping the annulus of area $\pi \sin^2(\pi/6)$ between its incircle and the unit circle.

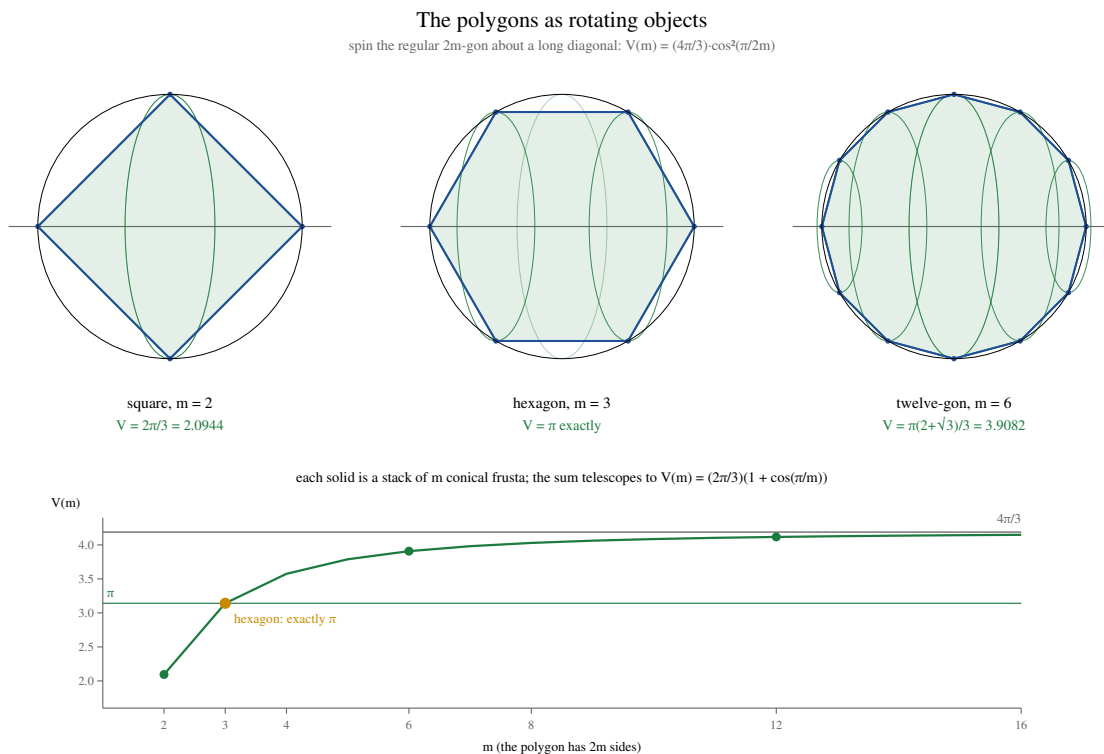


Figure 24: The polygons as rotating objects. Top: the square, the hexagon and the twelve-gon spun about a long diagonal, drawn with the circles of revolution through each vertex; each solid is a stack of m conical frusta (Lemma 18.9). Bottom: the volume $V(m) = \frac{4\pi}{3} \cos^2 \frac{\pi}{2m}$ of Theorem 18.10, rising to the volume $\frac{4\pi}{3}$ of the ball; the hexagon, $m = 3$, sits exactly on the line $V = \pi$.

3. the polygon is rigid: $|v_n(t, k+1) - v_n(t, k)| = 2 \sin \frac{\pi}{n}$ for $n \geq 3$, so its perimeter is $2b_n$ at every instant, and every edge of the hexagon has length exactly 1, the radius (`chord_eq`, `perimeter_eq`, `hexagon_chord`).

Proof. (1) is $|e^{i\theta}| = 1$. For (2), write $n = md$; then $\frac{\pi}{2} + t + \frac{2\pi j}{m} = \frac{\pi}{2} + t + \frac{2\pi(jd)}{md}$, so $v_m(t, j) = v_n(t, jd)$. For (3), the two vertices differ by the factor $e^{2\pi i/n}$,

$$v_n(t, k+1) - v_n(t, k) = v_n(t, k)(e^{2\pi i/n} - 1),$$

and $|e^{ia} - 1|^2 = (\cos a - 1)^2 + \sin^2 a = 2 - 2 \cos a = 4 \sin^2 \frac{a}{2}$ (`norm_exp_sub_one`); with $a = 2\pi/n$ this is $2 \sin \frac{\pi}{n} \geq 0$. Multiplying by n gives $2n \sin \frac{\pi}{n} = 2b_n$, and $2 \sin \frac{\pi}{6} = 1$. $\square$

So the animation neither stretches nor shears the figure: what turns on the portrait is Archimedes' own polygon, carrying its own lower bound $b_n < \pi$ of Theorem 18.2 with it. The next statement says how long the film is.

Theorem 19.2 (The loop). *Let $n \geq 1$. Then $V_n(s) = V_n(t)$ if and only if $s = t + \frac{2\pi k}{n}$ for some $k \in \mathbb{Z}$. In particular the n -gon repeats itself after a rotation by $2\pi/n$, and not before: the loop of the plate is one n -th of a turn (`spinVertexSet_eq_iff`, `spinVertexSet_rot`).*

Proof. If $s = t + 2\pi k/n$ then $v_n(s, j) = v_n(t, j+k)$, a reindexing of the ring. Conversely, if the two rings agree then $v_n(s, 0) \in V_n(t)$, so $\frac{\pi}{2} + s = \frac{\pi}{2} + t + \frac{2\pi j}{n} + 2\pi m$ for some $j, m \in \mathbb{Z}$ (`spinVertex_eq_iff`, which is $e^x = e^y$ up to $2\pi i\mathbb{Z}$), and $s = t + \frac{2\pi(j-mn)}{n}$. $\square$

Theorem 19.3 (S runs the animation backwards). *Let $S(z) = -1/z$. Then $S(v_n(t, k)) = v_n(-t, -k)$, hence $S(V_n(t)) = V_n(-t)$: the modular involution reverses the frame. Consequently, for $n \geq 1$,*

$$S(V_n(t)) = V_n(t) \iff t \in \frac{\pi}{n}\mathbb{Z},$$

so the spinning ring is S -symmetric at exactly two instants of each loop (`S_spinVertex`, `S_image_spinVertexSet`, `S_stable_iff`).

Proof. On the unit circle S is $\theta \mapsto \pi - \theta$ (`neg_inv_circle`), and $\pi - (\frac{\pi}{2} + t + \frac{2\pi k}{n}) = \frac{\pi}{2} - t - \frac{2\pi k}{n}$, which is the angle of $v_n(-t, -k)$. By Theorem 19.2, $V_n(-t) = V_n(t)$ iff $-t = t + 2\pi k/n$ for some k , i.e. $t = -\pi k/n$. Within one loop, of length $2\pi/n$, the set $\frac{\pi}{n}\mathbb{Z}$ has exactly two points. $\square$

At $t = 0$ the two fixed vertices are the anchor i and its antipode; for $n = 12$ this is the frame of Theorem 18.11, in which two vertices are the corners ρ and $\rho + 1$ of the modular fundamental domain. Half a step later, at $t = \pi/12$, the ring is symmetric again but no vertex sits at i : the mirror then pairs the two vertices adjacent to i .

Finally, the reason the rings were put on the portrait rather than beside it. Every vertex is on the arc, so the arc corollary of Section 14 applies at each of them.

Theorem 19.4 (The phase lock along the ring). *Let $v = v_n(t, k)$ and suppose $\text{Im } v > 0$. Then*

$$v^{12} \Delta(v)^2 = |\Delta(v)|^2,$$

so the argument of Δ at a vertex is determined, up to half turns, by the vertex angle alone (`Delta_spinVertex_phase`). For $n = 12$ the factor is the same at all twelve vertices,

$$v_{12}(t, k)^{12} = e^{12it} \quad \text{for every } k,$$

so $e^{12it}\Delta(v)^2 = |\Delta(v)|^2$ at each of them (*spinVertex_twelve_pow*, *Delta_spinVertex_twelve_phase*): the whole ring turns its phase in unison, twelve times per turn of the polygon, and returns to its initial lock after exactly one loop $t \mapsto t + \pi/6$. Moreover the two vertices exchanged by S carry the same modulus, $|\Delta(v_n(-t, -k))| = |\Delta(v_n(t, k))|$ (*norm_Delta_S_pair*).

Proof. The first identity is the arc corollary of Section 14 applied to v , which has $|v| = 1$ by Proposition 19.1(1). For the second, 12 times the angle $\frac{\pi}{2} + t + \frac{2\pi k}{12}$ is $6\pi + 12t + 2\pi k$, and $e^{i(6\pi+2\pi k)} = 1$. The last claim is $\Delta(-1/z) = z^{12}\Delta(z)$ with $|z| = 1$, using $S(v_n(t, k)) = v_n(-t, -k)$. $\square$

Remark 19.5 (What is drawn, and what is proved). The coloured background of Figures 26–28 is a raster of the truncated product $q \prod_{m \leq 160} (1 - q^m)^{24}$, evaluated pixel by pixel; the theorems above are statements about the polygons and about the true $\Delta = \eta^{24}$ of Section 14, and none of them depends on the picture. Conversely the picture is not evidence: it is the same drawing routine as Figure 3, re-plotted on a window centred so that the whole unit circle fits. Below the real axis there is nothing to draw — Δ lives on the upper half plane — so the lower halves of the polygons turn over a blank field.

The animation. The rotating version is `paper/archimedes_delta_spin.svg`, a plain SVG with SMIL animation: opened in any browser it turns the four rings continuously, one full turn in twelve seconds, so that the twelve-gon closes its loop once a second and the triangle once every four. The dial beside it turns twelve times as fast — the factor e^{12it} of Theorem 19.4 — and the vertical mirror flashes at each of the S -symmetric instants of Theorem 19.3. A printed page can only show one frame of it, which is Figure 28; Figure 27 is a filmstrip of six frames of one loop. All three plates, and the animation, are generated by `paper/make_archimedes_delta.py`, which recomputes the coefficients $\tau(n)$ and refuses to run unless they agree with the Lean-certified table.

20 Dissection: Euclid, OpenCV, and the character of Δ

The portrait of Δ has been drawn, mirrored, spun and overlaid with polygons. This section takes it apart. A computer-vision pipeline finds the shapes the picture actually contains; Euclid’s ternary dissection cuts each shape into nine congruent parts, and each of those into nine again; the value of Δ is read inside every part; the readings are normalised into a matrix of grey levels — a *character*, in the typographical sense — and Frobenius is then applied to it in both senses of the word. The Lean files are `RequestProject/DeltaDissection.lean` and `RequestProject/DeltaFrobenius.lean`; the plates are Figures 29 and 30, generated by `paper/make_delta_dissection.py`.

The dissection

Definition 20.1 (Ternary cells). For $k \in \mathbb{N}$ let the *address* of z at level k be $\iota_k(z) = (\lfloor 3^k \operatorname{Re} z \rfloor, \lfloor 3^k \operatorname{Im} z \rfloor)$ (*cellIndex*), and let

$$C_k(i, j) = \{z : \iota_k(z) = (i, j)\}$$

be the half-open square of side 3^{-k} with that address (*cell*).

Proposition 20.2 (The dissection is a partition). *For every k , every point of the plane lies in exactly one cell $C_k(i, j)$ (`exists_unique_cell`); distinct addresses give disjoint cells (`cell_disjoint`) and the cells of a level cover the plane (`iUnion_cell`). Moreover $z \in C_{k+1}(i, j)$ if and only if $3z \in C_k(i, j)$ (`mem_cell_succ_iff`), and any two points of a level- k cell are at distance at most $2 \cdot 3^{-k}$ (`dist_le_of_mem_cell`), a bound that tends to zero (`tendsto_cell_diam`).*

Proof. Membership is the equation $\iota_k(z) = (i, j)$, so existence and uniqueness are the definition of the floor function, and the two set statements follow. The similarity statement is $3^{k+1}x = 3^k(3x)$. For the diameter, the defining inequalities give $|\operatorname{Re} z - \operatorname{Re} w| \leq 3^{-k}$ and likewise for the imaginary parts, and $|u| \leq |\operatorname{Re} u| + |\operatorname{Im} u|$. $\square$

Theorem 20.3 (Nine, and nine again). *Every cell is the union of the nine cells one level down whose addresses are $(3i + a, 3j + b)$ with $0 \leq a, b < 3$ (`cell_subdivide`). Iterating, the level- $(k + m)$ descendants of $C_k(i, j)$ are indexed by the block $[3^m i, 3^m i + 3^m] \times [3^m j, 3^m j + 3^m]$ (`children`), there are exactly 9^m of them (`card_children`), and their union is the cell (`cell_eq_iUnion_children`).*

Proof. For any real u one has $3\lfloor u \rfloor \leq \lfloor 3u \rfloor < 3\lfloor u \rfloor + 3$, so $\lfloor 3u \rfloor = 3\lfloor u \rfloor + a$ with $a \in \{0, 1, 2\}$ (`floor_three_mul`); applying this to $u = 3^k \operatorname{Re} z$ and to $u = 3^k \operatorname{Im} z$ gives the nine-fold subdivision. For the iterate, $3^m i \leq \lfloor 3^{k+m} x \rfloor < 3^m i + 3^m$ is equivalent to $i \leq 3^k x < i + 1$ because $3^{k+m} x = 3^m(3^k x)$; the count is $3^m \cdot 3^m = 9^m$. $\square$

So a shape of the picture, once boxed, carries 9, 81, 729, ... organised parts as one goes down. Panel B of Figure 29 draws the first two of those levels.

The delta inside the parts

Definition 20.4 (The reading). The centre of $C_k(i, j)$ is $c_k(i, j) = 3^{-k}((i + \frac{1}{2}) + (j + \frac{1}{2})i)$ (`center`, and `center_mem_cell` says it lies in the cell), and the *reading* of Δ inside the part is

$$\mu_k(i, j) = \|\Delta(c_k(i, j))\|$$

(`mass`), where $\Delta = \eta^{24}$ is the honest infinite product of Section 14.

Proposition 20.5 (The two symmetries, in cell coordinates). *For $j \geq 0$ the reading is strictly positive (`mass_pos`); it is invariant under a shift of the address by a whole period, $\mu_k(i + 3^k, j) = \mu_k(i, j)$ (`mass_period`); and the cells with addresses i and $-1 - i$ are mirror images in the wall $x = 0$ and carry equal readings, $\mu_k(-1 - i, j) = \mu_k(i, j)$ (`mass_mirror`).*

Proof. Positivity is $\Delta \neq 0$ on the upper half plane (Definition 14.1). The centre of $C_k(i + 3^k, j)$ is the centre of $C_k(i, j)$ plus one, so periodicity is Proposition 14.6. The centre of $C_k(-1 - i, j)$ is $-c_k(i, j)$, so the mirror is $\Delta(-\bar{z}) = \overline{\Delta(z)}$ (Proposition 14.8) followed by $\|\bar{w}\| = \|w\|$. $\square$

Normalising into a character

Fix a level k , a base row $j_0 \geq 0$ and a half-width m . The block of readings has $m + 1$ rows and $2m + 2$ columns, placed symmetrically about the wall $x = 0$:

$$R(r, c) = \mu_k(c - (m + 1), j_0 + r), \quad 0 \leq r \leq m, \quad 0 \leq c \leq 2m + 1$$

(**raw**). Because $\|\Delta\|$ collapses with height, a single normalising constant would flatten every row but the lowest; so each row is divided by its own maximum $P(r) = \max_c R(r, c)$ (**rowPeak**):

$$G(r, c) = \frac{R(r, c)}{P(r)}$$

(**glyph**). This matrix of grey levels is the *character* of Δ at (k, j_0, m) . Quantising it to the three levels of the ternary dissection,

$$\pi(r, c) = \begin{cases} 0, & G(r, c) < \frac{1}{3}, \\ 1, & \frac{1}{3} \leq G(r, c) < \frac{2}{3}, \\ 2, & G(r, c) \geq \frac{2}{3}, \end{cases} \quad \text{read in } \mathbb{Z}/3$$

(**level**, **pix**), gives the three-shade glyph printed beside it.

Theorem 20.6 (The character). *Let $j_0 \geq 0$. Then $0 \leq G(r, c) \leq 1$ everywhere (**glyph_nonneg**, **glyph_le_one**); every row contains a pixel with $G(r, c) = 1$ (**glyph_row_eq_one**); and the character is a palindrome:*

$$G(r, 2m + 1 - c) = G(r, c)$$

for all r, c (**mirrorCol**, **glyph_mirror**), hence also $\pi(r, 2m + 1 - c) = \pi(r, c)$ (**pix_mirror**).

Proof. The row maximum is attained and positive by Proposition 20.5, which gives the bounds and the black pixel in each row. Column $2m + 1 - c$ has address $-1 - (c - (m + 1))$, so the palindrome is exactly $\mu_k(-1 - i, j) = \mu_k(i, j)$, and quantisation is a function of G . $\square$

Frobenius, twice

For blocks of pixels put

$$\langle A, B \rangle_F = \sum_{r, c} A(r, c) B(r, c), \quad \|A\|_F = \sqrt{\langle A, A \rangle_F}$$

(**frobInner**, **frobNorm**), and let $(\mathcal{M}A)(r, c) = A(r, 2m + 1 - c)$ be the mirror (**mirrorMat**).

Theorem 20.7 (The character in the Frobenius metric). *The mirror is an isometry, $\|\mathcal{M}A\|_F = \|A\|_F$ for every block (**frobInner_mirror**, **frobNorm_mirror**); the character of Δ is fixed by it, $\mathcal{M}G = G$ (**mirrorMat_glyph**); consequently the character and its mirror image have Frobenius cosine similarity exactly one,*

$$\langle G, \mathcal{M}G \rangle_F = \|G\|_F^2$$

(**frobInner_glyph_mirror**). Its size is pinned by

$$\sqrt{m + 1} \leq \|G\|_F \leq \sqrt{(m + 1)(2m + 2)}$$

(**sqr_rows_le_frobNorm_glyph**, **frobNorm_glyph_le**).

Proof. Reversal of columns is a permutation of the index set, so it preserves the sum defining $\langle \cdot, \cdot \rangle_F$; $\mathcal{M}G = G$ is Theorem 20.6. The lower bound adds up one black pixel per row, the upper bound bounds every pixel by one. $\square$

That is Frobenius in the sense of linear algebra. In the sense of arithmetic, quantisation puts the character in $\mathbb{Z}/3$, where the Frobenius endomorphism is the identity.

Theorem 20.8 (Frobenius fixes the three-shade glyph). $\pi(r, c)^3 = \pi(r, c)$ for every pixel, i.e. the Frobenius endomorphism $x \mapsto x^3$ of $\mathbb{Z}/3$ fixes the quantised character pointwise (`pix_frobenius`, `frobenius_pix`).

Proof. $x^p = x$ on $\mathbb{Z}/p$. □

The same endomorphism acts on the arithmetic side of Δ , where the datum attached to Frobenius at a prime p is the trace $\tau(p)$ (`frobTrace`).

Theorem 20.9 (The trace of Frobenius, and what Frobenius does to it). *For every prime $p \leq 30$ with $p \neq 3$:*

1. $\tau(p) \equiv p^2 + p^9 \pmod{27}$ — the trace is a sum of two power characters whose exponents add to $11 = k - 1$, the weight of Δ minus one (`frobTrace_mod_27`, `frobTrace_zmod_27`);
2. applying $x \mapsto x^3$ twice in $\mathbb{Z}/3$ collapses p^9 to p , so $\tau(p) \equiv p(p + 1) \pmod{3}$ (`frobTrace_zmod_3`);
3. hence $3 \mid \tau(p)$ — equivalently, the digital root of $|\tau(p)|$ lies in $\{3, 6, 9\}$ — exactly when $p \not\equiv 1 \pmod{3}$ (`three_dvd_tau_iff_not_one_mod_three`).

Finally, the row of normalised traces $\tau(p)/p^{11/2}$ over the ten primes below thirty has Frobenius norm at most $2\sqrt{10}$ (`frobNorm_traceRow_le`): the Ramanujan–Deligne bound, read as a length.

Proof. (1) is the congruence $\tau(n) \equiv n^2 \sigma_7(n) \pmod{27}$ of Proposition 12.5, checked in the kernel on $n \leq 30$, together with $\sigma_7(p) = 1 + p^7$. For (2), reduce (1) modulo 3 and use $p^9 = (p^3)^3 = p$ in $\mathbb{Z}/3$; (3) is $p(p + 1) \equiv 0 \pmod{3}$ unless $p \equiv 1$, where it is 2, plus Proposition 12.5. The last statement is $\tau(p)^2 \leq 4p^{11}$ at each of the ten primes, so the sum of squares is at most 40. □

Remark 20.10 (What is drawn and what is proved). The contours of panel A are found by OpenCV in a *raster* of a truncated product: the picture is thresholded at five levels, the connected components of each threshold are traced, and each trace is reduced to a polygon by the Douglas–Peucker algorithm. No theorem here reads that raster; the shapes are a way of pointing at parts of the picture, and everything asserted about those parts — the nine-fold subdivision, the readings, the character, the two Frobeniuses — is stated and proved for the exact objects of Definitions 20.1 and 20.4. On the arithmetic side, the congruences of Theorem 20.9 are verified for $p \leq 30$, the range on which the coefficients themselves are certified here; they are not proved for all p . The generating script recomputes $\tau(1), \dots, \tau(30)$ by two independent expansions, refuses to draw unless both agree with the certified table, and re-checks period, mirror, normalisation, palindromy, the Frobenius identities and the congruences numerically before emitting a single pixel.

21 Triangles: every frame cut to pieces, and sorted by colour

Section 19 set Archimedes’ polygons spinning on the portrait of Δ ; Section 20 cut the plane into squares Euclid’s way. This section does both at once, and on the moving picture: each frame

of the spinning n -gon is cut into n triangles by the fan from the centre, each triangle is cut into *nine* by trisecting its sides, that cut is iterated d times, and every one of the resulting $n \cdot 9^d$ parts is painted with the value of Δ at its centroid. The parts of a frame are then sorted by that colour. The Lean file is `RequestProject/DeltaTriangles.lean`; the plates are Figures 31 and 32, generated by `paper/make_delta_triangles.py`.

Triangles and their nine parts

Definition 21.1 (Triangles, parametrised). A triangle is an ordered triple $T = (a, b, c)$ of complex numbers (**Tri**). With the standard parameter triangle $\Sigma = \{(u_1, u_2) : u_1, u_2 \geq 0, u_1 + u_2 \leq 1\}$ (**stdTri**) and the barycentric parametrisation $p_T(u) = a + u_1(b - a) + u_2(c - a)$ (**para**), the filled triangle is $\text{sh}(T) = p_T(\Sigma)$ (**shape**). Its *size* is $\sigma(T) = |b - a| + |c - a|$ (**size**), and its centroid is $(a + b + c)/3$ (**centroidT**), which lies in it (**centroid_mem_shape**).

Definition 21.2 (Euclid's ternary cut of a triangle). For $(\alpha, \beta, s) \in \mathbb{R}^3$ let $A_{\alpha\beta s}(u) = (\alpha + su_1, \beta + su_2)$ (**aff**); then $p_T \circ A_{\alpha\beta s} = p_{T'}$ for the triangle $T' = (p_T(\alpha, \beta), p_T(\alpha + s, \beta), p_T(\alpha, \beta + s))$ (**affTri**, **para_aff**). The nine parts of T are the triangles $\chi_i(T)$, $i < 9$ (**child**), obtained from the nine parameter triples (**childP**)

$$\left(0, 0, \frac{1}{3}\right), \left(\frac{1}{3}, 0, \frac{1}{3}\right), \left(\frac{2}{3}, 0, \frac{1}{3}\right), \left(0, \frac{1}{3}, \frac{1}{3}\right), \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \left(0, \frac{2}{3}, \frac{1}{3}\right), \left(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}\right), \left(\frac{2}{3}, \frac{1}{3}, -\frac{1}{3}\right), \left(\frac{1}{3}, \frac{2}{3}, -\frac{1}{3}\right) :$$

six with $s = \frac{1}{3}$, the parts that point the same way as T , and three with $s = -\frac{1}{3}$, the parts that point the other way. For a word $w \in \{0, \dots, 8\}^*$ write $\chi_w(T)$ for the iterate (**desc**).

Theorem 21.3 (Nine parts, and 9^d of them). *For every triangle T :*

1. $\text{sh}(T) = \bigcup_{i < 9} \text{sh}(\chi_i(T))$ (**shape_eq_iUnion_child**);
2. $\sigma(\chi_i(T)) = \sigma(T)/3$ for every i (**size_child**), hence $\sigma(\chi_w(T)) = \sigma(T)/3^{|w|}$ (**size_desc**);
3. any two points of $\text{sh}(T)$ are at distance at most $\sigma(T)$ (**dist_le_size**);
4. for every d , $\text{sh}(T) = \bigcup_{|w|=d} \text{sh}(\chi_w(T))$ (**shape_eq_iUnion_desc**), a union of exactly 9^d parts (**card_words**) of size $\sigma(T)/3^d$.

Proof. Each $A_{\alpha\beta s}$ of Definition 21.2 maps Σ into Σ (**childMap_mapsTo**), which with $p_T \circ A = p_{\chi_i(T)}$ gives the inclusion $\supseteq$ in (1). For $\subseteq$, let $(x, y) \in \Sigma$ and put $a = 3x$, $b = 3y$, so $a, b \geq 0$ and $a + b \leq 3$. If $a, b \leq 1$ then (x, y) is the image of (a, b) under the first part when $a + b \leq 1$, and of $(1 - a, 1 - b)$ under the seventh when $a + b > 1$; the remaining cases $a > 1$, $b > 1$, and a or b beyond 2 are the same computation shifted, and exhaust Σ (**stdTri_subset_iUnion_childMap**). This is the trisection of the three sides: six upward parts and three downward ones. For (2), the two side vectors of $\chi_i(T)$ are $s(b - a)$ and $s(c - a)$ with $|s| = \frac{1}{3}$. For (3), a difference of two parametrised points is $(u_1 - v_1)(b - a) + (u_2 - v_2)(c - a)$ with $|u_j - v_j| \leq 1$. Item (4) is (1) and (2) by induction on d , the count being 9^d because a word of length d is a function $\{0, \dots, d - 1\} \rightarrow \{0, \dots, 8\}$. $\square$

One dissection for the whole animation

Definition 21.4 (The fan of a frame). For the frame t of the spinning n -gon of Section 19, the k -th triangle of the fan from the centre is $F_n(t, k) = (0, v_n(t, k), v_n(t, k + 1))$ (**fanTri**). Its size is 2: two unit radii (**size_fanTri**), so its parts d levels down have size $2/3^d$ (**size_desc_fanTri**).

Theorem 21.5 (The frames are one frame, turned). *Let $M_{a,b}(z) = az + b$ act on triangles vertex-wise (`mapTri`). Then the cut commutes with every complex affine map, $\chi_i(M_{a,b}T) = M_{a,b}\chi_i(T)$ and $\chi_w(M_{a,b}T) = M_{a,b}\chi_w(T)$ (`child_mapTri`, `desc_mapTri`), and $\text{sh}(M_{a,b}T) = M_{a,b}\text{sh}(T)$ (`shape_mapTri`). Since $F_n(t, k) = M_{e^{it}, 0}F_n(0, k)$ (`fanTri_rot`), every part of every frame is the corresponding part of the frame $t = 0$, rotated by t :*

$$\chi_w(F_n(t, k)) = M_{e^{it}, 0} \chi_w(F_n(0, k)) \quad (\text{desc_fanTri_rot}).$$

The same holds for the vertical mirror: $\chi_i(\overline{T}^-) = \overline{\chi_i(T)}^-$ where $\overline{T}^-$ is T with each vertex sent to $-\bar{z}$ (`conjTri`, `child_conjTri`, `desc_conjTri`).

Proof. $p_{M_{a,b}T}(u) = ap_T(u) + b$ and $z \mapsto -\bar{z}$ is $\mathbb{R}$ -affine, so both statements are the identity $p_{\varphi T} = \varphi \circ p_T$ applied at the nine fixed parameter triples; the iterate follows by induction on the word. For the fan, $v_n(t, k) = e^{it}v_n(0, k)$. $\square$

The reading, and the palette

Definition 21.6 (Reading a part, and sorting the frame). The colour of a part is $\rho(T) = \|\Delta(\text{centroid } T)\|$ (`reading`). The frame t of the n -gon, cut d levels down, carries the multiset

$$R_n(t, d) = \sum_{k < n} \{ \rho(\chi_w(F_n(t, k))) : |w| = d \} \quad (\text{frameReadings, readings}),$$

and sort $R_n(t, d)$ is its *palette* (`sortedReadings`).

Theorem 21.7 (The palette loops with the animation). *For $n \geq 1$ the frame carries exactly $n \cdot 9^d$ colours (`card_readings`), and*

$$R_n\left(t + \frac{2\pi}{n}, d\right) = R_n(t, d), \quad \text{sort } R_n\left(t + \frac{2\pi}{n}, d\right) = \text{sort } R_n(t, d)$$

(`readings_rot`, `sortedReadings_rot`): *after one loop of the animation the frame shows not merely a similar palette but the same sorted list. Moreover the mirror does not change a colour: if the centroid of T lies in the upper half plane then $\rho(\overline{T}^-) = \rho(T)$ (`reading_conjTri`).*

Proof. Rotating the frame by $2\pi/n$ carries the k -th triangle of the fan to the $(k+1)$ -st, $F_n(t + 2\pi/n, k) = F_n(t, k+1)$ (`fanTri_shift`), so it carries each part to a part; and the index wraps, $v_n(t, k+n) = v_n(t, k)$ (`spinVertex_add_nat`). A sum over $k = 0, \dots, n-1$ of a function with $f(n) = f(0)$ is therefore unchanged by the shift $k \mapsto k+1$ (`sum_range_shift`), which is the first identity; sorting equal multisets gives equal lists. The last statement is $\Delta(-\bar{z}) = \overline{\Delta(z)}$ (`DeltaModular.Delta_neg_conj`) at the centroid, whose mirror is the centroid of the mirrored part. $\square$

How the polygons approximate the curve

Proposition 21.8 (Chords, and the arc). *For all real a, b , $\|e^{ia} - e^{ib}\| \leq |a - b|$: a chord is at most its arc (`norm_exp_sub_exp_le`). Consequently, at every instant t and for every θ there is a vertex of the frame within π/n of $e^{i\theta}$,*

$$\exists k \in \mathbb{Z}, \quad \|e^{i\theta} - v_n(t, k)\| \leq \frac{\pi}{n} \quad (\text{dist_circle_spinVertexSet_le}),$$

and $\pi/n \rightarrow 0$ (*tendsto_dist_circle*), while the perimeters $2n \sin(\pi/n)$ rise to 2π (Theorem 18.4). So the frames of the animation approximate the circle they are inscribed in, in position as well as in length.

Proof. $\|e^{ic} - 1\|^2 = 2 - 2 \cos c \leq c^2$, the inequality being $1 - c^2/2 \leq \cos c$; multiplying by the unit e^{ib} gives the first claim. For the second, the vertex angles are $c + 2\pi k/n$ with $c = \pi/2 + t$; taking k the nearest integer to $(\theta - c)n/2\pi$ makes $|\theta - (c + 2\pi k/n)| = (2\pi/n) |x - \text{round } x| \leq \pi/n$, and chords are at most arcs. $\square$

Remark 21.9 (What is drawn and what is proved). The plates paint a part only when its centroid lies in the upper half plane: Δ is the eta product, defined there and nowhere else, and the frames of the animation dip below the real axis. Those parts are drawn in flat grey and are not read, so the palette drawn is that of the readable parts — which is again invariant under the loop, because a rotation by $2\pi/n$ permutes the parts of the frame among themselves and so maps readable parts to readable parts. The colour scale is a percentile-clipped logarithm of $\|\Delta\|$, chosen so that the structure is visible; nothing is claimed about it. The generating script recomputes $\tau(1), \dots, \tau(30)$, refuses to draw unless they match the certified table, and then re-checks numerically that the nine parts cover a triangle, that each cut divides the size by three, that the parts of a frame t are the parts of the frame 0 rotated, that the cut commutes with the mirror and the mirror preserves the reading, that the palette is unchanged after a turn of $2\pi/n$, and that every point of the arc is within π/n of a vertex, before emitting a single polygon.

22 Filling the plane with Euclid’s triangles, and its character table

Section 21 cut one frame of the spinning n -gon into $n \cdot 9^d$ triangles and left the rest of the plane untouched. This section keeps cutting. The same fan is placed on a ladder of circles of radii 3^j , $j = 0, 1, 2, \dots$; each of its n sectors is cut Euclid’s way to depth d ; and the parts — all of them triangles, all of them similar to the first — cover the whole plane (Theorem 22.3). Every part is then read by Δ at its centroid, the 9^d readings of a sector are added into one number, and the resulting matrix of *palette traces* — rings against sectors — is expanded in the character table of the cyclic group C_n of sectors, in the way Frobenius tabulated characters (Definition 22.7). The Lean files are `RequestProject/EuclidTiling.lean` and `RequestProject/PaletteCharacterTable.lean`; the plate is Figure 33 and the two tables are displayed on page 72, all generated by `paper/make_palette_character.py`.

The ring ladder

Definition 22.1 (Rings and tiles). For a level $j \in \mathbb{N}$, a time $t \in \mathbb{R}$ and a sector $k \in \mathbb{Z}$ put

$$R_{j,k}(t) = 3^j \cdot F_k(t) \quad (\text{ringTri}),$$

the k -th triangle $F_k(t)$ of the fan of Definition 21.4 scaled by 3^j : ring j is the fan of the inscribed n -gon of circumradius 3^j , and ring 0 is the frame of Section 21. For a word $w \in \{0, \dots, 8\}^*$ the *tile*

$$\Theta_{j,k,w}(t) = \chi_w(R_{j,k}(t)) \quad (\text{tile})$$

is the part of that sector addressed by w in Euclid’s ternary cut of Definition 21.2.

Proposition 22.2 (Sizes and counts). $\sigma(R_{j,k}(t)) = 2 \cdot 3^j$ (*size_ringTri*) and $\sigma(\Theta_{j,k,w}(t)) = 2 \cdot 3^j / 3^{|w|}$ (*size_tile*). In particular along the diagonal $|w| = j$ every tile has size exactly 2 (*size_tile_diag*): cutting each ring as many times as the ring is large gives one single scale for the whole filling. A ladder of N rings of n sectors cut to depth d carries exactly $N \cdot n \cdot 9^d$ tiles (*card_tiles*).

Theorem 22.3 (The triangles fill the plane). Let $n \geq 3$ and let $t \in \mathbb{R}$. Then

1. the fan of the unit n -gon covers the closed disc of radius $\cos \frac{\pi}{n}$: if $|z| \leq \cos \frac{\pi}{n}$ then $z \in \text{sh}(F_k(t))$ for some k (*mem_shape_fanTri*); hence ring j covers the disc of radius $3^j \cos \frac{\pi}{n}$ (*mem_shape_ringTri*);
2. $\cos \frac{\pi}{n} \geq \frac{1}{2}$ (*half_le_cos_pi_div*), so ring j covers the disc of radius $3^j/2$ (*closedBall_subset_ring*) and the rings exhaust the plane, $\bigcup_{j,k} \text{sh}(R_{j,k}(t)) = \mathbb{C}$ (*iUnion_ring_eq_univ*);
3. for every depth d every point of the plane lies in a tile of depth d (*exists_tile*), that is,

$$\bigcup_{j \in \mathbb{N}} \bigcup_{k \in \mathbb{Z}} \bigcup_{|w|=d} \text{sh}(\Theta_{j,k,w}(t)) = \mathbb{C} \quad (\text{iUnion_tiles_eq_univ}).$$

Proof. For (1), write $z = re^{i\varphi}$ with $r = |z| \leq \cos \frac{\pi}{n}$ and choose the sector k with φ in the angular interval of $F_k(t)$, of width $2\pi/n$. In the barycentric parametrisation of Definition 21.1 the point z has coordinates obtained by solving the two-by-two real system with determinant $\sin \frac{2\pi}{n} > 0$; both coordinates are nonnegative because φ lies between the two vertex angles, and their sum is at most 1 exactly because the distance from the centre to the edge — the inradius $\cos \frac{\pi}{n}$ — is at least r . Scaling by 3^j is Definition 22.1, which gives the statement for ring j . For (2), $\pi/n \leq \pi/3$ and the cosine decreases on $[0, \pi]$, so $\cos \frac{\pi}{n} \geq \cos \frac{\pi}{3} = \frac{1}{2}$; given z , any j with $3^j > 2|z|$ puts z in ring j , and such a j exists since $3^j \rightarrow \infty$. Part (3) is (2) followed by Theorem 21.3(4), which cuts each $R_{j,k}(t)$ into its 9^d parts of depth d without losing a point. $\square$

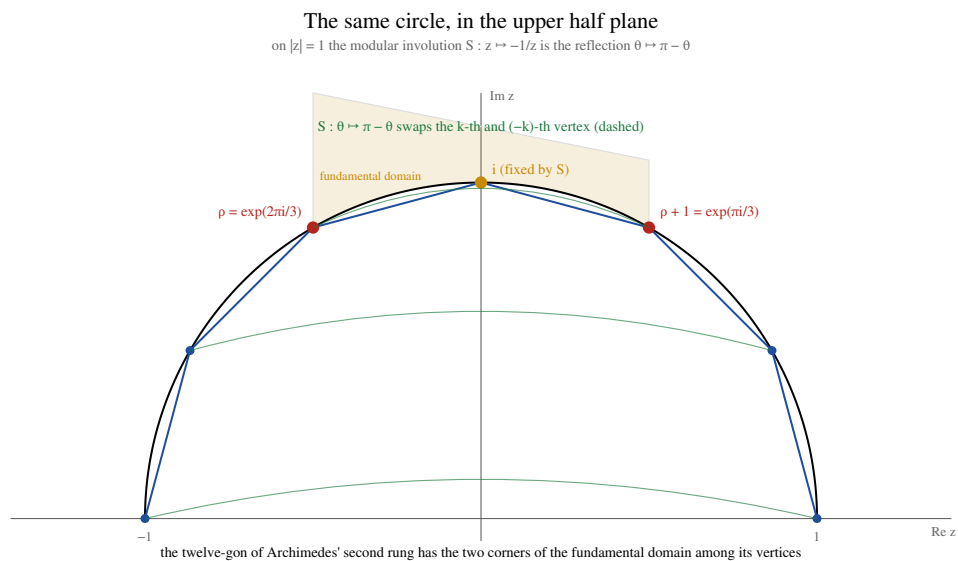
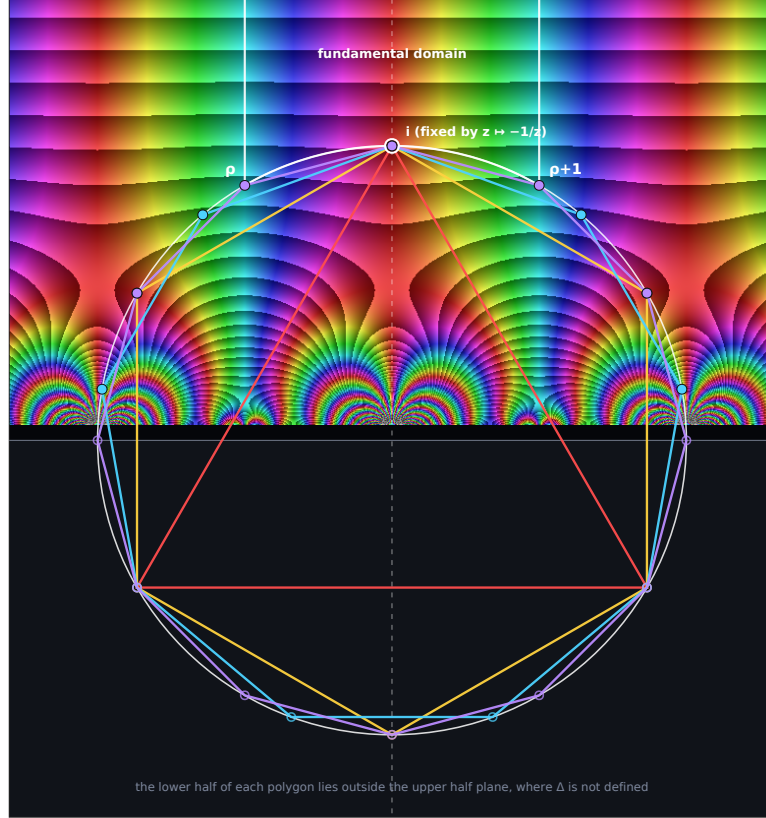


Figure 25: The same circle in the upper half plane. The regular twelve-gon anchored at i , drawn on the arc $|z| = 1$ of Section 14; the dashed arcs are the action of $S : \theta \mapsto \pi - \theta$, which swaps the k -th and $(-k)$ -th vertex and fixes i (Theorem 18.11). Two of the twelve vertices are the corners ρ and $\rho + 1$ of the modular fundamental domain (shaded).

Archimedes' polygons on the portrait of Δ

hue = $\arg \Delta(z)$, brightness banded by $|\Delta(z)|$; the 3-, 6-, 9- and 12-gons inscribed in the circle $|z| = 1$
the circle is the mirror of the portrait: $\Delta(1/\bar{z}) = \text{conj}(z^{12}\Delta(z))$ (Lean: `DeltaModular.Delta_inv_conj`)



ring	edge $2 \sin(\pi/n)$	perimeter $2\text{-lower } n$	exact value
— $n = 3$	1.73205	5.19615	$3\sqrt{3} / 2$
— $n = 6$	1.00000	6.00000	3 (edge = radius)
— $n = 9$	0.68404	6.15636	$3.077\dots < b_9 < 3.079\dots$
— $n = 12$	0.51764	6.21166	$3(\sqrt{6} - \sqrt{2})$

lower $n = n \sin(\pi/n) \rightarrow \pi$ (Lean: `Archimedes369.tendsto_lower`); lower $6 = 3$ exactly (Archimedes369.lower_six).

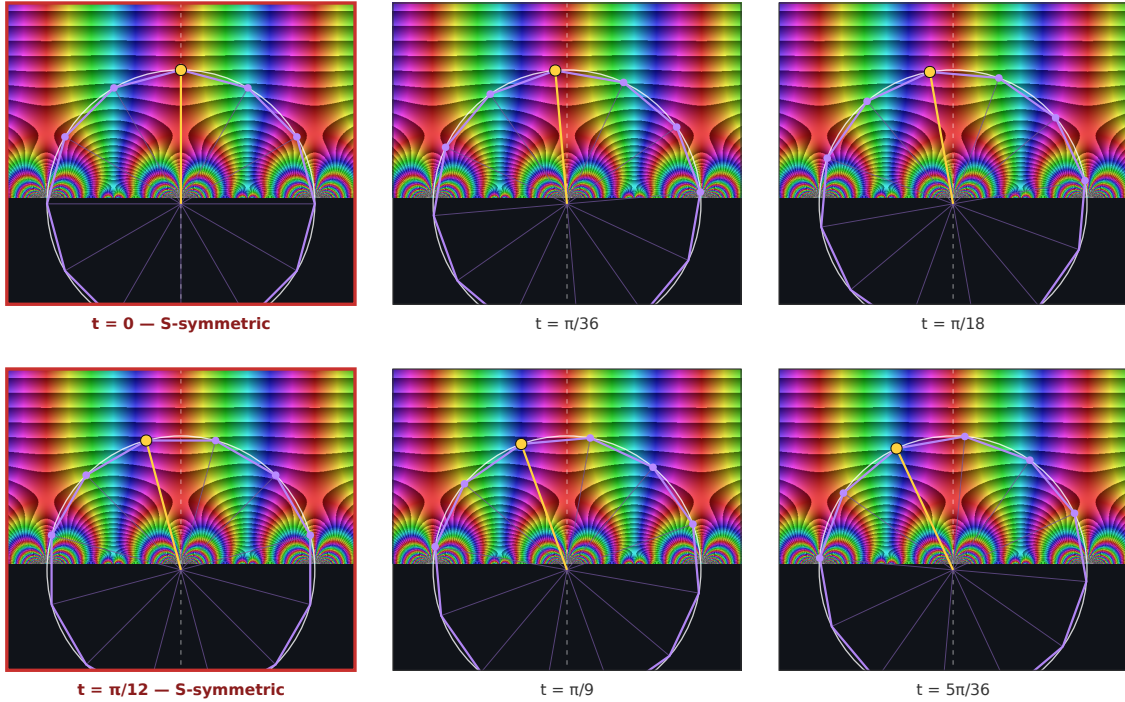
Every vertex has modulus 1 (ArchimedesDelta.norm_spinVertex); the rings nest because $3 \mid 6 \mid 12$ and $3 \mid 9$ (spinVertexSet_subset_of_dvd).

On each vertex in the upper half plane the phase is locked: $z^{12}\Delta(z)^2 = |\Delta(z)|^2$ (ArchimedesDelta.Delta_spinVertex_phase).

Figure 26: Archimedes' polygons on the portrait of Δ : the regular 3-, 6-, 9- and 12-gons inscribed in the circle $|z| = 1$, at the frame $t = 0$, over the phase portrait of the discriminant form. Every vertex lies on the circle (Proposition 19.1(1)), which is exactly the mirror of the portrait, $|\Delta(1/\bar{z})| = |z|^{12}|\Delta(z)|$ (Section 14); the rings are nested because $3 \mid 6 \mid 12$ and $3 \mid 9$ (Proposition 19.1(2)); two vertices of the twelve-gon are the corners ρ and $\rho + 1$ of the fundamental domain (Theorem 18.11). The table gives the edge length $2 \sin \frac{\pi}{n}$ and the perimeter $2b_n$ of each ring.

The twelve-gon turning on the portrait of Δ

six equally spaced frames of one loop: the ring at t and at $t + \pi/6$ is the same (Lean: ArchimedesDelta.spinVertexSet_eq_iff)
boxed frames are the instants $t = \pi k/12$ at which the ring is stable under $z \mapsto -1/z$ — exactly two per loop (Lean: ArchimedesDelta.S_stable_iff)



The ring is rigid as it turns: every edge keeps the length $2 \sin(\pi/12) = 0.51764\dots$ (Lean: ArchimedesDelta.chord_eq),
so the perimeter is $2 \cdot \text{lower } 12 = 6(\sqrt{6} - \sqrt{2}) = 6.21165\dots < 2\pi$ (Archimedes369.lower_twelve, twelve_gon_bounds).
At every vertex $e^{12it}\Delta(z)^2 = |\Delta(z)|^2$: the whole ring turns its phase in unison (ArchimedesDelta.Delta_spinVertex_twelve_phase).
The yellow spoke is the anchor vertex $k = 0$, which starts at i , the fixed point of $z \mapsto -1/z$ (ArchimedesDelta.spinVertex_zero).

Figure 27: Six equally spaced frames of one loop of the spinning twelve-gon, $t = 0, \frac{\pi}{36}, \frac{\pi}{18}, \frac{\pi}{12}, \frac{\pi}{9}, \frac{5\pi}{36}$; the seventh frame would be the first again (Theorem 19.2). The yellow spoke is the anchor vertex $k = 0$, which starts at i . The two boxed frames, $t = 0$ and $t = \pi/12$, are the instants at which the ring is stable under $S : z \mapsto -1/z$ (Theorem 19.3) — exactly two per loop. The figure is rigid as it turns (Proposition 19.1(3)).

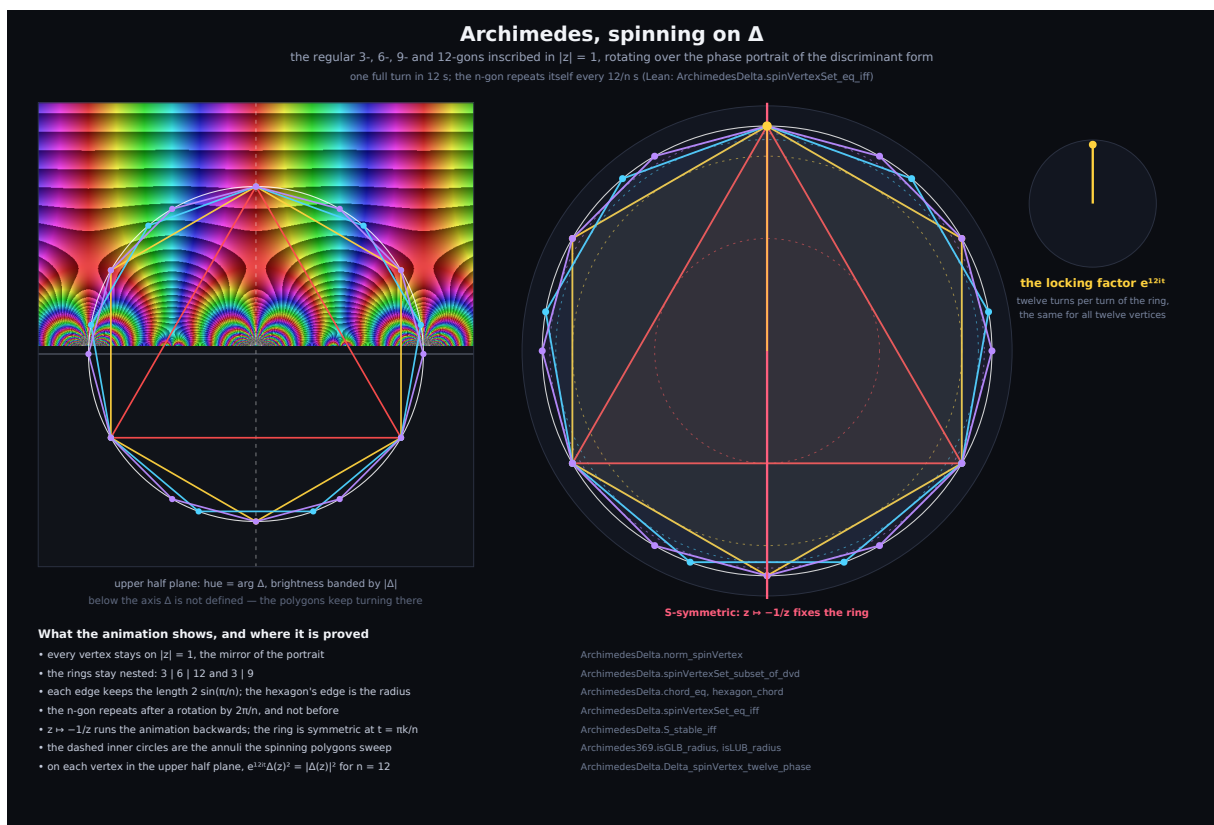
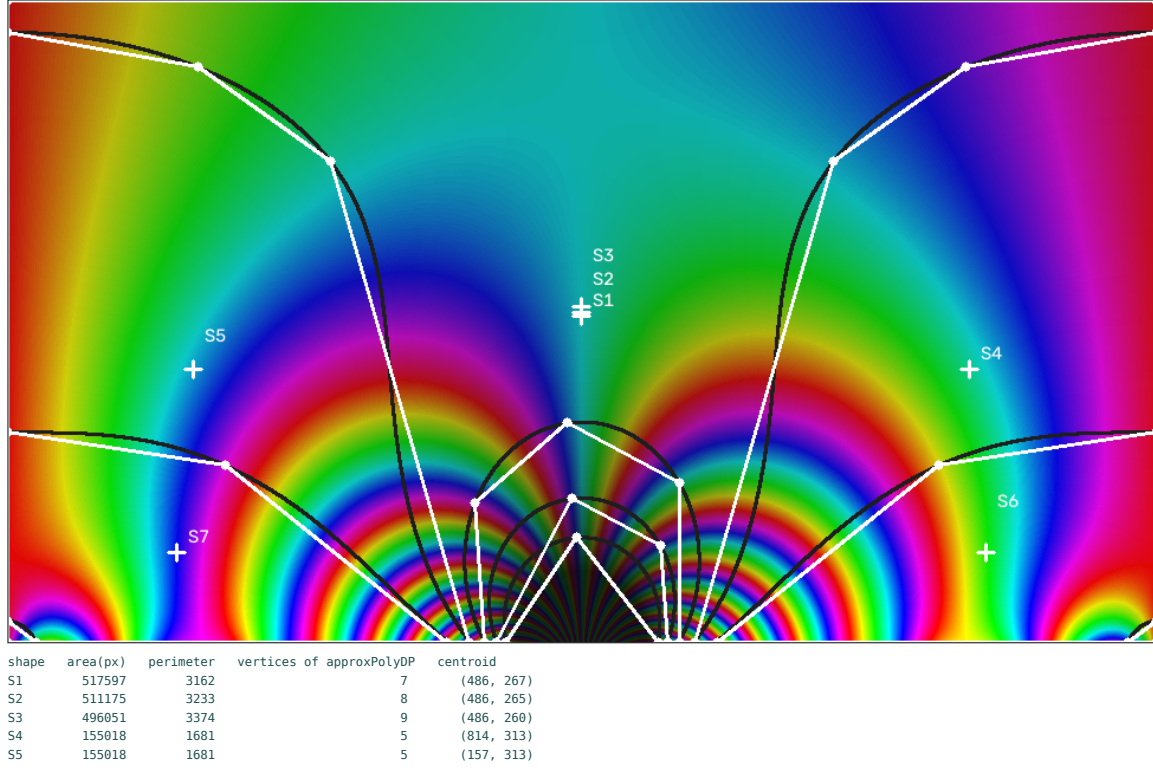


Figure 28: One frame of the animation `archimedes_delta_spin.svg`. Left: the rings turning over the portrait, on the window of Figure 26. Right: the same rings on the bare disk, with the dashed incircles $|z| = \cos \frac{\pi}{n}$ bounding the annulus each spinning polygon sweeps (Theorem 18.8), the mirror line that lights up at the S -symmetric instants (Theorem 19.3), and the dial of the locking factor e^{12it} (Theorem 19.4).

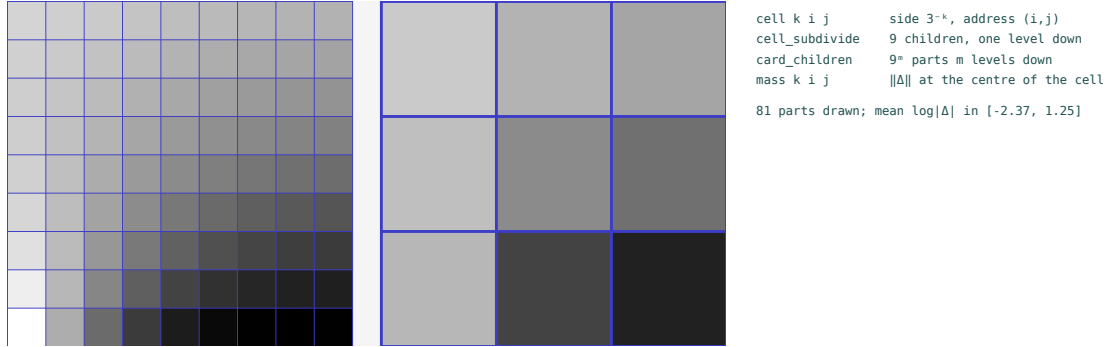
Euclid, OpenCV and Frobenius on the portrait of Δ

the shapes of the picture, cut into nine, read, normalised into a character, and hit with Frobenius

A the dissection: OpenCV contours and their Euclidean polygons



B Euclid: the shape cut into 9, and into $9^2 = 81$ organised parts; each part shaded by the mean of $\log|\Delta|$ inside it



C the character: readings normalised to [0,1] (left) and quantised to the three grey levels of $\mathbb{Z}/3$ (right)

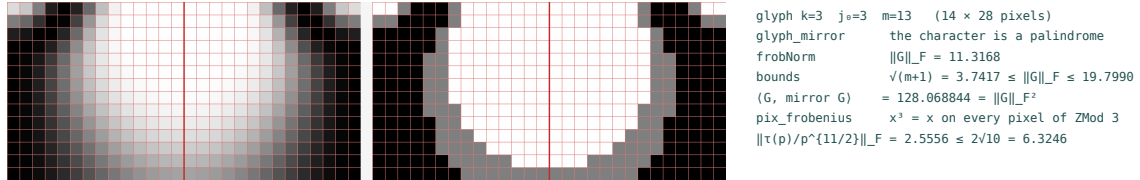
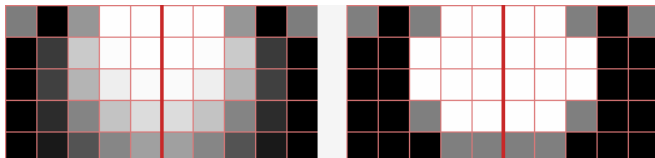


Figure 29: The dissection plate. **A**: the portrait on $-\frac{1}{2} \leq x \leq \frac{1}{2}$, $0.06 \leq y \leq 0.62$, with the OpenCV contours of five magnitude thresholds and the polygons of `approxPolyDP` drawn on them, centroids marked; the table lists area, perimeter and vertex count. **B**: one shape, cut into 9 and into $9^2 = 81$ organised parts (Theorem 20.3), each part shaded by the mean of $\log|\Delta|$ inside it. **C**: the character G (left) and its quantisation π to the three grey levels of $\mathbb{Z}/3$ (right), with the wall $x = 0$ marked; the palindromy of Theorem 20.6 is visible as the exact left-right symmetry, and the panel lists $\|G\|_F$ against the bounds of Theorem 20.7.

The character sheet of Δ

one glyph per dissection level: $\|\Delta\|$ read on the cells, each row divided by its own maximum, then hit with Frobenius

level $k = 2$ base row $j_0 = 1$ half-width $m = 4$ — 5×10 pixels of side 3^{-2}

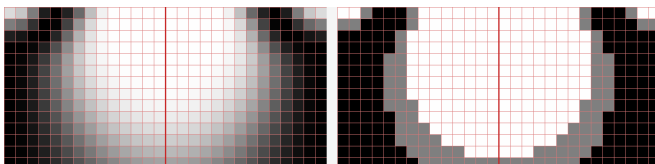


```

||G||_F          = 4.31744
||G||_F upper bound = 7.07107
(G, mirror G)    = 18.64027
palindromic      = yes (max |G - mirror G| < 1e-12)
x^3 = x on ZMod 3 = True
black pixels     = 20 of 50

```

level $k = 3$ base row $j_0 = 3$ half-width $m = 13$ — 14×28 pixels of side 3^{-3}



```

||G||_F          = 11.31675
||G||_F upper bound = 19.79899
(G, mirror G)    = 128.06084
palindromic      = yes (max |G - mirror G| < 1e-12)
x^3 = x on ZMod 3 = True
black pixels     = 130 of 392

```

Frobenius traces $\tau(p)$ on the ten primes below thirty:

p	$\tau(p)$	$\tau(p) \bmod 27$	$p^2 + p^9 \bmod 27$	$\tau(p) \bmod 3$	$p(p+1) \bmod 3$	$\tau(p)/p^{(11/2)}$
2	-24	3	3	0	0	-0.5303
3	252	9	9	0	0	+0.5987
5	4830	24	24	0	0	+0.6912
7	-16744	23	23	2	2	-0.3765
11	534612	12	12	0	0	+1.0009
13	-577738	8	8	2	2	-0.4316
17	-6905934	18	18	0	0	-1.1797
19	10661420	11	11	2	2	+0.9878
23	18643272	15	15	0	0	+0.6040
29	128406630	3	3	0	0	+1.1625

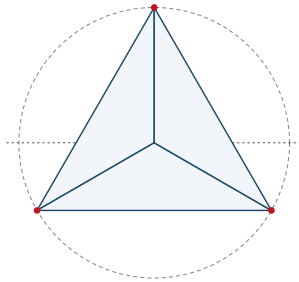
Frobenius norm of the row of normalised traces = $2.55555 \leq 2\sqrt{10} = 6.32456$ (Ramanujan-Deligne)

Figure 30: The character sheet: the glyph of Δ at two dissection levels, $k = 2$ with 5×10 pixels of side 3^{-2} and $k = 3$ with 14×28 pixels of side 3^{-3} , each shown as grey levels and as its three-shade $\mathbb{Z}/3$ quantisation, with the Frobenius data beside it. Below, the traces of Frobenius on the ten primes under thirty: $\tau(p)$ against $p^2 + p^9$ modulo 27 and $p(p+1)$ modulo 3 (Theorem 20.9), and the Frobenius norm of the row of normalised traces against $2\sqrt{10}$.

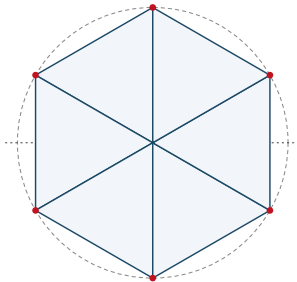
Every frame, cut into triangles, and read

the polygon $\rightarrow$ its fan of triangles $\rightarrow$ Euclid's ternary cut $\rightarrow \|\Delta\|$ on each part $\rightarrow$ the parts sorted by colour

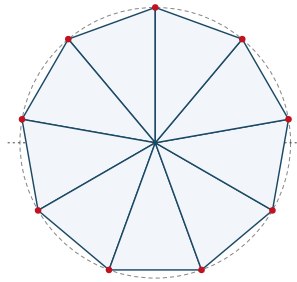
A the frames: the 3-, 6- and 9-gon in $|z| = 1$, each cut into triangles by the fan from the centre



n = 3: 3 triangles, each of size 2
arc approximated to $\pi/3 = 1.047$
perimeter $2n \sin(\pi/n) = 5.1962$



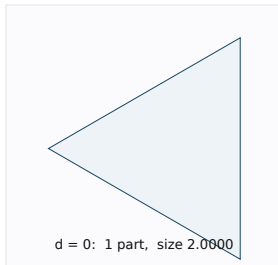
n = 6: 6 triangles, each of size 2
arc approximated to $\pi/6 = 0.524$
perimeter $2n \sin(\pi/n) = 6.0000$



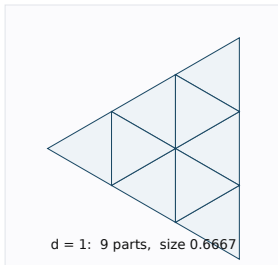
n = 9: 9 triangles, each of size 2
arc approximated to $\pi/9 = 0.349$
perimeter $2n \sin(\pi/n) = 6.1564$

fanTri n t k dist_circle_spinVertexSet_le : every point of the arc is within π/n of a vertex; the perimeters rise to 2π

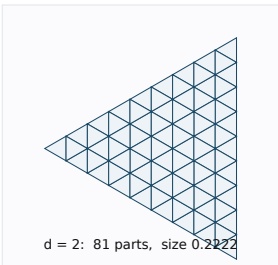
B Euclid on a triangle: trisecting the sides cuts it into nine — six one way up, three inverted; d cuts give 9^d parts



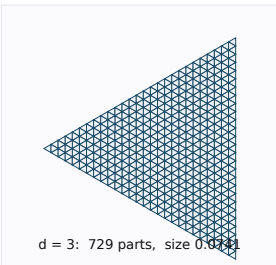
d = 0: 1 part, size 2.0000



d = 1: 9 parts, size 0.6667



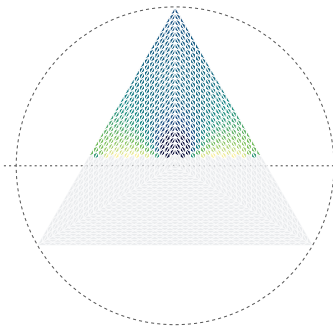
d = 2: 81 parts, size 0.2222



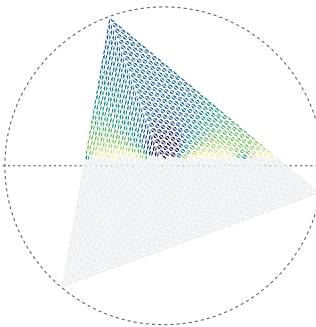
d = 3: 729 parts, size 0.0741

child i T (i < 9) shape_eq_iUnion_child : shape T = $\bigcup_i$ shape (child i T)
desc w T (|w| = d) shape_eq_iUnion_desc, size_desc : size (desc w T) = size T / 3^d , 9^d parts

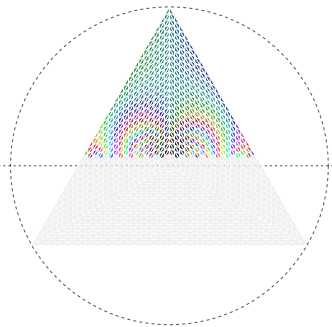
C the Δ diagram as those triangles: each part painted from Δ at its centroid, grey where it leaves the half plane



$\|\Delta\|$, frame t = 0
2187 parts, 866 of them read



$\|\Delta\|$, frame t = 0.35 (turned)
2187 parts, 910 of them read



arg Δ , frame t = 0
2187 parts, 866 of them read

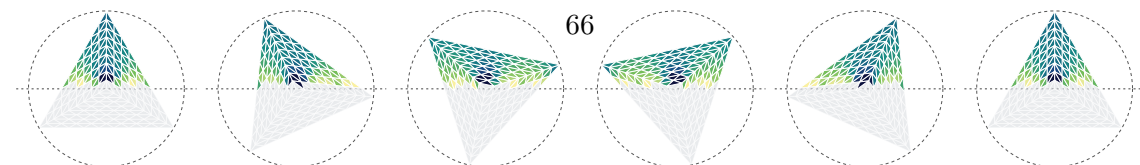
reading T = $\|\Delta(\text{centroid } T)\|$ centroid_mem_shape : the centroid lies in the part
desc_fanTri_rot : the parts of frame t are those of frame 0, turned — so turning the triangle turns every part

D sorted by colour: the palette of the two magnitude frames above (DeltaTriangles.sortedReadings)



readings_rot / sortedReadings_rot : after a turn of $2\pi/n$ the frame shows exactly the same sorted list of colours
child_conjTri / reading_conjTri : the mirror $z \mapsto -z^*$ commutes with the cut and leaves each colour unchanged

E turning the triangle: one loop of the n = 3 frame, t = 0 ... $2\pi/3$, with the palette of each instant



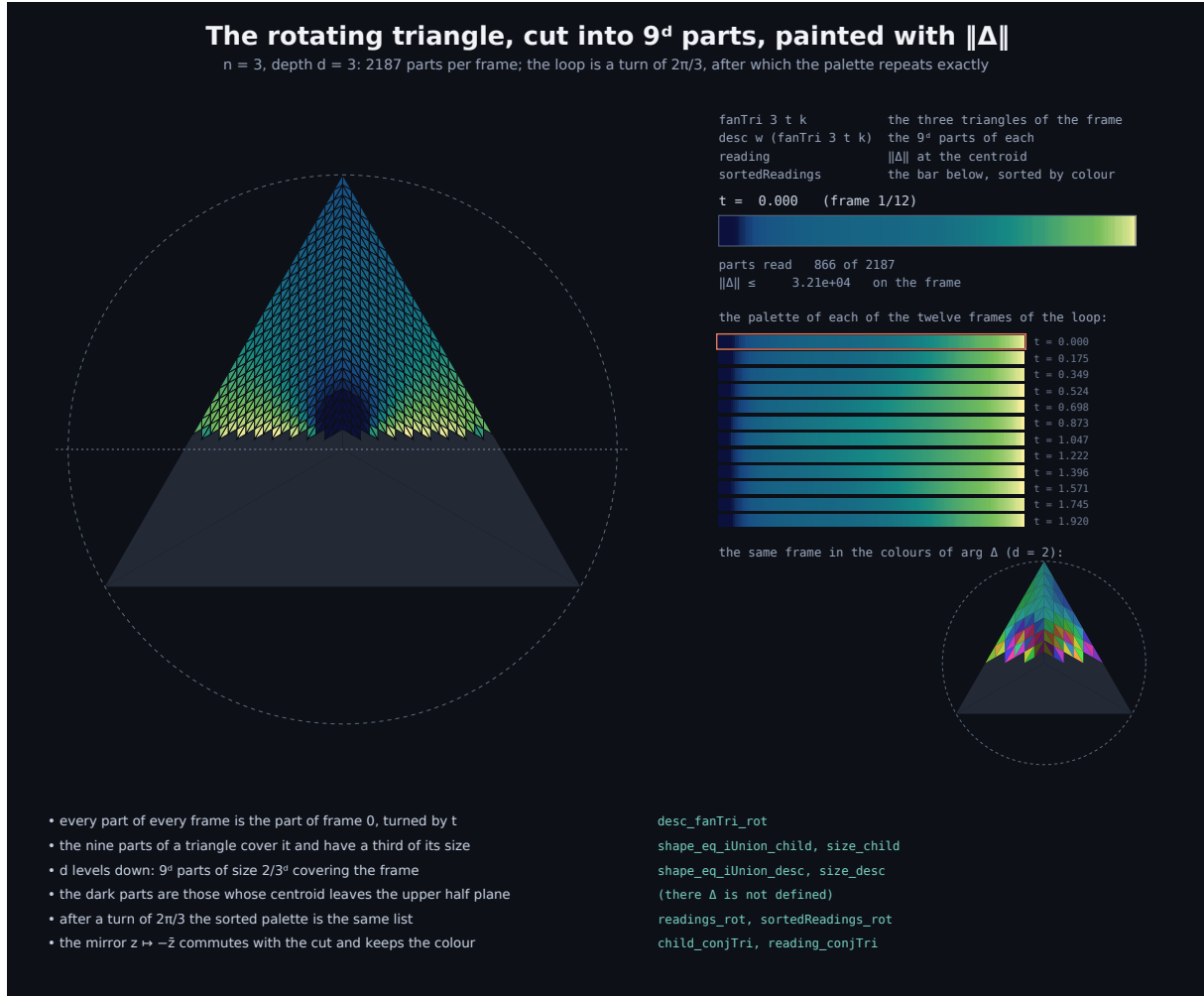


Figure 32: The animation `delta_triangles_spin.svg`, shown at its first frame: the rotating triangle, cut into $3 \cdot 9^3$ parts, each repainted from Δ at every frame. Opening the file in a browser turns the triangle through one loop $t \in [0, 2\pi/3)$ in twelve steps; the bar under the legend is the palette of the current frame, the stack beside it the palettes of all twelve, and after the loop the palette returns to the first exactly (Theorem 21.7). The small disc shows the same frame in the colours of $\arg \Delta$. The PDF copy is one frame.

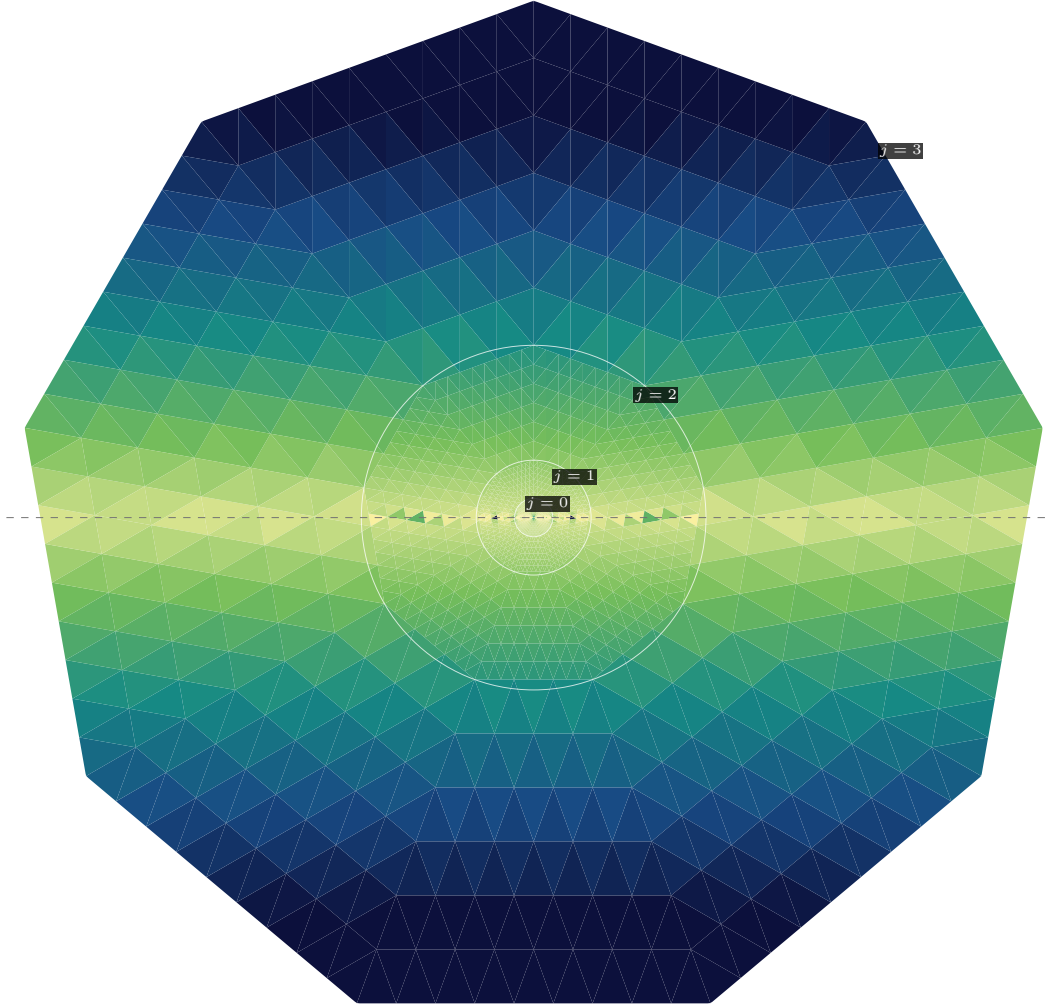


Figure 33: The filling of the plane by Euclid's triangles, painted by Δ . Four rings $j = 0, 1, 2, 3$ of the ladder of Definition 22.1 for $n = 9$ at $t = 0$, each of nine sectors cut twice into nine, so $4 \cdot 9 \cdot 81 = 2916$ triangles; the picture is drawn to the scale of the outer ring, so rings 0 and 1 are the small figure at the centre. Each triangle carries the value $\log |\Delta|$ at its centroid, on a clipped scale chosen for legibility, the lower half plane being read through the mirror $z \mapsto \bar{z}$ of Definition 22.4. The circles are the circumscribed circles of radii 3^j . Successive rings are the same picture scaled by three, which is the self-similarity of the filling.

Two features of the filling deserve a word. It is not a tiling in the strict sense: neighbouring rings overlap, ring j containing the disc of radius $3^j/2$ and therefore all of ring $j - 1$. And it is self-similar by construction — multiplication by 3 carries ring j onto ring $j + 1$ sector for sector — so the picture of the whole plane is the picture of one annulus, repeated at every power of three.

Reading Δ on a tile, anywhere

The reading of Section 21 is defined only above the real axis, where Δ lives; the filling covers the plane. The gap is closed by the mirror.

Definition 22.4 (The reflected reading). Let $\rho(z) = z$ if $\text{Im } z > 0$ and $\rho(z) = \bar{z}$ otherwise (`reflUp`), and put

$$\text{rd}(T) = \begin{cases} 0, & \text{Im}(\text{cen } T) = 0, \\ |\Delta(\rho(\text{cen } T))|, & \text{otherwise} \end{cases} \quad (\text{readingRef1}),$$

where $\text{cen } T$ is the centroid. Off the axis ρ really lands in the upper half plane (`im_reflUp_pos`), above the axis rd is the old reading (`readingRef1_eq_reading`), and ρ is blind to conjugation (`reflUp_conj`).

Definition 22.5 (Palette trace and trace matrix). The *palette trace* of sector k of ring j at depth d is the total reading of its parts,

$$f_t^{(j)}(k) = \sum_{|w|=d} \text{rd}(\Theta_{j,k,w}(t)) \quad (\text{paletteTrace}),$$

a sum of 9^d nonnegative numbers (`paletteTrace_nonneg`). The *trace matrix* of the ladder is $(f_t^{(j)}(k))_{j < N, k < n}$ (`traceMatrix`).

Proposition 22.6 (What the trace matrix remembers). *Let $n \geq 1$. Then, for all j, d, t and k :*

1. $f_t^{(j)}(k + n) = f_t^{(j)}(k)$: the sector index lives in $\mathbb{Z}/n$ (`ringTri_periodic`, `paletteTrace_periodic`);
2. $f_{t+2\pi/n}^{(j)}(k) = f_t^{(j)}(k + 1)$: one step of the animation is one step of the cyclic group of sectors, on every ring at once (`paletteTrace_rot`);
3. at $t = 0$ the palette is a palindrome, $f_0^{(j)}(-1 - k) = f_0^{(j)}(k)$ (`paletteTrace_mirror`).

Proof. (1) and (2) are the corresponding statements for the fan (`spinVertex_add_nat`, `fanTri_shift`) transported through the scaling of Definition 22.1, which does not touch the sector index. For (3): the mirror $z \mapsto -\bar{z}$ sends the still frame's vertex k to vertex $-k$, hence exchanges the sector k with the sector $-1 - k$ and, on the way, reverses the orientation of the triangle (`conjTri_fanTri`, `ringTri_mirror`); the reversal $T \mapsto (a, c, b)$ (`swapTri`) permutes the nine children of a triangle by the involution

$$\sigma = (0)(13)(25)(4)(6)(78) \quad (\text{sigma9}, \text{sigma9_involutive}, \text{child_swapTri}),$$

so it permutes the words of depth d letter by letter (`wordPerm`), a bijection of the index set of the sum. Finally $|\Delta(-\bar{z})| = |\Delta(z)|$ (Proposition 13.1), and reflecting the centroid does not change its reading (`readingRef1_conjTri`), so the two sums have the same terms. $\square$

The character table of the sectors

The group acting is C_n , the cyclic group of sectors, and it acts on the whole ladder at once by Proposition 22.6(2). Its irreducible characters are one-dimensional, and Frobenius' table of them is the following $n \times n$ matrix.

Definition 22.7 (The character table of C_n). With $\zeta_n = e^{2\pi i/n}$ (**zeta**) put $\chi_a(k) = \zeta_n^{ak}$ (**chi**) for $a, k \in \mathbb{Z}$, and let the table be

$$X_n = (\chi_a(k))_{0 \leq a, k < n} \quad (\text{charTable}).$$

Every entry has modulus 1 (**norm_charTable**), the row $a = 0$ is the trivial character (**charTable_zero_left**), and the table is symmetric, $X_n = X_n^T$ (**charTable_symm**) — for an abelian group the characters and the elements are indexed by the same $\mathbb{Z}/n$.

Theorem 22.8 (Frobenius' two orthogonality relations). For $n \geq 1$ and $a, b, k, l < n$,

$$\sum_{k < n} \chi_a(k) \overline{\chi_b(k)} = \begin{cases} n, & a = b \\ 0, & a \neq b \end{cases} \quad (\text{charTable_row_orthogonality}),$$

$$\sum_{a < n} \chi_a(k) \overline{\chi_a(l)} = \begin{cases} n, & k = l \\ 0, & k \neq l \end{cases} \quad (\text{charTable_col_orthogonality}).$$

Equivalently $X_n \overline{X_n}^T = nI$: the table is $\sqrt{n}$ times a unitary matrix.

Proof. Both reduce to the geometric sum $\sum_{k < n} \zeta_n^{ck} = n$ if $n \mid c$ and 0 otherwise (**sum_chi**), applied to $c = a - b$; for indices in $[0, n)$ the divisibility $n \mid a - b$ is equality (**dvd_iff_eq**). The column relation is the row relation after the symmetry $\chi_a(k) = \chi_k(a)$ (**chi_symm**). $\square$

Proposition 22.9 (3, 6, 9 inside the table). For $m \geq 1$ and all a, k , $\chi_{ma}^{(mn)}(k) = \chi_a^{(n)}(k)$ (**chi_scale**); in particular

$$\chi_{3a}^{(9)}(k) = \chi_a^{(3)}(k) \quad (\text{chi_nine_three}).$$

So rows 3 and 6 of the character table of C_9 are the two nontrivial characters of C_3 , and row $9 \equiv 0$ is the trivial one: the character table of the nine-gon's sectors contains the character table of the three-gon's, and the sub-table is indexed by exactly the trinity $\{3, 6, 9\}$ of Section 3 read in $\mathbb{Z}/9$.

The palette, read in the table

Definition 22.10 (Character coefficients of a ring). For a ring j set

$$c_j(a) = \sum_{k < n} f_t^{(j)}(k) \overline{\chi_a(k)} \quad (\text{fourierCoeff, paletteCoeff}),$$

and let the *character table of the filling* be the $N \times n$ complex matrix $(c_j(a))_{j < N, a < n}$ (**paletteChar**): one row per ring, one column per irreducible character of the sector group.

Theorem 22.11 (The table is a complete, isometric record of the palette). Let $n \geq 1$. Then

1. (Inversion.) $\sum_{a < n} c_j(a) \chi_a(k) = n f_t^{(j)}(k)$ for every $k < n$ (*fourier_inversion*, *paletteTrace_eq_sum*): nothing is lost in passing to the table;

2. (Plancherel.) $\sum_{a < n} |c_j(a)|^2 = n \sum_{k < n} (f_t^{(j)}(k))^2$ (*sum_normSq_fourierCoeff*, *sum_normSq_paletteCoeff*), hence in Frobenius norm

$$\| (c_j(a)) \|_F = \sqrt{n} \| (f_t^{(j)}(k)) \|_F \quad (\text{frobNormC_paletteChar});$$

3. (The trivial character is the total mass.) $c_j(0) = \sum_{k < n} f_t^{(j)}(k)$, the sum of all $n \cdot 9^d$ readings of the ring (*paletteCoeff_zero*).

Proof. (1) Expand and exchange the two sums; the inner sum is $\sum_{a < n} \chi_{k-l}(a)$, which is n for $l = k$ and 0 otherwise by Theorem 22.8. (2) is the same computation applied to $c_j(a) \overline{c_j(a)}$, and the norm statement follows by summing over the rings and taking square roots, the Frobenius norm being the one of Section 20 (*DeltaFrobenius.frobNorm*). (3) is the case $a = 0$ of the definition. $\square$

Theorem 22.12 (The character table diagonalises the spin). *Let $n \geq 1$. Turning the whole filling by one step of the animation multiplies the a -th column of its character table by the single scalar ζ_n^a :*

$$c_j(a) \Big|_{t+2\pi/n} = \zeta_n^a c_j(a) \Big|_t \quad \text{for every ring } j \quad (\text{paletteCoeff_rot}),$$

so the moduli $|c_j(a)|$ are constants of the animation (*norm_paletteCoeff_rot*). In the sector basis the spin is the cyclic shift of Proposition 22.6(2), a full $n \times n$ permutation matrix; in the character basis it is the diagonal matrix $\text{diag}(1, \zeta_n, \dots, \zeta_n^{n-1})$.

Proof. By Proposition 22.6(2) the row at time $t + 2\pi/n$ is the row at time t shifted by one, and by (1) that row is n -periodic; shifting the summation index in the definition of $c_j(a)$ pulls out exactly one factor $\chi_a(1) = \zeta_n^a$ (*sum_range_shift_int*, *fourierCoeff_shift*). The modulus statement is $|\zeta_n^a| = 1$ (*norm_chi*). $\square$

Theorem 22.13 (The palindrome, in the table). *At $t = 0$ every coefficient satisfies*

$$c_j(a) = \zeta_n^a \overline{c_j(a)} \quad (\text{fourierCoeff_mirror}, \text{paletteCoeff_mirror}),$$

that is, $c_j(a)$ lies on the real line through the origin of direction $\zeta_n^{a/2}$; in particular $c_j(0)$ is real and positive, and $|c_j(a)|$ is the absolute value of a single real number, not of a genuinely complex one.

Proof. Reflect the summation index by $k \mapsto -1 - k$, which is a bijection of $\mathbb{Z}/n$ and preserves the n -periodicity (*sum_range_reflect_int*); the palindrome of Proposition 22.6(3) leaves the values alone, while $\overline{\chi_a(-1-k)} = \chi_a(1)\chi_a(k)$ turns the sum into ζ_n^a times the conjugate of itself. $\square$

Remark 22.14 (What the numbers say). Evaluated at $n = 9$, $t = 0$, depth $d = 2$ and $N = 4$ rings — $4 \cdot 9 \cdot 81 = 2916$ triangles, the plate of Figure 33 — the normalised coefficients $|c_j(a)|/c_j(0)$ are the second table on page 72. They are palindromic in a , as Theorem 22.13 forces, and they are the *same* for all four rings to four decimals: the self-similarity noted after Theorem 22.3 is visible in the character table, the ring index contributing only an overall scale. The numbers themselves, 1, 0.1736, 0.9397, 0.5000, 0.7660, 0.7660, 0.5000, 0.9397, 0.1736, are $|\cos(4\pi a/9)|$ to all four decimals, which is exactly what a palette concentrated on the two sectors $k = 2$ and $k = 6$ gives: for $f = \delta_2 + \delta_6$ one has $c(a) = \zeta_9^{-2a} + \zeta_9^{-6a} = 2\zeta_9^{-4a} \cos(4\pi a/9)$. No theorem here claims these numbers; they are reported as a computation with a truncated Δ , not as a proof.

$\chi_a(k)$	$k = 0$	$k = 1$	$k = 2$
$a = 0$	1	1	1
$a = 1$	1	ζ_3^1	ζ_3^2
$a = 2$	1	ζ_3^2	ζ_3^1

$ c_j(a) /c_j(0)$	$a = 0$	$a = 1$	$a = 2$	$a = 3$	$a = 4$	$a = 5$	$a = 6$	$a = 7$	$a = 8$
$j = 0$	1.0000	0.1736	0.9397	0.5000	0.7660	0.7660	0.5000	0.9397	0.1736
$j = 1$	1.0000	0.1736	0.9397	0.5000	0.7660	0.7660	0.5000	0.9397	0.1736
$j = 2$	1.0000	0.1736	0.9397	0.5000	0.7660	0.7660	0.5000	0.9397	0.1736
$j = 3$	1.0000	0.1736	0.9397	0.5000	0.7660	0.7660	0.5000	0.9397	0.1736

Table 1: Above: the character table of C_3 of Definition 22.7, the sub-table of C_9 picked out by the trinity $\{3, 6, 9\}$ in Proposition 22.9. Below: the normalised character coefficients $|c_j(a)|/c_j(0)$ of the palette of the plate, for the four rings $j = 0, 1, 2, 3$ at $n = 9$, $t = 0$, $d = 2$ (Remark 22.14).

23 Zeta and moonshine: the trinity in ζ , in j , and in the McKay–Thompson series $1A$, $2A$, $3A$

Everything so far has read the trinity off one modular object, the discriminant $\Delta = q \prod_{m \geq 1} (1 - q^m)^{24}$, whose coefficients $\tau(n)$ carry a digital root 9 at every multiple of three (Proposition 12.1). Two obvious neighbours invite the same treatment, and they answer in opposite ways. The Riemann zeta function has *no* coefficients to read — its Dirichlet series is $\sum n^{-s}$, every coefficient a bare 1 — and the trinity has to be looked for in the indices, where it turns out to be exact. Klein’s j -invariant, on the other hand, is nothing but coefficients, and there the trinity is loud; comparing it with the McKay–Thompson series of the Monster at the classes $2A$ and $3A$ shows precisely which prime is responsible.

Zeta: the trinity lives in the indices

Definition 23.1 (Trinity indices). $\text{TI} = \{n \in \mathbb{N} : n > 0, \text{dr}(n) \in \{3, 6, 9\}\}$ and $\text{TI}^* = \{n \in \mathbb{N} : n > 0, \text{dr}(n) \notin \{3, 6, 9\}\}$. By Theorem 3.1 these are the positive multiples of three and the positive non-multiples.

Theorem 23.2 (The trinity carries a scaled copy of ζ). *For $\text{Re } s > 1$,*

$$\sum_{n \in \text{TI}} \frac{1}{n^s} = 3^{-s} \zeta(s), \quad \sum_{n \in \text{TI}^*} \frac{1}{n^s} = (1 - 3^{-s}) \zeta(s).$$

Proof. The map $n \mapsto 3(n+1)$ is a bijection $\mathbb{N} \rightarrow \text{TI}$, and $(3(n+1))^{-s} = 3^{-s}(n+1)^{-s}$ because both factors are positive reals; summing gives $3^{-s} \sum_{m \geq 1} m^{-s} = 3^{-s} \zeta(s)$. For the second sum, the Dirichlet series is absolutely convergent, so it splits over TI and its complement in $\mathbb{N}$; the index $n = 0$ contributes $0^{-s} = 0$ and may be moved from one side to the other freely, so the complementary sum is $\zeta(s) - 3^{-s} \zeta(s)$. $\square$

The proof used nothing whatever about the number three: the same bijection gives $\sum_{n > 0, m|n} n^{-s} = m^{-s} \zeta(s)$ for every $m \geq 1$, and the trinity is the case $m = 3$ — in keeping with Section 9, where the trinity of an arbitrary base turned out to be equally unremarkable.

Both halves are worth a second look. The first says the trinity is not merely a third of the integers: on the level of the zeta function it is a perfect, self-similar copy of the whole, shrunk by 3^{-s} — at $s = 1$ the shrinking factor is exactly $\frac{1}{3}$, the density. The second says the non-trinity part is $\zeta(s)$ with Euler's factor at the prime three removed, $(1 - 3^{-s}) \zeta(s) = \prod_{p \neq 3} (1 - p^{-s})^{-1}$. So the only trace the trinity leaves in ζ is the single Euler factor at 3; there is nothing else to find, because there are no coefficients in which to find it. Concretely, at $s = 2$,

$$\sum_{n \in \text{TI}} \frac{1}{n^2} = \frac{1}{9} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{54}.$$

Zeta values: where the constants of the forms come from

There is, however, a second place where three, six and nine appear in ζ , and it is the place that ties this section to all the others. Euler's even values

$$\zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90}, \quad \zeta(6) = \frac{\pi^6}{945}$$

have denominators 6, 90, 945 with digital roots 6, 9, 9; and at the negative odd integers,

$$\zeta(-1) = -\frac{1}{12}, \quad \zeta(-3) = \frac{1}{120}, \quad \zeta(-5) = -\frac{1}{252},$$

so that

$$-\frac{2}{\zeta(-1)} = 24, \quad \frac{2}{\zeta(-3)} = 240, \quad -\frac{2}{\zeta(-5)} = 504.$$

Theorem 23.3 (The modular constants are zeta values). *The three numbers just displayed are exactly the constants this project has been using: 24 is the exponent in $\Delta = q \prod (1 - q^m)^{24}$, 240 is the multiplier of $E_4 = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n$, which builds $j = E_4^3 / \Delta$, and 504 is the multiplier of E_6 . Their digital roots are 6, 6, 9.*

That is not a coincidence but the standard normalisation $E_k = 1 - \frac{2k}{B_k} \sum_{n \geq 1} \sigma_{k-1}(n) q^n$ read through $\zeta(1 - k) = -B_k/k$; it is recorded here because it is the honest reason the trinity keeps turning up in the coefficients below. One further curiosity, verified rather than explained: the denominator of $\zeta(-5)$ is $252 = \tau(3)$, the third coefficient of Δ . And the zeta function has a trivial zero at -6 .

Klein's j , and the Monster's $2A$ and $3A$

Definition 23.4 (Three hauptmoduls). Let $c(n)$ be the coefficients of Klein's invariant

$$j = \frac{E_4^3}{\Delta} = q^{-1} + 744 + 196884q + 21493760q^2 + \cdots,$$

whose non-constant part is the McKay–Thompson series of class $1A$ of the Monster. Let T_{2A} and T_{3A} be the McKay–Thompson series of classes $2A$ and $3A$, given by their standard eta quotients

$$T_{2A} = \left(\frac{\eta(z)}{\eta(2z)} \right)^{24} + 2^{12} \left(\frac{\eta(2z)}{\eta(z)} \right)^{24} + 24, \quad T_{3A} = \left(\frac{\eta(z)}{\eta(3z)} \right)^{12} + 3^6 \left(\frac{\eta(3z)}{\eta(z)} \right)^{12} + 12.$$

All three are computed in the formalisation as truncated integer power series, by the algorithm that computes τ .

Proposition 23.5 (The coefficients). $j = q^{-1} + 744 + 196884q + 21493760q^2 + 864299970q^3 + 20245856256q^4 + \cdots$; $T_{2A} = q^{-1} + 4372q + 96256q^2 + 1240002q^3 + 10698752q^4 + \cdots$; $T_{3A} = q^{-1} + 783q + 8672q^2 + 65367q^3 + 371520q^4 + \cdots$; both McKay–Thompson series have vanishing constant term.

Theorem 23.6 (The trinity in the coefficients). *On the computed range:*

1. $3^5 = 243$ divides $c(3n)$ and $2^{11} = 2048$ divides $c(2n)$ — *Lehner's congruences*. In particular every third coefficient of j has digital root 9, and so does every third coefficient of T_{2A} and of T_{3A} , exactly as for τ .
2. $c(n) \equiv T_{3A}(n) \pmod{3^6}$ for $1 \leq n \leq 12$; consequently j and its $3A$ shadow have identical digital roots, term by term: 9, 5, 9, 9, 2, 9, 9, 3, 9, 9, 6, 9.
3. $c(n) \equiv T_{2A}(n) \pmod{2^{12}}$ for $1 \leq n \leq 12$ — *the two-adic companion* — but here the digital roots part company at once: $\text{dr}(4372) = 7$ while $\text{dr}(196884) = 9$.

The three statements together say something clean. Read modulo powers of two, j and the $2A$ series are the same object; read modulo powers of three, j and the $3A$ series are the same object. The digital root is a three-adic instrument — it is the residue mod 9 — so it sees the second congruence and is blind to the first. That is why the trinity of j is reproduced exactly by T_{3A} and not at all by T_{2A} : not numerology, but the observation that dr is a function of $n \bmod 9$ and $729 = 3^6$ is divisible by 9.

Remark 23.7 (What is proved, and on what range). The zeta statements of this section, Theorems 23.2 and 23.3, are theorems for all s and all the values named. The moonshine statements are kernel computations on the coefficients that the formalisation actually computes: the tables and every divisibility and congruence in Theorem 23.6 are checked for the indices displayed — $n \leq 12$ for the congruences, $n \leq 4$ for the multiples of three, $n \leq 6$ for the multiples of two — and are not proved for all n here. Lehner's congruences and the p -adic relations between j and its pA shadows are classical theorems; what is machine-checked here is their instance on the computed range, together with the digital-root consequences drawn from it. The identification of the two eta quotients of Definition 23.4 with the McKay–Thompson series of classes $2A$ and $3A$ is also imported, not proved: what is proved is that those quotients have the stated coefficients.

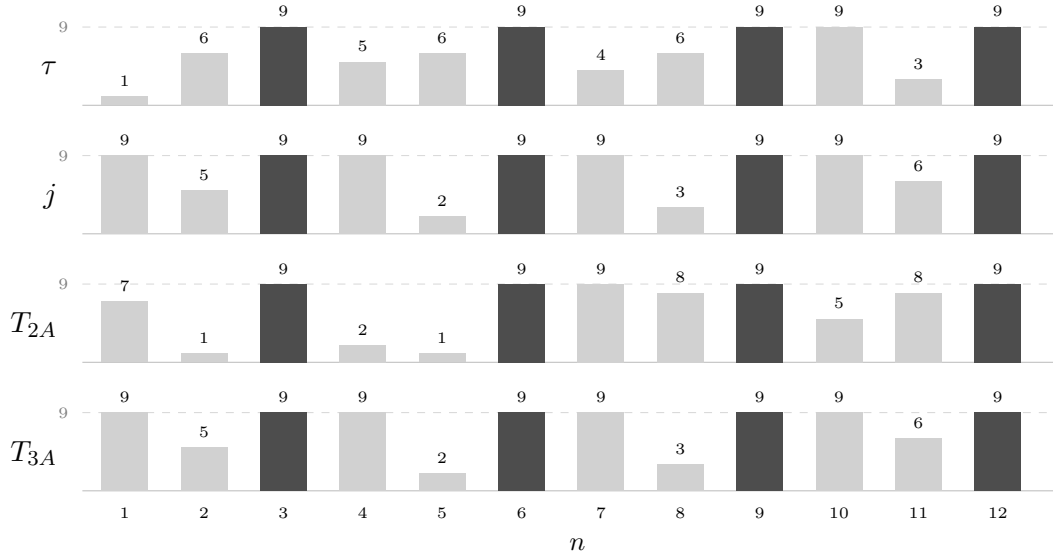


Figure 34: The digital roots of the first twelve coefficients of Δ , of Klein’s j , and of the McKay–Thompson series of classes $2A$ and $3A$; bars over the multiples of three are dark. Every dark bar reaches 9 in all four rows (Theorem 23.6(1)), and the rows j and T_{3A} are identical, which is the congruence mod 3^6 of Theorem 23.6(2). The ζ row is missing on purpose: all of its coefficients are 1, so its bars would all have height one, and its trinity is in the indices instead (Theorem 23.2).

One tiling, ten functions: the standalone plates

The ring ladder of Section 22 does not care which function is read on its tiles. Definition 22.1 builds the tiles out of nothing but a centre, a radius and Euclid’s ternary cut; Definition 22.5 adds the readings of a sector; the character table of Definition 22.7 expands the result. Only the reading itself mentions Δ . Replacing it by another function therefore costs nothing, and gives a portrait of that function in the same coordinates — the same $4 \cdot 9 \cdot 9^3 = 26244$ triangles, the same sectors, the same C_9 .

Ten functions are drawn this way, each in its own standalone file. Five live on the upper half plane and are read through the mirror of Definition 22.4: the discriminant Δ , Klein’s j , and the three McKay–Thompson hauptmoduls $T_{1A} = j - 744$, T_{2A} and T_{3A} of Definition 23.4. Five live on the whole s -plane and are read directly: the Riemann ζ , its completion $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, and the Dirichlet L -functions of the odd characters mod 3 and mod 4 and of the even character mod 5 — three more members of the same family as ζ , and the first of them is the L -function of the trinity’s own character.

The conventions are the same in every plate. The colour of a tile is the *rank* of its reading in the sorted palette of the plate (Definition 21.6), not the reading itself: j runs over sixty orders of magnitude on these rings, and a linear or logarithmic ramp would show nothing but the cusp. Ring j paints the annulus its predecessor is not proved to cover, the disc of radius $\cos(\pi/n)3^{j-1}$ of Theorem 22.3, so the rings leave no gap. Tiles that are not read — those meeting the real axis, where a form of the upper half plane has no value — are grey. Under each disc are the two panels of Section 22: the palette traces of the nine sectors, ring by ring, and the same data read in the character table of C_9 . For the five L -functions the ladder is centred at $s = \frac{1}{2}$, so that the

critical line is the axis of the picture; for the five forms it is centred at 0 with radii $\frac{1}{9}, \frac{1}{3}, 1, 3$, which is where their structure is.

Each function also gets the animation of Section 21: one triangle of the fan, cut into $3 \cdot 9^3$ parts, turning through a loop $t \in [0, 2\pi/3)$ in nine frames, with the palette of every frame stacked beside it. The files are

<code>paper/delta_palette.svg, paper/delta_spin.svg</code>	Δ
<code>paper/j_palette.svg, paper/j_spin.svg</code>	j
<code>paper/t1a_palette.svg, paper/t1a_spin.svg</code>	T_{1A}
<code>paper/t2a_palette.svg, paper/t2a_spin.svg</code>	T_{2A}
<code>paper/t3a_palette.svg, paper/t3a_spin.svg</code>	T_{3A}
<code>paper/zeta_palette.svg, paper/zeta_spin.svg</code>	ζ
<code>paper/xi_palette.svg, paper/xi_spin.svg</code>	ξ
<code>paper/lchi3_palette.svg, paper/lchi3_spin.svg</code>	$L(s, \chi_{-3})$
<code>paper/lchi4_palette.svg, paper/lchi4_spin.svg</code>	$L(s, \chi_{-4})$
<code>paper/lchi5_palette.svg, paper/lchi5_spin.svg</code>	$L(s, \chi_5)$

together with the gallery `paper/modular_palettes.svg`, reproduced as Figure 35. All of them are emitted by `paper/make_modular_palettes.py`, on top of the evaluators of `paper/modular_functions.py`.

Remark 23.8 (What these plates are, and are not). The tiling, its sizes, its covering, the mirror, the rotation and the two orthogonality relations are the theorems of Sections 21 and 22, and the script re-checks them numerically before it draws anything. The *values* are numerics, and no theorem of this paper is asserted by a colour. The forms are evaluated from their q -expansions after reduction — by $z \mapsto z + 1$ and $z \mapsto -1/z$ for Δ and j , and by $z \mapsto z + 1$ and the Fricke involution $z \mapsto -1/(Nz)$ for T_{2A} ($N = 2$) and T_{3A} ($N = 3$) — and ζ and the L -functions by Euler–Maclaurin on the Hurwitz zetas of the residue classes, continued to the left half plane by the functional equation. Before emitting a file the generator checks that its own recomputed q -expansions agree with the tables certified in Lean (Propositions 12.1 and 23.5), that each evaluator satisfies the modular transformations just listed, that $j(i) = 1728$ and $j(\rho) = 0$, and that the zeta values of Theorem 23.3 come out right, along with ζ ’s functional equation, $L(1, \chi_{-4}) = \pi/4$, $L(1, \chi_{-3}) = \pi/3\sqrt{3}$ and the trivial zeros of all three L -functions. That is a check of the pictures, not a proof of anything in them.

24 A gallery from the LMFDB, and why all the palettes coincide

The plates of Section 23 raise a question one cannot un-see. Put j , T_{1A} , T_{2A} and T_{3A} side by side and they are, to the eye, *the same picture*; put Δ beside them and it is that picture with the colours reversed. The functions could hardly be more different — T_{2A} and T_{3A} live on different groups, their coefficients grow at different rates — and yet the plates agree. This section widens the experiment as far as the data allow, by reading a gallery of newforms pulled in from the LMFDB, and then proves that what one sees is forced.

The gallery

The functions are the 25 rational newforms listed in Table 2, fetched from the L-functions and Modular Forms Database (lmfdb.org) by `paper/fetch_lmfdb.py` through its public API —

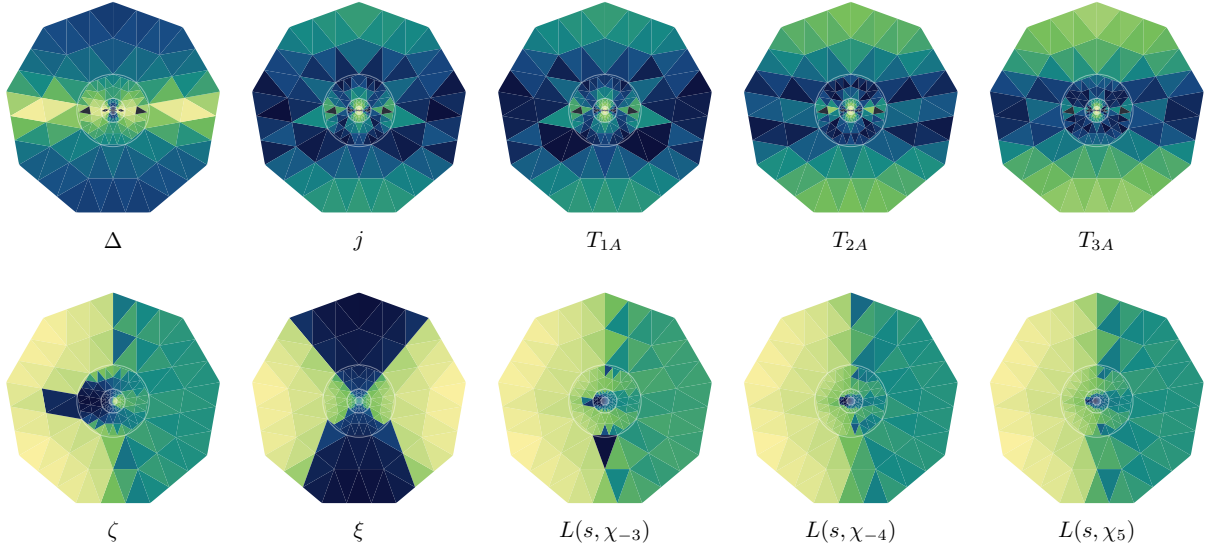


Figure 35: One tiling, ten functions: four rings of nine sectors, each sector cut once into nine similar parts, read by each function at the centroid of every part and coloured by the rank of the reading in the palette of that plate. Top row, on the upper half plane (radii $\frac{1}{9}, \frac{1}{3}, 1, 3$, the lower half read through the mirror of Definition 22.4): Δ , j , T_{1A} , T_{2A} , T_{3A} — the horizontal band of cusps along the real axis, and the mirror symmetry of Theorem 22.13. Bottom row, on the s -plane about $s = \frac{1}{2}$ (radii $1, 3, 9, 27$): ζ , ξ , $L(s, \chi_{-3})$, $L(s, \chi_{-4})$, $L(s, \chi_5)$ — the growth to the left given by the functional equation, the dark wedge of the trivial zeros along the negative real axis, and the notches on the critical line (dashed) where the nontrivial zeros are. The plate of ξ is symmetric under $s \mapsto 1 - s$ because ξ is. The full-resolution plates are the standalone files listed above, at 9^3 parts per sector.

the orbit invariants from the table `mf_newforms`, the Hecke eigenvalues from `mf_hecke_nf` — and cached verbatim in `paper/lmfdb_newforms.json` (fetched 2026-08-17). They include 5 forms with complex multiplication and 5 of positive analytic rank; `1.12.a.a` is Δ itself and `11.2.a.a` is $\eta(z)^2\eta(11z)^2$, the form of the elliptic curve of conductor 11. They are the plates of this section; the cache behind them is larger — it holds 109 rational newforms in all, weights 2 to 26 and levels 1 to 389 with 28 of them CM and 38 of positive rank, chosen from the database by the rule of `paper/select_lmfdb.py` — and every check below, and the whole of Section 25, runs over all of it.

Each form is evaluated from its own q -expansion

$$f(z) = \sum_{n \geq 1} a_n q^n, \quad q = e^{2\pi i z},$$

where the a_n are rebuilt from the eigenvalues a_p , $p \leq 2999$, by the Hecke recursions — $a_{mn} = a_m a_n$ for coprime m, n , $a_{p^{i+1}} = a_p a_{p^i} - p^{k-1} a_{p^{i-1}}$ at a good prime, $a_{p^i} = a_p^i$ at a bad one — and checked against the a_n the database itself lists. The series is truncated where Deligne's bound $|a_n| \leq d(n)n^{(k-1)/2}$ puts everything dropped below 10^{-12} of the leading term. The tiling is the ring ladder of Section 22: four rings of nine sectors about a centre in the upper half plane, each sector cut Euclid's way, each tile painted by the rank of its reading in the palette of the plate, exactly as in Section 23.

Two windows are used. *High up*, the ladder is centred at $\frac{5}{2}i$ and covers $\text{Im } z \in [1.30, 3.70]$; *low down*, it is centred at $\frac{3}{5}i$ and covers $[0.05, 1.15]$. Figure 36 is the high gallery, with the *hauptmoduls* j , T_{2A} , T_{3A} appended: every newform gives the same plate, and the three *hauptmoduls* give its negative. Figure 37 is the same four forms read high up, read low down, and read as *residuals* $h = f/q$, which is the plate with the universal factor divided out.

Why they coincide

Write $q(z) = e^{2\pi i z}$, so $\|q(z)\| = e^{-2\pi \text{Im } z}$.

Definition 24.1 (Normalised q -series, *hauptmodul* shape, palette rank). For $a : \mathbb{N} \rightarrow \mathbb{C}$ put $f_a(z) = \sum_{n \geq 0} a_n q(z)^n$ and, for $c : \mathbb{N} \rightarrow \mathbb{C}$, $h_c(z) = q(z)^{-1} + \sum_{n \geq 0} c_n q(z)^n$. Say the coefficients are *polynomially bounded* with constant C and degree k if $\|a_n\| \leq C(n+1)^k$ for all n , and call a *normalised* if in addition $a_0 = 0$ and $a_1 = 1$. For a finite sample $S \subset \mathbb{C}$ the *palette rank* of a function g at $z \in S$ is

$$\text{rank}_g(S, z) = \#\{w \in S : \|g(w)\| < \|g(z)\|\},$$

which is the colour the plates paint with.

Deligne's bound gives a polynomial bound for the coefficients of every newform in the gallery, and the coefficients of j , T_{1A} , T_{2A} , T_{3A} are polynomially bounded as well; that hypothesis is assumed below, not proved.

Lemma 24.2 (The leading term rules). *If a is normalised with constant C and degree k , and $\|q(z)\| \leq \frac{1}{2}$, then*

$$\|f_a(z) - q(z)\| \leq C K_k \|q(z)\|^2, \quad K_k = \sum_{n \geq 0} (n+3)^k 2^{-n}.$$

If only the polynomial bound is assumed, then $\|h_c(z) - q(z)^{-1}\| \leq C K_k$.

label	level	weight	rank	a_2	note
1.12.a.a	1	12	0	-24	Δ : the discriminant, $a_n = \tau(n)$
1.16.a.a	1	16	0	216	
1.18.a.a	1	18	1	-528	
1.20.a.a	1	20	0	456	
1.22.a.a	1	22	1	-288	
1.26.a.a	1	26	1	-48	
3.6.a.a	3	6	0	-6	
4.6.a.a	4	6	0	0	
5.4.a.a	5	4	0	-4	
6.4.a.a	6	4	0	-2	
7.4.a.a	7	4	0	-1	
9.4.a.a	9	4	0	0	CM by $\mathbb{Q}(\sqrt{-3})$
11.2.a.a	11	2	0	-2	$\eta(z)^2\eta(11z)^2$: the curve of conductor 11
14.2.a.a	14	2	0	-1	
15.2.a.a	15	2	0	-1	
17.2.a.a	17	2	0	-1	
19.2.a.a	19	2	0	0	
26.2.a.a	26	2	0	-1	
27.2.a.a	27	2	0	0	CM by $\mathbb{Q}(\sqrt{-3})$
32.2.a.a	32	2	0	0	CM by $\mathbb{Q}(i)$
36.2.a.a	36	2	0	0	CM by $\mathbb{Q}(\sqrt{-3})$
37.2.a.a	37	2	1	-2	the first elliptic curve of rank one
37.2.a.b	37	2	0	0	
49.2.a.a	49	2	0	1	CM by $\mathbb{Q}(\sqrt{-7})$
389.2.a.a	389	2	2	-2	the first elliptic curve of rank two

Table 2: The gallery: the 25 rational newforms of Figure 36, fetched from the LMFDB ([paper/lmfdb_newforms.json](#), fetched 2026-08-17), with the analytic rank and the first Hecke eigenvalue the database lists for each.

Proof. Both series converge absolutely for $\|q\| < 1$, by comparison with $\sum_n C(n+1)^k \|q\|^n$. Split off the first two terms: since $a_0 = 0$ and $a_1 = 1$, $f_a(z) - q(z) = \sum_{n \geq 0} a_{n+2} q^{n+2}$, and term by term $\|a_{n+2} q^{n+2}\| \leq C(n+3)^k \|q\|^n \|q\|^2 \leq C(n+3)^k 2^{-n} \|q\|^2$, using $\|q\| \leq \frac{1}{2}$ in the factor $\|q\|^n$. Summing gives the first bound; the second is the same computation without the two leading terms, using $(n+1)^k \leq (n+3)^k$. $\square$

So $\|f_a(z)\|$ lies between $\|q\|(1 - CK_k \|q\|)$ and $\|q\|(1 + CK_k \|q\|)$: high up, the reading is $e^{-2\pi \operatorname{Im} z}$ to within a relative error that tends to 0. That is the whole mechanism. Fix a gap $\delta > 0$ and set

$$Y(C, k, \delta) = \frac{-\log \rho}{2\pi}, \quad \rho = \min\left(\frac{1}{2}, \frac{(1-\theta)\theta}{2(CK_k + 1)}\right), \quad \theta = e^{-2\pi\delta}.$$

Theorem 24.3 (Reading by height). *Let $\operatorname{Im} w \geq Y(C, k, \delta)$ and $\operatorname{Im} z \geq \operatorname{Im} w + \delta$. Then for every normalised a with constant C and degree k ,*

$$\|f_a(z)\| < \|f_a(w)\|,$$

and for every polynomially bounded c with the same constants, $\|h_c(w)\| < \|h_c(z)\|$.

Proof. Write $r_z = \|q(z)\|$, $r_w = \|q(w)\|$, $B = CK_k$. The hypotheses give $r_w \leq \rho \leq \frac{1}{2}$ and $r_z \leq \theta r_w$. From $r_w \leq \rho$ one gets $Br_w < (1 - \theta)/(1 + \theta)$, i.e. $\theta(1 + Br_w) < 1 - Br_w$, and hence, by Lemma 24.2 twice,

$$\|f_a(z)\| \leq r_z(1 + Br_z) \leq \theta r_w(1 + Br_w) < r_w(1 - Br_w) \leq \|f_a(w)\|.$$

For the hauptmodul, $r_w \leq (1 - \theta)\theta/(2(B + 1))$ gives $r_w^{-1}(1 - \theta)/\theta \geq 2(B + 1)/\theta^2 > 2B$, so $r_z^{-1} - B \geq \theta^{-1}r_w^{-1} - B > r_w^{-1} + B$, and Lemma 24.2 bounds $\|h_c(z)\|$ below by the left side and $\|h_c(w)\|$ above by the right. $\square$

Corollary 24.4 (The palettes are the same picture). *Let S be a finite sample with $\text{Im } z \geq Y(C, k, \delta)$ for all $z \in S$ and $|\text{Im } z - \text{Im } w| \geq \delta$ for distinct $z, w \in S$. Then*

- (i) $\text{rank}_{f_a}(S, z) = \text{rank}_{f_b}(S, z)$ for all $z \in S$ and all normalised a, b with constant C and degree k ;
- (ii) $\text{rank}_{h_c}(S, z) = \text{rank}_{h_d}(S, z)$ for all $z \in S$ and all polynomially bounded c, d — the whole T series is one plate;
- (iii) $\|f_a(z)\| < \|f_a(w)\| \iff \|h_c(w)\| < \|h_c(z)\|$: the hauptmodul plate is the negative of the cusp form plate.

Proof. For distinct $z, w \in S$ one of $\text{Im } z \geq \text{Im } w + \delta$ or $\text{Im } w \geq \text{Im } z + \delta$ holds, and Theorem 24.3 decides the comparison $\|\cdot(z)\| < \|\cdot(w)\|$ the same way for every function of the class — with the order reversed between the two classes. The ranks are cardinalities of sets cut out by those comparisons, so they agree. $\square$

This is the content of `RequestProject/PaletteUniversality.lean`: `norm_qSeries_sub_qq` and `norm_hauptSeries_sub_inv` are Lemma 24.2, `qSeries_norm_lt_of_height` and `hauptSeries_norm_lt_of_height` are Theorem 24.3 with `qHeight` the explicit $Y(C, k, \delta)$ above, and `paletteRank_eq_of_normalized`, `paletteRank_eq_of_haupt`, `haupt_order_reverse` are the three parts of Corollary 24.4.

What the pictures measure

The theorem says the plates coincide above a height that depends on the coefficient bound; the gallery says how quickly. Over the 297 tiles of the high window the worst disagreement between the rank palettes of any two of the 25 newforms is 0.67% of the number of tiles, and between any two hauptmoduls 0.34%; that is one or two tiles, and the eye sees none of it. Low down the same measurement gives 99.0%, and the residual plates disagree by 99.7% — there the forms are entirely themselves. The palette traces of Definition 22.5 tell the same story numerically: normalised ring by ring, the worst deviation of a newform's trace from the mean of all of them is 7.7×10^{-3} high up and 3.2×10^{-1} low down (Figure 38).

The exact statement of Corollary 24.4 is also checked verbatim, on a sample where its hypotheses are visibly met: nine points with imaginary parts 1.5, 1.8, $\dots$, 3.9, so $\delta = 0.3$. There all 109 newforms of the cache give literally one and the same rank palette, 8, 7, $\dots$, 0, decreasing in the height; j , T_{1A} , T_{2A} , T_{3A} give literally its reverse; and the generator refuses to write a figure if this fails.

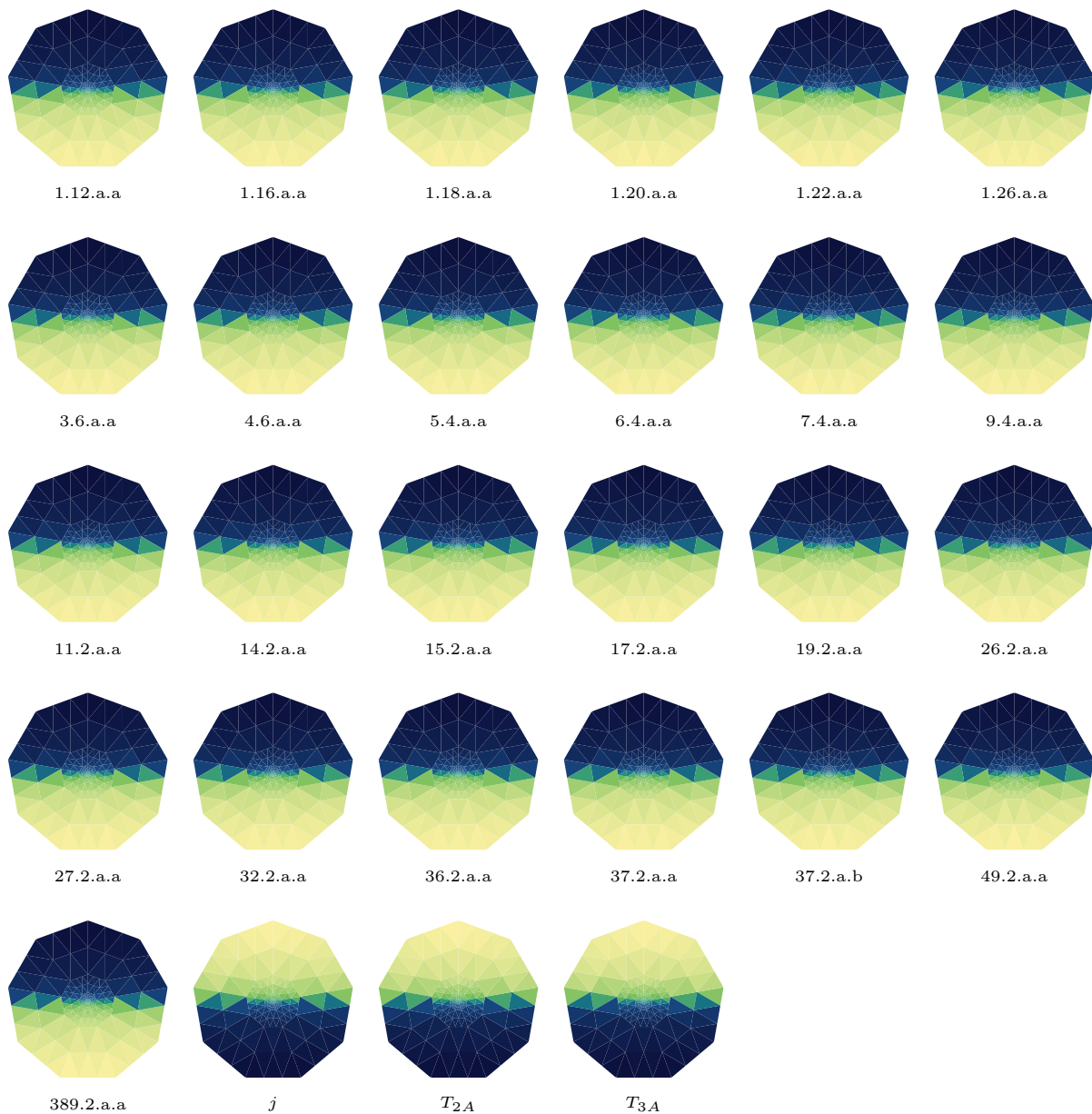


Figure 36: The gallery: 25 rational newforms of the LMFDB (5 with CM, 5 of positive analytic rank) and the Hauptmoduls j , T_{2A} , T_{3A} , all read on the same four rings of nine sectors, high up in the upper half plane ($\text{Im } z$ from 1.30 to 3.70), each tile coloured by the rank of its reading. Every newform gives the same plate and the Hauptmoduls give its negative, as Corollary 24.4 says they must. Full resolution: [paper/lmfdb_gallery.svg](https://paper.lmfdb_gallery.svg).

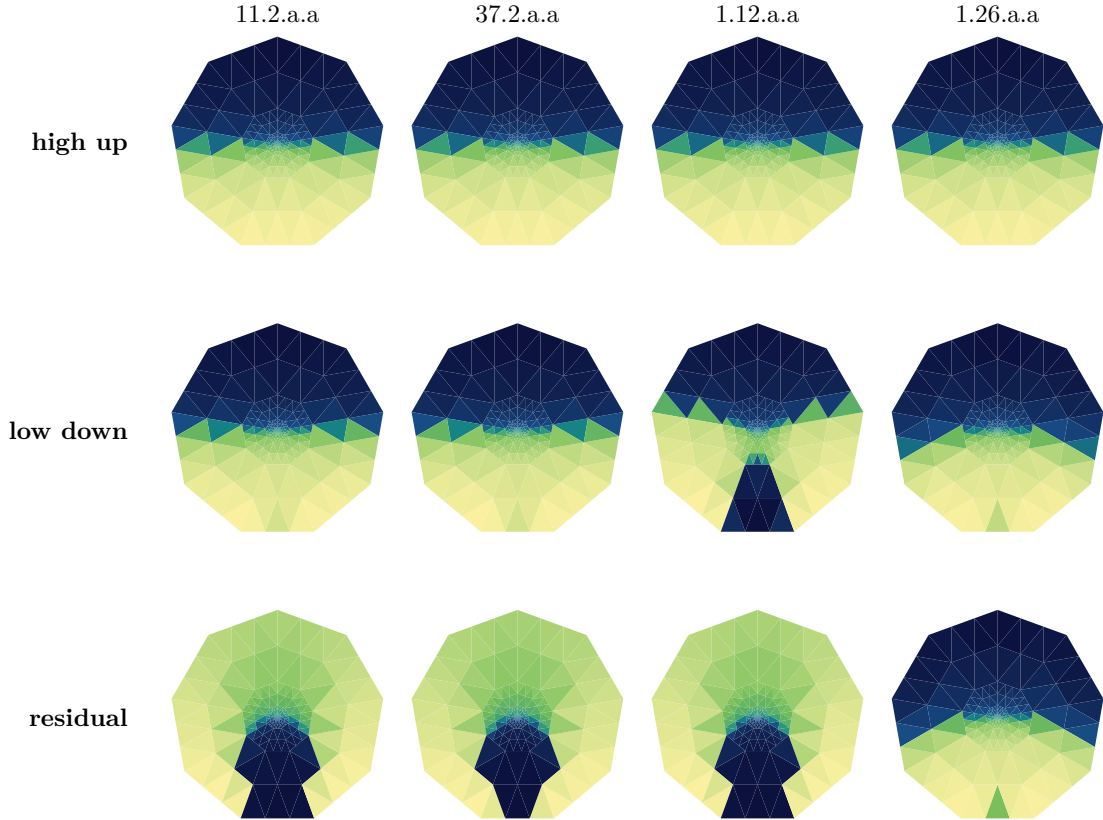


Figure 37: Four of the forms three ways. Top: high up, where the leading term rules and the plates coincide. Middle: low down ($\text{Im } z$ from 0.05 to 1.15), where each form is itself again. Bottom: the residual $h = f/q$ on the same low window — the universal factor divided out, leaving the form and not the cusp. Full resolution: [paper/lmfdb_hero.svg](#), [paper/lmfdb_shallow_gallery.svg](#), [paper/lmfdb_residual_gallery.svg](#).

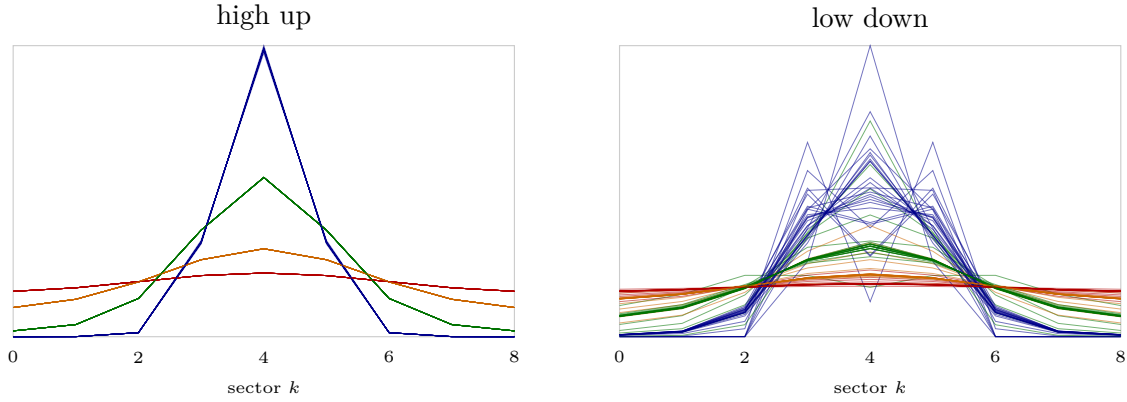


Figure 38: The palette traces — the T series of each plate. For every newform of the gallery and every ring j , the normalised trace $T_j(k)$ of Definition 22.5 is drawn against the sector k ; the four colours are the four rings. High up (left) all 25 newforms trace the same four curves, to 7.7×10^{-3} ; low down (right) they scatter, to 3.2×10^{-1} .

Remark 24.5 (What is proved, and what is drawn). Corollary 24.4 is a theorem about series with polynomially bounded coefficients, proved in Lean; that the Hecke eigenvalues of these particular newforms satisfy such a bound is Deligne’s theorem, which is imported, not proved here. The coefficients themselves are LMFDB data, cached in the repository, and the only claims made about them are the ones the generator checks: $a_1 = 1$, the Hecke recursions, Deligne’s bound at the primes in range, the agreement of 1.12.a.a with the τ table certified in Lean (Proposition 12.1) and with the independent Δ evaluator, and the agreement of 11.2.a.a with the eta product $\eta(z)^2\eta(11z)^2$. The colours are numerics; no theorem is asserted by one.

25 The best view of a form, and the tree of the sample

Corollary 24.4 is a negative result about looking: high in the upper half plane there is nothing to see, because every normalised q -series paints the same plate there. The positive question it leaves is *where to look*. This section answers it for every newform of the cache — a search over windows for the one in which the form is least like the universal picture — and then uses the winning window to sort the whole sample into a tree.

Views, individuality, and the score of a window

Definition 25.1 (View, palette distance). A *view* is a finite set $V \subset \mathbb{C}$ of points of the upper half plane: the sample a plate reads. Two readers f, g are *indistinguishable at V* when they paint every sample with the same rank, $\text{rk}_f(z) = \text{rk}_g(z)$ for all $z \in V$, where $\text{rk}_f(z) = \#\{w \in V : |f(w)| < |f(z)|\}$ is the rank of Definition 24.1; and the *palette distance* at V is the number of samples they paint differently,

$$d_V(f, g) = \#\{z \in V : \text{rk}_f(z) \neq \text{rk}_g(z)\}.$$

In Lean: `PaletteTaxonomy.Indist` and `PaletteTaxonomy.viewDist`, with `viewDist_self`, `viewDist_comm`, `viewDist_triangle` and `viewDist_eq_zero_iff` — d_V is a pseudometric which vanishes exactly on indistinguishable pairs.

The *individuality* of a form at a view is $d_V(f, q)/\#V$, its palette distance to the universal reader $q = e^{2\pi iz}$, which by Corollary 24.4 is the picture every normalised form shares high up; and the *separation score* of a view for a family $(f_i)_{i \in I}$ is the number of pairs it tells apart,

$$\text{score}(V) = \#\{(i, j) : f_i, f_j \text{ distinguishable at } V\},$$

which is `PaletteTaxonomy.sepScore`. A *best view* in a finite catalogue $\mathcal{V}$ is one of maximal score, `PaletteTaxonomy.IsBestView`. Three facts about that notion are proved in `RequestProject/PaletteTaxonomy.lean`.

Proposition 25.2 (The search terminates). *Over a finite non-empty catalogue of views a best view exists (`PaletteTaxonomy.exists_bestView`).*

Proposition 25.3 (A deep view is blind). *Call a view deep for the data C, k, δ when all its samples lie above the height $Y(C, k, \delta)$ of Theorem 24.3 and their imaginary parts are δ -separated (`PaletteTaxonomy.DeepView`). If every member of the family is a normalised q -series with the coefficient bound C, k , then a deep view separates no pair at all: its score is 0 (`PaletteTaxonomy.sepScore_eq_zero_of_deep`), all palette distances vanish, and the tree of the next subsection collapses to a single group already at threshold 0 (`PaletteTaxonomy.deep_cluster_eq_univ`).*

Corollary 25.4 (The best view is never deep). *If some candidate view distinguishes some pair of the family, then a best view is not a deep one (`PaletteTaxonomy.bestView_not_deep`). The search is pushed out of the universality regime — towards the real axis — by the theorem itself.*

The catalogue actually searched is a ladder of 92 windows: the ring ladder of Section 22 centred at $x_0 + iy_0$ with $x_0 \in \{0, \frac{1}{6}, \frac{1}{3}, \frac{1}{2}\}$, y_0 running geometrically from 0.07 to 2.74, and a half height of $0.35y_0$ or $0.7y_0$; a window is admissible only when its lowest tile stays above $\text{Im } z = 0.05$, where the truncation of the q -expansions is certified negligible. Each window is read not on all its tiles but on the greedy δ -separated subsample of them, δ being a fixed fraction of the window height: two tiles at exactly the same height are ordered by nothing but the tail of the expansion, and the hypothesis of Corollary 24.4 — and of Proposition 25.3 — is precisely that the sample be separated. So the sample read is one to which the theorems apply verbatim.

The scan is `paper/palette_taxonomy.py` and its picture is `paper/lmfdb_viewscan.svg`. Individuality collapses to 0 well before the windows reach the height at which the Lean bound bites: from $\text{Im } z_0 \approx 1$ upwards every form of the sample paints the universal plate exactly, which is Proposition 25.3 observed rather than proved. Downwards the palettes come alive, and the best view of every single form of the sample is the lowest admissible window. The common best view — the maximiser of the separation score — is

$$\text{Im } z \in [0.063, 0.357], \quad x_0 = 0.333,$$

which separates 5884 of the 5886 pairs of the sample on its 134 samples. Table 3 lists the best view and the individuality of every form, and Figures 39–41 draw every form on the ladder of its own best view.

From a distance table to a tree

A taxonomy is a tree: a family of groups any two of which are nested or disjoint. The tree of a table of distances is built in `RequestProject/UltrametricTaxonomy.lean`, for an arbitrary N -valued table d on a finite set — no symmetry, no triangle inequality assumed.

Definition 25.5 (Chain distance). i and j are *linked at t* when one can walk from i to j in steps of cost at most t (`UltrametricTaxonomy.Linked`), and the *chain distance* $s(i, j)$ is the least such t (`UltrametricTaxonomy.slink`). The *group of i at threshold t* is $C_t(i) = \{j : s(i, j) \leq t\}$ (`UltrametricTaxonomy.cluster`).

Theorem 25.6 (The chain distance is the subdominant ultrametric). $s \leq d$, and $s(i, k) \leq \max\{s(i, j), s(j, k)\}$ for all i, j, k ; and s is the largest ultrametric below d : any u with $u(a, a) = 0$, $u(a, c) \leq \max\{u(a, b), u(b, c)\}$ and $u \leq d$ satisfies $u \leq s$. In Lean: `slink_le_dist`, `slink_ultra`, `le_slink_of_ultra`.

Theorem 25.7 (The groups form a tree). If d is symmetric then for all thresholds t, s and all i, j the groups $C_t(i)$ and $C_s(j)$ are disjoint or nested (`cluster_laminar`); at a large enough threshold there is one group, the root (`cluster_top`); and if no two distinct objects are at distance 0 the groups at threshold 0 are the single objects, the leaves (`cluster_bot`).

Theorem 25.8 (The tree refines a classification). If an invariant ι cannot change across a single step of cost at most t — that is, if $d(a, b) \leq t$ implies $\iota(a) = \iota(b)$ — then ι is constant on every group at threshold t (`cluster_monochrome`, and `PaletteTaxonomy.palette_cluster_monochrome` for the palette distance of Definition 25.1).

The taxonomy of the sample

Feeding the palette distances at the common best view into that machinery gives the tree of Figure 42, also written out in `paper/lmfdb_tree.txt`. What makes it a taxonomy rather than a dendrogram is Theorem 25.8 together with the measured thresholds: over the sample, two forms of different weight are never closer than 39 tiles, two forms differing in complex multiplication never closer than 44, two of different analytic rank never closer than 29, and two whose levels are of different shape (one, prime, prime power, composite) never closer than 44. By Theorem 25.8 every group below those thresholds is therefore pure for the invariant in question: the palette tree *refines* the arithmetic classification, and does so with a wide margin for complex multiplication.

That fixes the ranks of the taxonomy. Rather than cut the tree at arbitrary heights we cut it where something happens, and name the five levels so obtained (Table 4):

- the three lowest cuts are the *certified* ones, placed exactly at the thresholds above. At the *family* cut (threshold 44) complex multiplication and the shape of the level are provably constant on each of the 107 taxa; at the *genus* cut (39) the weight joins them, on 108 taxa; at the *species* cut (29) all five invariants do, and the 109 taxa are the individual forms — no two newforms of the sample come within 30 tiles of one another, so at this view every form is its own species. The daggers in Table 4 mark the entries that Theorem 25.8 *proves*; the rest are measured;
- the *order* cut (73) is the *horizon* of complex multiplication: the coarsest cut at which CM-purity still reaches 90%. It gives 81 taxa, 93% CM-pure and 68% weight-pure. Above it the CM forms are swallowed by the spine, below it they stand apart — complex multiplication is the invariant this view sees furthest;
- the *kingdom* cut (102) shows the shape of the tree itself and guarantees nothing: 13 taxa, of which one is a spine of 96 forms chained into one another and 11 are single forms that never join it. Those *eccentrics* — Table 5 — are the forms whose palette at the best view

resembles no other's; the discriminant Δ itself is one of them, and so are 3 of the CM forms of the sample.

Reading the eight best windows together instead of one — a view, in the sense of Definition 25.1, is just a finite set of points, and a union of windows is one too — gives 242 samples and the same picture, with separations 79 for the weight and 112 for complex multiplication.

Remark 25.9 (What is proved, and what is measured). Theorems 25.6, 25.7 and 25.8, and Propositions 25.2 and 25.3, are proved in Lean for arbitrary distance tables and arbitrary views; what the data supply are the numbers that instantiate them: the palette distances of these particular newforms at this particular window, and hence the thresholds quoted above. The generator re-checks, before writing any figure, that the computed table is a pseudometric, that its chain closure is an ultrametric dominated by it, that the groups at every threshold are pairwise nested or disjoint, that the deep window of Section 24 separates nothing, and that the best view separates something. The hierarchy is the single-linkage one, the canonical choice singled out by Theorem 25.6. Like every single-linkage hierarchy it chains, and here it chains badly: at this view the palette distance nearly saturates — the mean distance over the 5886 pairs is close to the 134 samples available — so unrelated forms are all about equally far apart and the upper tree is a spine with eccentrics hanging off it, not a balanced dendrogram. That is itself the honest reading of the data: the arithmetic is legible in the *near* neighbourhoods, where it is certified, and in which forms refuse to join the spine at all; it is not legible in the coarse blocks. We tried to make the coarse blocks speak — reading the ladder at a dozen other windows, and coarsening the palette to 3, 6, 9, 18 and 27 bands of rank — and none of those readings raised the purity of a cut into a few dozen taxa above about 30%. So the ranks are named for what they do prove.

label	lvl	wt	CM	rk	best view Im z	ind
11.2.a.a	11	2		0	0.063-0.357	1.00
14.2.a.a	14	2		0	0.078-0.162	1.00
15.2.a.a	15	2		0	0.104-0.216	1.00
17.2.a.a	17	2		0	0.104-0.216	1.00
19.2.a.a	19	2		0	0.078-0.162	1.00
26.2.a.a	26	2		0	0.058-0.121	1.00
27.2.a.a	27	2	yes	0	0.058-0.121	1.00
32.2.a.a	32	2	yes	0	0.058-0.121	0.97
36.2.a.a	36	2	yes	0	0.058-0.121	1.00
37.2.a.a	37	2		1	0.182-0.378	1.00
37.2.a.b	37	2		0	0.078-0.162	1.00
43.2.a.a	43	2		1	0.063-0.357	1.00
49.2.a.a	49	2	yes	0	0.078-0.162	0.83
389.2.a.a	389	2		2	0.078-0.162	1.00
5.4.a.a	5	4		0	0.084-0.476	1.00
6.4.a.a	6	4		0	0.137-0.283	1.00
7.4.a.a	7	4		0	0.063-0.357	1.00
8.4.a.a	8	4		0	0.063-0.357	0.97
9.4.a.a	9	4	yes	0	0.104-0.216	0.97
13.4.a.a	13	4		1	0.058-0.121	1.00
17.4.a.a	17	4		1	0.058-0.121	1.00
32.4.a.b	32	4	yes	0	0.104-0.216	1.00
3.6.a.a	3	6		0	0.111-0.629	1.00
4.6.a.a	4	6		0	0.137-0.283	1.00
5.6.a.a	5	6		0	0.137-0.283	1.00
6.6.a.a	6	6		0	0.240-0.499	0.97
7.6.a.a	7	6		1	0.078-0.162	0.97
10.6.a.a	10	6		1	0.240-0.499	1.00
27.6.a.a	27	6	yes	1	0.182-0.378	1.00
32.6.a.b	32	6	yes	1	0.137-0.283	1.00
2.8.a.a	2	8		0	0.111-0.629	1.00
3.8.a.a	3	8		0	0.137-0.283	1.00
5.8.a.a	5	8		1	0.104-0.216	1.00
6.8.a.a	6	8		0	0.078-0.162	0.97
7.8.a.a	7	8		1	0.104-0.216	1.00
8.8.a.b	8	8		0	0.240-0.499	1.00
27.8.a.a	27	8	yes	0	0.182-0.378	0.97
32.8.a.a	32	8	yes	1	0.104-0.216	0.97
2.10.a.a	2	10		0	0.084-0.476	1.00
3.10.a.a	3	10		1	0.137-0.283	1.00
3.10.a.b	3	10		0	0.325-0.675	1.00
4.10.a.a	4	10		0	0.104-0.216	1.00
5.10.a.a	5	10		1	0.137-0.283	0.97
6.10.a.a	6	10		0	0.058-0.121	1.00
9.10.a.b	9	10	yes	1	0.240-0.499	0.83
32.10.a.a	32	10	yes	0	0.078-0.162	0.77
1.12.a.a	1	12		0	0.111-0.629	1.00
3.12.a.a	3	12		0	0.104-0.216	0.97
5.12.a.a	5	12		1	0.058-0.121	1.00
6.12.a.a	6	12		0	0.058-0.121	1.00
6.12.a.c	6	12		0	0.058-0.121	1.00
27.12.a.a	27	12	yes	1	0.104-0.216	0.83
32.12.a.a	32	12	yes	0	0.058-0.121	0.83
2.14.a.a	2	14		1	0.063-0.357	1.00
2.14.a.b	2	14		0	0.429-0.891	1.00

label	lvl	wt	CM	rk	best view Im z	ind
3.14.a.a	3	14		1	0.325-0.675	1.00
4.14.a.a	4	14		0	0.058-0.121	1.00
6.14.a.a	6	14		0	0.240-0.499	0.77
9.14.a.a	9	14		0	0.325-0.675	0.93
27.14.a.a	27	14	yes	0	0.058-0.121	0.83
32.14.a.a	32	14	yes	1	0.137-0.283	0.90
1.16.a.a	1	16		0	0.111-0.629	1.00
2.16.a.a	2	16		0	0.063-0.357	0.97
3.16.a.a	3	16		0	0.240-0.499	0.97
3.16.a.b	3	16		1	0.078-0.162	0.97
4.16.a.a	4	16		1	0.058-0.121	0.97
6.16.a.a	6	16		0	0.572-1.188	0.70
9.16.a.b	9	16	yes	0	0.429-0.891	0.83
32.16.a.a	32	16	yes	1	0.137-0.283	0.77
1.18.a.a	1	18		1	0.111-0.629	1.00
2.18.a.a	2	18		0	0.058-0.121	1.00
3.18.a.a	3	18		1	0.078-0.162	1.00
6.18.a.b	6	18		0	0.572-1.188	0.87
6.18.a.c	6	18		0	0.325-0.675	0.80
9.18.a.a	9	18		0	0.572-1.188	0.77
27.18.a.a	27	18	yes	1	0.058-0.121	0.67
32.18.a.a	32	18	yes	0	0.104-0.216	0.87
1.20.a.a	1	20		0	0.084-0.476	1.00
2.20.a.a	2	20		0	0.104-0.216	1.00
2.20.a.b	2	20		1	0.104-0.216	1.00
3.20.a.a	3	20		1	0.058-0.121	1.00
6.20.a.a	6	20		0	0.760-1.579	0.80
6.20.a.c	6	20		0	0.325-0.675	0.97
27.20.a.a	27	20	yes	0	0.104-0.216	0.63
32.20.a.a	32	20	yes	0	0.137-0.283	0.83
1.22.a.a	1	22		1	0.182-0.378	1.00
2.22.a.a	2	22		1	0.078-0.162	1.00
2.22.a.b	2	22		0	0.078-0.162	1.00
6.22.a.a	6	22		0	0.182-0.378	0.90
6.22.a.b	6	22		0	0.240-0.499	0.90
9.22.a.a	9	22		0	0.760-1.579	0.70
9.22.a.b	9	22	yes	1	0.104-0.216	0.53
32.22.a.a	32	22	yes	1	0.137-0.283	0.80
2.24.a.a	2	24		0	0.078-0.162	0.97
3.24.a.a	3	24		1	0.058-0.121	1.00
6.24.a.a	6	24		0	0.058-0.121	0.93
6.24.a.b	6	24		1	0.760-1.579	0.77
18.24.a.b	18	24		0	0.137-0.283	0.77
18.24.a.c	18	24		0	0.182-0.378	0.80
27.24.a.a	27	24	yes	1	0.240-0.499	0.50
32.24.a.a	32	24	yes	1	0.182-0.378	0.67
1.26.a.a	1	26		1	0.137-0.283	1.00
2.26.a.a	2	26		1	0.078-0.162	1.00
6.26.a.b	6	26		0	1.008-2.093	0.73
6.26.a.c	6	26		0	0.058-0.121	0.70
9.26.a.a	9	26		0	0.182-0.378	0.83
16.26.a.a	16	26		0	0.572-1.188	0.80
27.26.a.a	27	26	yes	0	0.058-0.121	0.53
32.26.a.a	32	26	yes	0	0.182-0.378	0.67

Table 3: The best view of each newform: the window, among the 92 candidates, in which its palette differs most from the universal one, and the individuality it achieves there (the fraction of the sample painted differently from q). Computed by `paper/palette_taxonomy.py`.

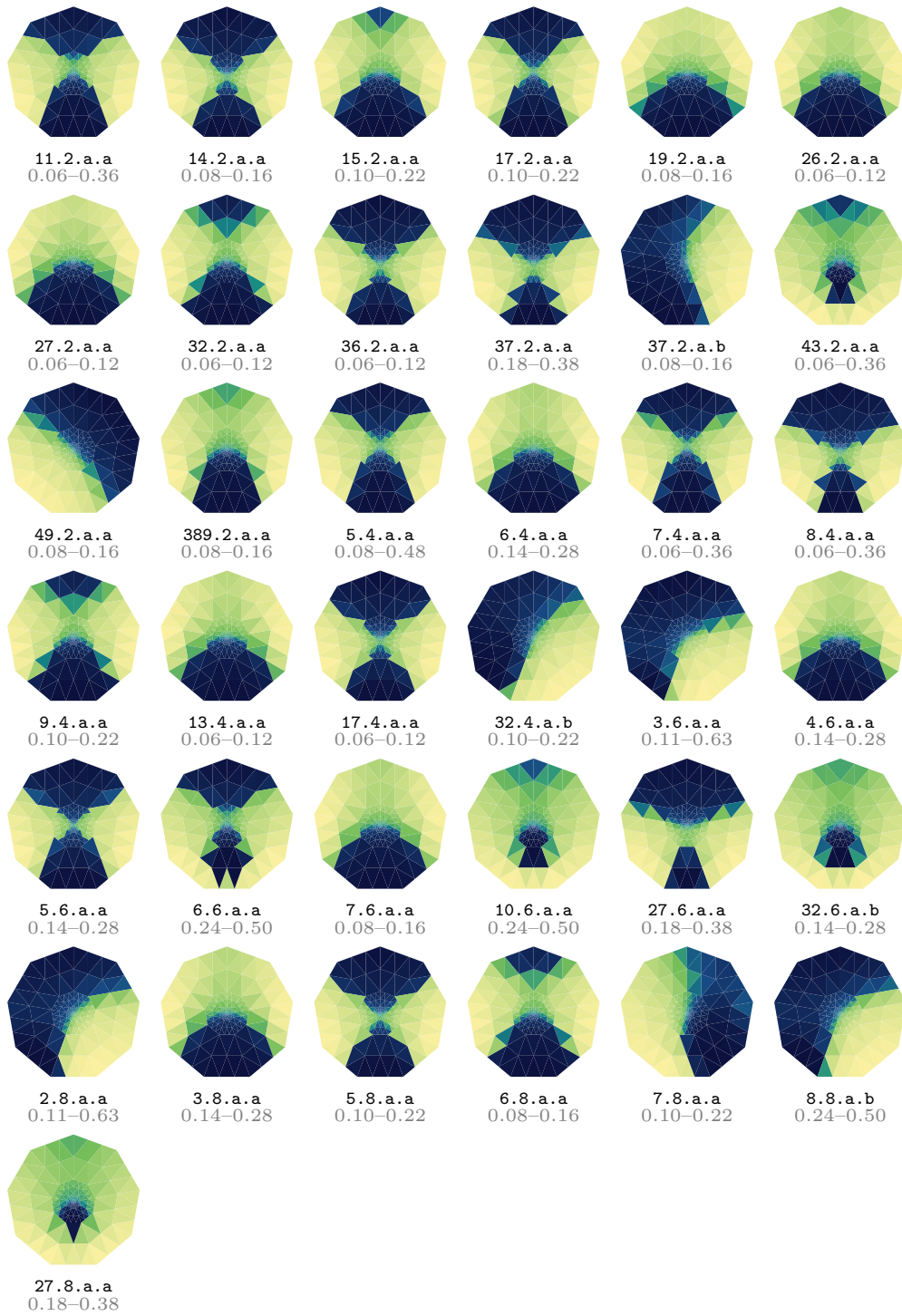


Figure 39: Every newform of the cache on the ring ladder of *its own* best view; under each plate its label and the window $\text{Im } z$. Sorted by weight, then level; first third of the sample, continued in Figures 40 and 41. Full resolution, all 109 plates in one image: [paper/lmfdb_bestview_gallery.svg](#); the scan that chose the windows is [paper/lmfdb_viewscan.svg](#).

rank	threshold	taxa	weight	CM	analytic rank	level kind
kingdom	102	13	10%	12%	12%	12%
order	73	81	68%	93%	72%	75%
family	44	107	98%	100% [†]	98%	100% [†]
genus	39	108	100% [†]	100% [†]	98%	100% [†]
species	29	109	100% [†]	100% [†]	100% [†]	100% [†]

Table 4: The five ranks of the taxonomy: the threshold of the cut, the number of taxa, and the percentage of the sample lying in a taxon on which the weight, complex multiplication, the analytic rank and the shape of the level are constant. A dagger marks a purity that is not measured but *proved* at that cut by Theorem 25.8, the threshold being at most the monochrome threshold of that invariant.

form	level	weight	CM	rank	d to nearest	nearest neighbour
27.8.a.a	27	8	yes	0	117	6.8.a.a
9.26.a.a	9	26		0	116	2.18.a.a
32.6.a.b	32	6	yes	1	114	32.8.a.a
5.10.a.a	5	10		1	114	5.6.a.a
32.8.a.a	32	8	yes	1	113	3.10.a.b
1.12.a.a	1	12		0	111	5.4.a.a
6.8.a.a	6	8		0	109	3.16.a.b
8.8.a.b	8	8		0	109	3.12.a.a
5.6.a.a	5	6		0	107	2.14.a.b
4.16.a.a	4	16		1	107	3.10.a.a
7.8.a.a	7	8		1	104	27.12.a.a

Table 5: The eccentrics: the 11 newforms that at the coarsest cut (threshold 102) are alone in their taxon — their palette at the common best view is further from every other palette of the sample than the spine’s forms are from one another. The last two columns give the nearest palette and its distance, out of 134 samples.

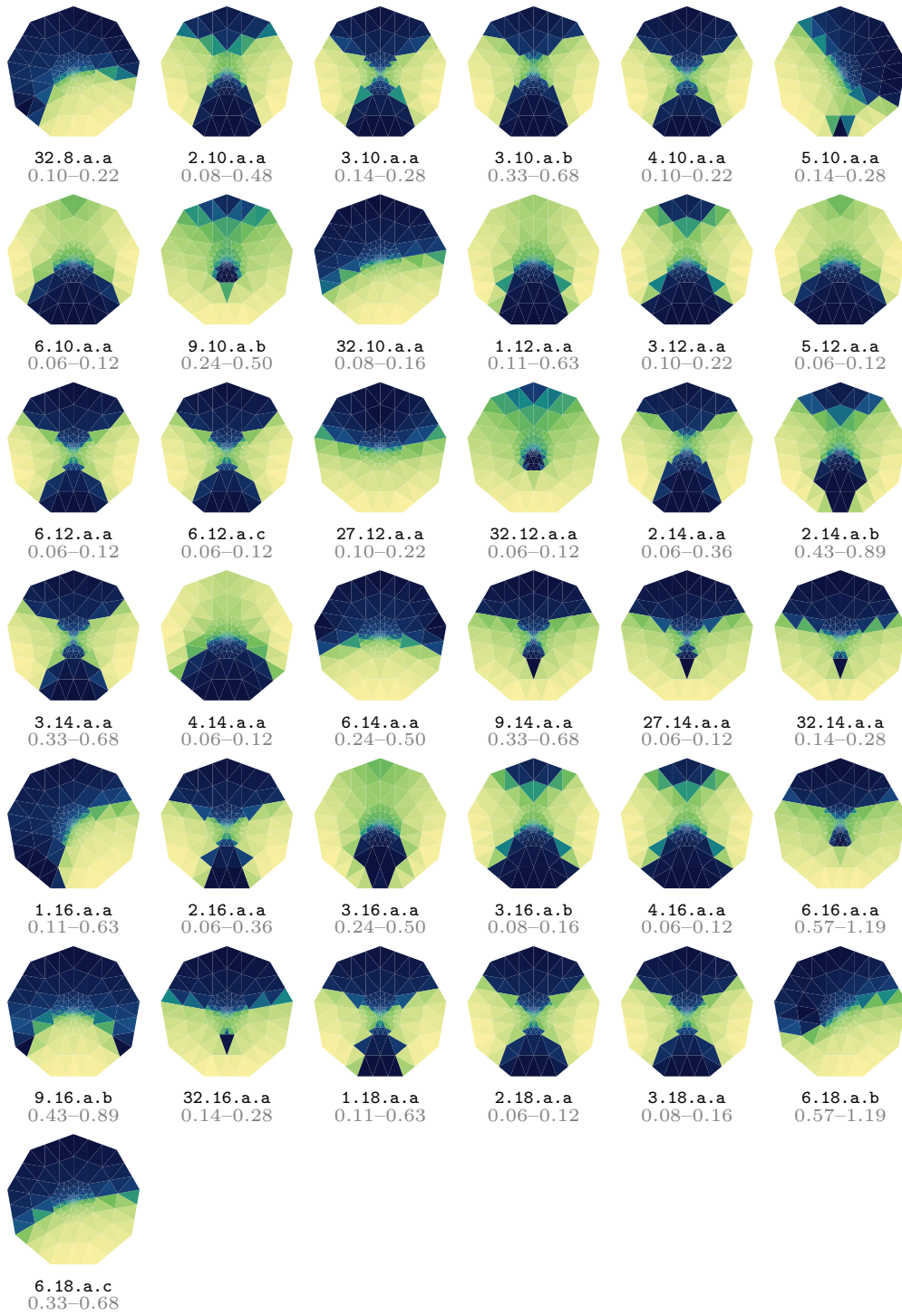


Figure 40: The best view of each newform, continued: the second third of the sample.

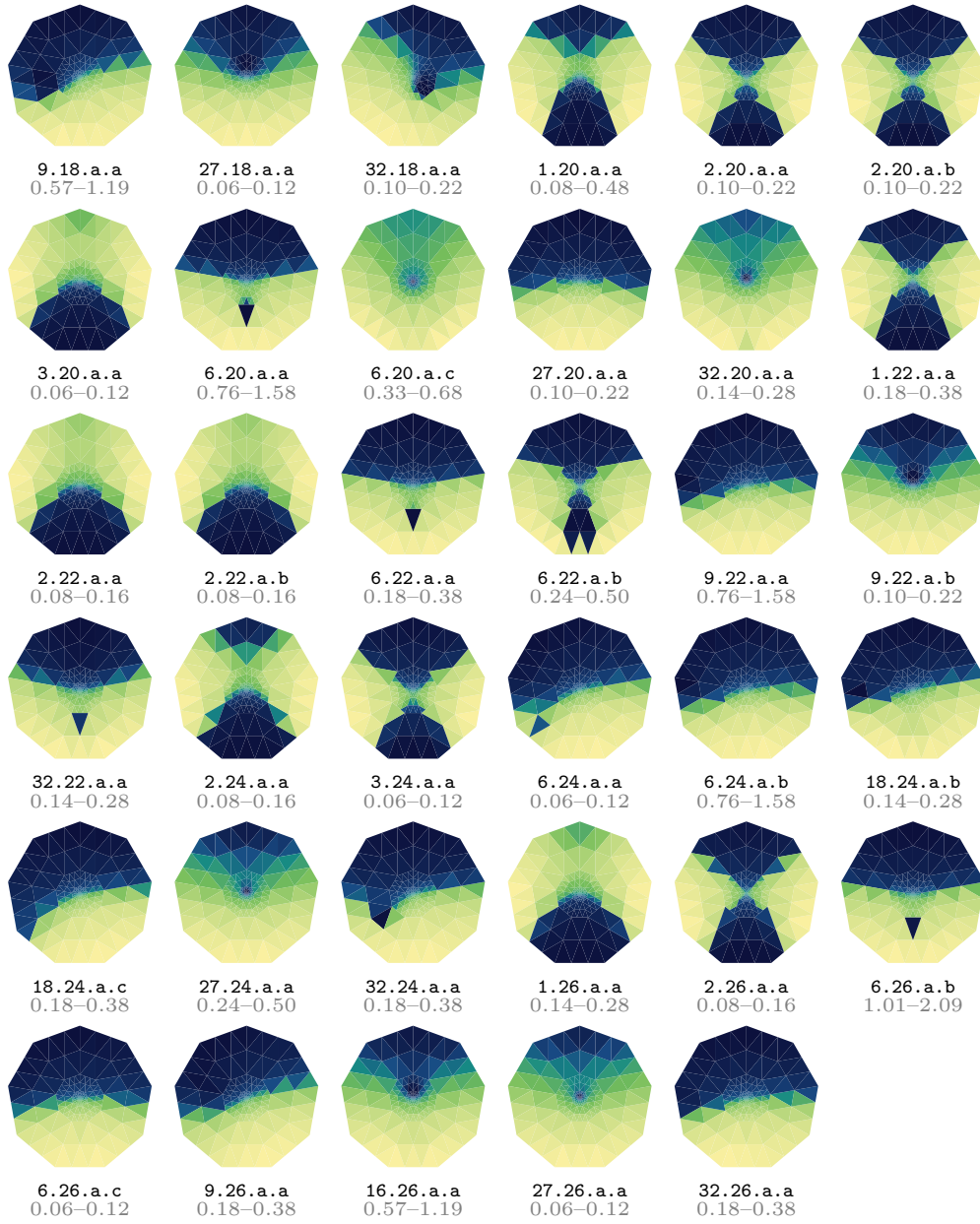


Figure 41: The best view of each newform, concluded: the last third of the sample.

26 Surveying the LMFDB along the lines of 3, 6, 9

The gallery of Section 24 looked at the newforms of the sample as *pictures*, and the taxonomy of Section 25 sorted those pictures by how far apart they are. This section sorts the same 109 newforms by *arithmetic*, and along the only lines this paper cares about: it reads each form through its digital roots and organises the sample into a tree whose three levels are 3, 6 and 9.

The spectrum of a form

Definition 26.1 (Digital-root spectrum). Let $f = \sum a_n q^n$ be a rational newform of level N and weight k . Its *spectrum* is the list of nine counts

$$c_r(f) = \#\{p \leq 997 : p \nmid N, p \neq 3, a_p \equiv r \pmod{9}\}, \quad r \in \mathbb{Z}/9,$$

i.e. how often each digital root occurs among the sampled Hecke eigenvalues (`LmfdbSpectrum369.counts`, and `Spectrum369.Spec` in the abstract).

The nine residues are not an unstructured index set: they are the enneagram. The trinity $T = \{0, 3, 6\}$ — the digital roots 9, 3, 6 — is the subgroup $3\mathbb{Z}/9\mathbb{Z}$, and the six remaining residues form the doubling circuit $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 7 \rightarrow 5 \rightarrow 1$ of Section 10. In `RequestProject/Spectrum369.lean`:

Theorem 26.2 (The enneagram splits). *T and the circuit are disjoint and together exhaust $\mathbb{Z}/9$; the circuit is exactly the unit group of $\mathbb{Z}/9$ and exactly the doubling orbit of 1; T is exactly the set of multiples of 3, is closed under addition, and absorbs multiplication: $xt \in T$ for every x and every $t \in T$ (`trinity_union_circuit`, `mem_circuit_iff_isUnit`, `circuit_eq_orbit`, `trinity_absorbing`). Doubling fixes 9 and swaps 3 and 6, and six doublings return every residue to itself (`dbl_iterate_six`).*

Theorem 26.3 (The geometry of the nine vertices). *Place the residue r at the vertex $v_r = e^{2\pi i r/9}$. Then $\sum_r v_r = 0$; the three trinity vertices are exactly the cube roots of unity, an equilateral triangle inscribed in the enneagram, and they too sum to zero; hence so do the six vertices of the circuit (`vertex_sum_univ`, `trinity_vertices`, `vertex_sum_trinity`, `vertex_sum_circuit`).*

So a spectrum splits canonically into its *triangle part*, the mass on T , and its *circuit part*, the mass on the hexagon; and the survey has three natural layers:

Definition 26.4 (The three layers). *Layer 9* of a form is the set of residues mod 9 that occur, *layer 6* is the set of occupied circuit vertices, *layer 3* is the set of residues mod 3 that occur (`supp9`, `hexad`, `supp3`; on the data, `supp9L`, `hexadL`, `supp3L`).

Theorem 26.5 (The layers are a tree). *Layer 9 determines layer 6 and layer 3, so the groups of forms cut out at the three layers are pairwise nested or disjoint — a laminar family, i.e. a tree (`Spectrum369.spectral_tree_laminar`, `hexad_determined`, via the general `block_nested_or_disjoint`). Refining a layer further by the chain distance of Definition 25.5 applied to the ℓ^1 distance of spectra is again a tree (`spectral_cluster_laminar`).*



Figure 42: The taxonomy: the single-linkage tree of the palette distances of the 109 newforms at the common best view ($\text{Im } z \in [0.063, 0.357]$, 134 samples). Two forms join at the least number of tiles their palettes must differ by along a chain; the dashed lines are the five cuts of Table 4, and the leaves are coloured by weight. Theorem 25.7 is what makes the drawing a tree at all. Full resolution: [paper/lmfdb_tree.svg](#), and as text [paper/lmfdb_tree.txt](#).

What the sample does

Run the 109 newforms of the sample through Definition 26.1 — the computation is `paper/spectrum369_survey.py`, and its result is re-checked by the Lean kernel in `RequestProject/LmfdbSpectrum369.lean` from the residues stored in `RequestProject/LmfdbSpectrum369Data.lean`. Three things happen.

Layer 3 is a dichotomy, and it is a congruence. Every form of the sample has layer 3 equal to $\{0, 2\}$ or to $\{0, 1, 2\}$: there is no third possibility, and there are 78 of the first kind against 31 of the second (`layer_three_dichotomy`, `layer_three_counts`). The split is not cosmetic:

Theorem 26.6 (The 3-line is the Eisenstein congruence). *A form of the sample misses the digital roots 1, 4, 7 — i.e. has layer 3 equal to $\{0, 2\}$ — exactly when $a_p \equiv 1 + p^{k-1} \pmod{3}$ at every sampled prime (`eisenstein_three_iff_deficient`), and that happens for 78 of the 109 forms. One direction holds in general and for every k : if $a \equiv 1 + p^{k-1} \pmod{3}$ with p unramified, then $a \not\equiv 1$, because p^{k-1} is a unit (`Spectrum369.eisenstein_three_avoids_one`).*

Layer 9 has only seven values. Of the 512 subsets of $\mathbb{Z}/9$ that could occur, exactly 7 do (Table 6), with sizes 44, 27, 12, 11, 9, 4, 2 (`layer_nine_classes`, `layer_nine_sizes`). The thinnest of them is remarkable: for 11 forms of the sample *every* sampled eigenvalue is $\equiv 0$ or $2 \pmod{9}$, so the whole spectrum sits on one trinity vertex and one circuit vertex. Again there is a reason, and the reason is six-fold periodicity in the weight:

Theorem 26.7 (The thin branches, explained). *Let p be a residue on the circuit and suppose $a \equiv 1 + p^{k-1} \pmod{9}$. If $k \equiv 4 \pmod{6}$ then $a \equiv 0$ or 2 ; and if k is even with $k \not\equiv 4 \pmod{6}$ then $a \in \{0, 2, 3, 5, 6, 8\}$ (`Spectrum369.eisenstein_nine_weight_four`, `Spectrum369.eisenstein_nine_even`). The weight matters only modulo 6 because the circuit has six vertices: $p^6 = 1$ for every unit mod 9 (`circuit_pow_six`).*

Corollary 26.8 (Seen in the data). *In the sample the converse holds too: the thin class is exactly the set of forms satisfying the congruence mod 9 with $k \equiv 4 \pmod{6}$ (`thin_class_iff`), namely the 11 forms `9.4.a.a`, `3.10.a.a`, `3.10.a.b`, `9.10.a.b`, `1.16.a.a`, `3.16.a.a`, `3.16.a.b`, `9.16.a.b`, `1.22.a.a`, `9.22.a.a`, `9.22.a.b` (`thin_class_labels`); and each of the 28 forms obeying the congruence mod 9 has its layer 9 inside one of the two sets Theorem 26.7 allows (`eisenstein_nine_supports`). 10 forms of the sample obey it mod 27 as well.*

The trinity carries the mass. Every form of the sample puts at least a third of its sampled eigenvalues on the triangle $\{9, 3, 6\}$ — the smallest triangle mass in the sample is 33.7%, attained by the conductor-11 curve's form — and every form with complex multiplication puts at least half of it there, the smallest being 51.8% (`trinity_at_least_a_third`, `cm_trinity_at_least_a_half`). The digital root 9 itself occurs for every single form (`nine_always_occurs`).

The tree

Stacking the three layers and then refining each layer-9 class by single linkage on the ℓ^1 distance of the (per-mille normalised) spectra gives the tree of Figure 43, written out in full in `paper/lmfdb_spectrum369_tree.txt`; Theorem 26.5 is what makes it a tree. Its shape is worth reading aloud: the root splits by the mod-3 trinity datum into 78 and 31 forms; the deficient branch splits again by which circuit vertices are lit — $\{2, 5, 8\}$, or $\{2\}$ alone, or $\{2, 8\}$

— and only then by the full enneagram; the other branch is the forms whose eigenvalues visit all nine roots, with the CM forms of level 32 hanging off it on the four-vertex arc $\{2, 4, 5, 7\}$.

The leaves of that tree are not the individual forms. The spectrum is coarse: the 109 newforms carry only 51 distinct spectra (`distinct_spectra_card`), and all 11 forms of the thin branch — weights 4, 10, 16, 22, levels 1, 3, 9 — share literally the same nine counts (`thin_class_one_spectrum`). That is the arithmetic counterpart of the universality of Section 24: up there every form painted the same plate because the leading term dominates; here many forms carry the same spectrum because a congruence, not the form, decides the digital roots. Figure 44 shows all 109 spectra as enneagram wheels, and the coincidences are visible as repeated glyphs.

Remark 26.9 (What is proved, and what is data). Theorems 26.2, 26.3, 26.5, 26.7 and the general half of Theorem 26.6 are proved in Lean for arbitrary residues, weights and spectra. Everything said about the sample — the dichotomy at layer 3, the seven classes at layer 9, the counts, the thin class and its labels, the mass on the trinity, the 51 distinct spectra — is a statement about the 109 cached newforms over the primes $p \leq 997$ with $p \nmid N$ and $p \neq 3$ (between 165 and 167 primes per form), and each such statement is checked by the Lean kernel on the residues embedded in `RequestProject/LmfdbSpectrum369Data.lean`, not merely computed by the script. Nothing here is claimed for all primes: these are congruences observed throughout a finite sample, and Theorem 26.7 says only what follows *if* such a congruence holds.

mod 9 support	mod 3	#forms	CM	Eis. mod 9	weights
$\{0, 2, 3, 5, 6, 8\}$	$\{0, 2\}$	44	0	17	2, 4, 6, 8, 12, 14, 18, 20, 24, 26
$\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$	$\{0, 1, 2\}$	27	10	0	2, 4, 6, 8, 10, 12, 14, 18, 20, 24, 26
$\{0, 2, 5, 8\}$	$\{0, 2\}$	12	10	0	2, 6, 8, 12, 14, 18, 20, 24, 26
$\{0, 2\}$	$\{0, 2\}$	11	4	11	4, 10, 16, 22
$\{0, 2, 3, 6\}$	$\{0, 2\}$	9	0	0	4, 10, 16, 22
$\{0, 2, 4, 5, 7\}$	$\{0, 1, 2\}$	4	4	0	4, 10, 16, 22
$\{0, 2, 8\}$	$\{0, 2\}$	2	0	0	10, 16

Table 6: The seven digital-root spectra of the LMFDB sample: which residues $a_p \bmod 9$ occur at the sampled good primes ($p \leq 997$, $p \nmid N$, $p \neq 3$: between 165 and 167 primes per form). “Eis. mod 9” counts the forms of the class satisfying $a_p \equiv 1 + p^{k-1} \pmod{9}$ throughout the sample.

27 The Pascal board: which cells are lit, and why

The last picture of this paper is not a modular form but a triangle of numbers. Colour the cell (n, k) of Pascal’s triangle *lit* when a fixed prime p does not divide $\binom{n}{k}$, and one gets the familiar fractal (Figure 45A, B). Colour instead only those cells whose Pascal entry can be the index of a k -set stabilizer in a group of a given order, and one gets a very different picture: a narrow band at the two ends of the row, with a dark middle (Figure 45C). Both patterns, and the exact place where the second one stops, are theorems; this section states them and gives the proofs. The connection with the rest of the paper is the same trinity as before: the mod-3 board is where 3, 6, 9 do the colouring, and row 24 turns out to have exactly nine lit cells, sitting exactly at the multiples of three.



Figure 43: The 3-6-9 spectral tree of the 109 newforms: the root, then layer 3 (the residues mod 3 that occur), layer 6 (the occupied vertices of the doubling circuit), layer 9 (the residues mod 9). Each node carries its enneagram, drawn with the trinity triangle in gold and each vertex sized by the mass the node's forms put on it. Full resolution: [paper/lmfdb_spectrum369_tree.svg](#); the whole tree, including the single-linkage refinement below layer 9, is [paper/lmfdb_spectrum369_tree.txt](#).

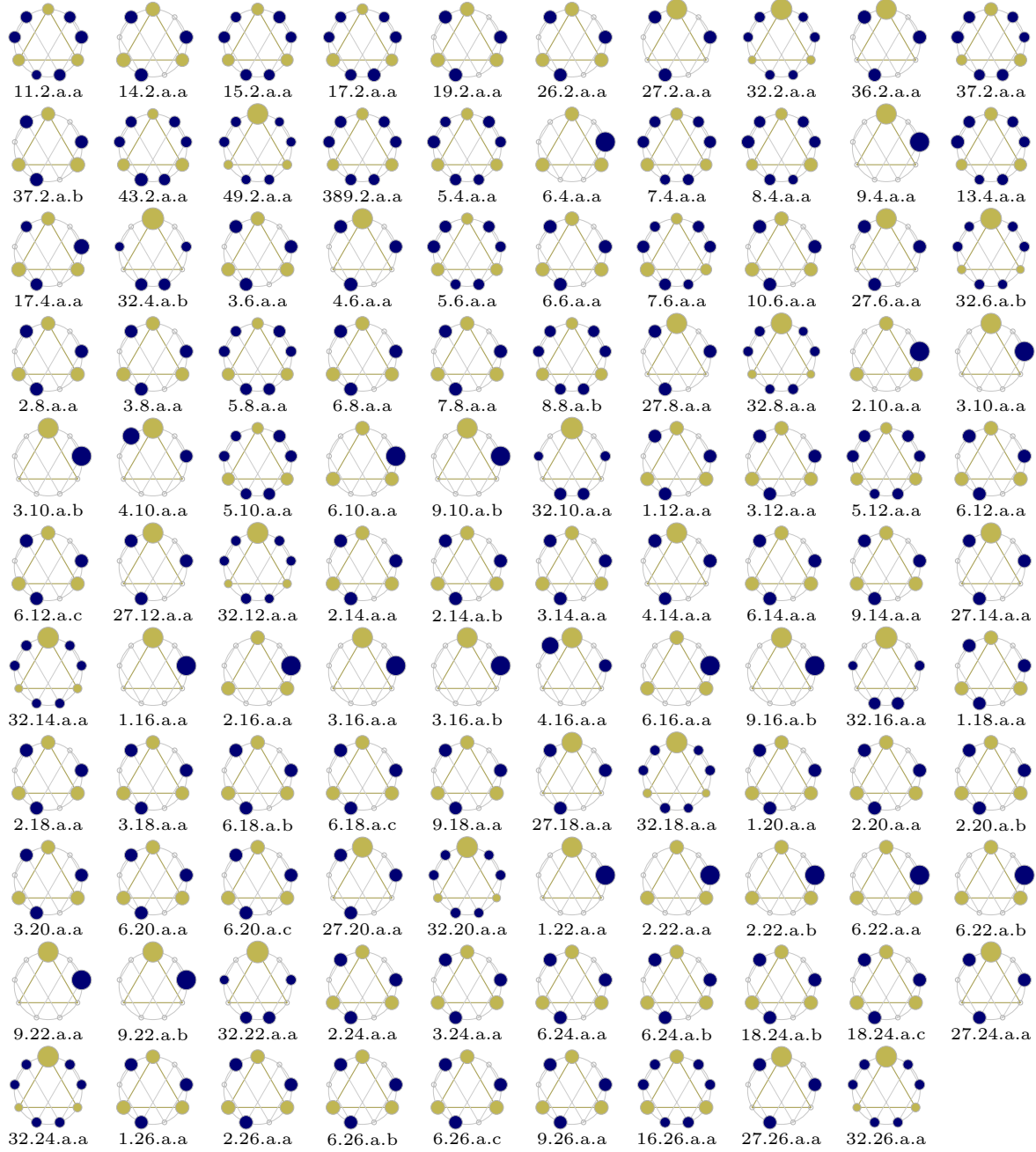


Figure 44: Every newform of the sample as an enneagram wheel: vertex r sized by the number of sampled primes with $a_p \equiv r \pmod{9}$, trinity in gold. Forms with identical wheels share a spectrum exactly; there are only 51 distinct wheels among the 109 forms.

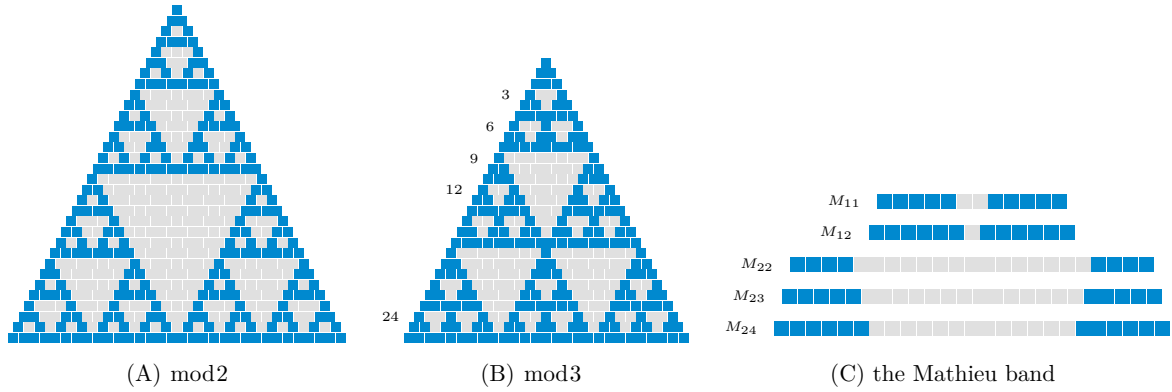


Figure 45: The three layers of the board. (A) mod 2: lit iff $\binom{n}{k}$ is odd, i.e. iff $k \& n = k$ (Theorem 27.3). (B) mod 3, with the rows 3, 6, 9, 12, 24 marked; row 24 is lit exactly at the multiples of 3 and has exactly nine lit cells (Proposition 27.7). (C) the Mathieu band: for $n = 11, 12, 22, 23, 24$ a cell is lit when $\binom{n}{k}$ divides the order of M_n — which by Theorem 27.10 is necessary for any group of that order to be transitive on k -subsets. The middle of every row is dark, and dark for an arithmetic reason.

The Frobenius layer: Lucas, Kummer, and the row count

Definition 27.1 (The mod- p board). For a prime p , the cell (n, k) is *lit* if $p \nmid \binom{n}{k}$. Write $d_i(n) = \lfloor n/p^i \rfloor \bmod p$ for the i -th base- p digit of n , and say that n *dominates* k if $d_i(k) \leq d_i(n)$ for every i .

Theorem 27.2 (Lucas's criterion). *For a prime p , the cell (n, k) is lit if and only if n dominates k .*

Proof. Choose a with $n, k < p^a$; Lucas's theorem gives $\binom{n}{k} \equiv \prod_{i < a} \binom{d_i(n)}{d_i(k)} \pmod{p}$. Since p is prime it divides the product if and only if it divides one factor. A factor with $d_i(k) > d_i(n)$ is 0, hence divisible. A factor with $d_i(k) \leq d_i(n) < p$ is not: from $\binom{x}{y} y! (x-y)! = x!$ and $p \nmid x!$ for $x < p$ one gets $p \nmid \binom{x}{y}$. Finally, for $i \geq a$ both digits vanish, so the condition over all i is the condition over $i < a$. $\square$

Theorem 27.3 (The Sierpiński gasket). *$\binom{n}{k}$ is odd if and only if $k \& n = k$, i.e. every binary digit of k is a binary digit of n .*

Proof. Base-2 digits are 0 or 1, so domination says precisely that a set bit of k is a set bit of n ; and that is the bitwise statement $k \& n = k$. $\square$

Kummer's theorem describes the same board from the other side: the dark cells.

Theorem 27.4 (Kummer's criterion). *Let p be prime and $k \leq n$. The cell (n, k) is dark if and only if adding k and $n - k$ in base p produces at least one carry, i.e. $p^i \leq (k \bmod p^i) + ((n - k) \bmod p^i)$ for some $i \geq 1$. Consequently domination and carry-freeness are the same condition.*

Proof. The p -adic valuation of $\binom{n}{k}$ is the number of such i below any bound exceeding $\log_p n$ (Kummer); the entry is divisible by p exactly when that count is nonzero, and no carry can occur at an i with $p^i > n$. The last sentence is Theorem 27.2 combined with this. $\square$

The dark inverted triangles inside the fractal are therefore not empty space: they are exactly the cells whose base- p addition carries, and grading them by how many carries occur is the heat map of the p -adic valuation.

Theorem 27.5 (Self-similarity and the row count). *Let $L_p(n)$ be the number of lit cells in row n . Then $L_p(0) = 1$ and*

$$L_p(n) = ((n \bmod p) + 1) L_p(\lfloor n/p \rfloor), \quad \text{hence} \quad L_p(n) = \prod_{i < a} (d_i(n) + 1) \quad (n < p^a).$$

Proof. Domination splits as a condition on the last digit and a condition on the shifted numbers: n dominates k iff $k \bmod p \leq n \bmod p$ and $\lfloor n/p \rfloor$ dominates $\lfloor k/p \rfloor$. The map $k \mapsto (k \bmod p, \lfloor k/p \rfloor)$ is therefore a bijection from the lit cells of row n onto $\{0, \dots, n \bmod p\}$ times the lit cells of row $\lfloor n/p \rfloor$, with inverse $(r, q) \mapsto r + pq$; a dominated k automatically satisfies $k \leq n$, so no boundary condition is lost. The product formula follows by induction on a . $\square$

Corollary 27.6. *Every cell of row $p^m - 1$ is lit: those rows have all digits equal to $p - 1$, so they dominate everything below them, and $L_p(p^m - 1) = p^m$.*

Proposition 27.7 (The trinity on the board). *On the mod-3 board: row 3 is lit exactly at $k = 0, 3$; row 6 exactly at the multiples of 3, i.e. $k = 0, 3, 6$; row 9 only at its two ends $k = 0, 9$; row 12 exactly at $k = 0, 3, 9, 12$; and row 24 exactly at the multiples of 3. The corresponding counts are $L_3(3) = 2$, $L_3(6) = 3$, $L_3(9) = 2$, $L_3(12) = 4$ and $L_3(24) = 9$.*

Proof. In base three $24 = (220)_3$, so domination forces $k \equiv 0 \bmod 3$ and imposes nothing else below 27; the count is $(0 + 1)(2 + 1)(2 + 1) = 9$ by Theorem 27.5. Likewise $12 = (110)_3$ gives $1 \cdot 2 \cdot 2 = 4$ cells, $9 = (100)_3$ gives $1 \cdot 1 \cdot 2 = 2$, $6 = (20)_3$ gives $1 \cdot 3 = 3$ and $3 = (10)_3$ gives $1 \cdot 2 = 2$. $\square$

So the row that carries M_{24} carries, on the trinity's own board, exactly nine lit cells — one for each vertex of the enneagram of Section 26 — and they are the cells whose column index is a multiple of three.

The group layer: why a cell can carry a group at all

Definition 27.8. A group G acting on a finite set Ω with $|\Omega| = n$ is k -homogeneous if it is transitive on the k -element subsets of Ω .

Theorem 27.9 (Orbit-stabilizer on the board). *If G is k -homogeneous on Ω and $|S| = k$, the setwise stabilizer G_S has index $[G : G_S] = \binom{n}{k}$.*

Proof. The index of a stabilizer is the size of the orbit; the orbit of S consists of k -subsets, and by homogeneity is all of them, and there are $\binom{n}{k}$ of those. $\square$

This is why the cells of such a board are Pascal entries at all. It also gives the obstruction that shapes Figure 45C.

Theorem 27.10 (The arithmetic obstruction). *If a finite group G is k -homogeneous on n points, then $\binom{n}{k}$ divides $|G|$. Equivalently: if $\binom{n}{k} \nmid |G|$, no action of G on n points is transitive on k -subsets.*

Proof. An index divides the order. $\square$

Corollary 27.11 (The middle of row 24 is dark). *No group of order $|M_{24}| = 244\,823\,040$ is k -homogeneous on 24 points for $6 \leq k \leq 18$; indeed $\binom{24}{k}$ divides 244 823 040 exactly when $k \leq 5$ or $k \geq 19$. Similarly no group of order $|M_{12}| = 95\,040$ is 6-homogeneous on 12 points: $\binom{12}{6} = 924$ does not divide 95 040.*

The band in Figure 45C is exactly this divisibility, computed in the kernel for the five Mathieu orders. It reproduces the visible feature of the picture — the left edge of each row lights up for a few steps and then the row goes dark — without knowing anything about the Mathieu groups except their orders. The width of the lit edge is the arithmetic shadow of the degree of transitivity: five for M_{12} , M_{24} , four for M_{11} , M_{23} , three for M_{22} .

The converse question — what *can* fill a cell — has a completely general answer.

Theorem 27.12 (Every cell is a group index). *For every n and k the symmetric group S_n is k -homogeneous, and the setwise stabilizer of a k -subset has index exactly $\binom{n}{k}$.*

Proof. Given subsets S, T of equal size, a bijection $S \rightarrow T$ extends to a permutation of the whole set (the complements have equal size too), and it carries S to T . Now apply Theorem 27.9. $\square$

So the board is completely fillable by ordinary symmetry: nothing sporadic is needed to realise a Pascal entry as the index of a set stabilizer. What is rare, and what the original diagram highlights, is a stabilizer that is *maximal* in a sporadic group — a property this paper does not formalise.

The synthetic layer: keep the geometry, drop the group

Row 24 lights up at $k = 8$ for a different reason than at $k \leq 5$: no group of order $|M_{24}|$ is 8-homogeneous (Corollary 27.11), so the subgroup living there is not the stabilizer of an arbitrary 8-set but of a distinguished one, an octad of the Steiner system $S(5, 8, 24)$. Dropping the group and keeping the blocks is the “synthetic” move, and it is enough for the counting.

Definition 27.13 (Steiner system). A *Steiner system* $S(t, k, n)$ on a point set Ω with $|\Omega| = n$ is a family $\mathcal{B}$ of k -element blocks such that every t -element subset of Ω lies in exactly one block.

Theorem 27.14 (The block count). *In an $S(t, k, n)$,*

$$|\mathcal{B}| \cdot \binom{k}{t} = \binom{n}{t}.$$

In particular $\binom{k}{t}$ divides $\binom{n}{t}$.

Proof. Count the t -subsets twice. Each lies in exactly one block, so the sets $\{T \subseteq B : |T| = t\}$, for $B \in \mathcal{B}$, are pairwise disjoint and cover all of them; each contributes $\binom{k}{t}$. $\square$

Corollary 27.15 (The octads, and where the pattern stops). *An $S(5, 8, 24)$ has exactly $\binom{24}{5}/\binom{8}{5} = 42\,504/56 = 759$ blocks. There is no $S(5, 8, 25)$: $\binom{8}{5} = 56$ does not divide $\binom{25}{5} = 53\,130$.*

That second sentence is the honest form of the question “can we extend the pattern past row 24?”. For the octad geometry the answer is no, and the reason is a divisibility, not the classification of finite simple groups. (The classification is what rules out an M_{25} ; this paper proves neither, and asserts neither.)

Theorem 27.16 (What lights the cell (24, 8)). *Let a group G permute the blocks of an $S(t, k, n)$, transitively. Then the stabilizer of a block B satisfies $[G : G_B] \binom{k}{t} = \binom{n}{t}$. For $S(5, 8, 24)$ this gives $[G : G_B] = 759$, whereas the Pascal entry of that cell is $\binom{24}{8} = 735\,471$.*

Proof. The orbit of B is all of $\mathcal{B}$, so the index is $|\mathcal{B}|$, and Theorem 27.14 converts that into the displayed identity. $\square$

Remark 27.17 (What is and is not claimed here). The Mathieu groups themselves are not constructed in this development, and neither is the Golay code or the Steiner system $S(5, 8, 24)$. Everything above is either unconditional arithmetic (Theorems 27.2–27.12, Corollary 27.11) or conditional on the existence of the design (Theorem 27.14 onwards): *if* an $S(5, 8, 24)$ exists it has 759 blocks, and *no* $S(5, 8, 25)$ exists. The degenerate system with $t = k$ is constructed, so the definition is not vacuous. Maximality of the highlighted subgroups — the actual selection rule of the original diagram — is not formalised.

28 What is true, and what is not

Summarising the formal content:

- **True.** $\{3, 6, 9\}$ is exactly the set of digital roots of multiples of 3; it is closed under addition, absorbs multiplication, and generates $3\mathbb{Z}$. The doubling cycle 1, 2, 4, 8, 7, 5 has period six and never meets it. Every prime beyond 3 avoids it.
- **True, and geometric.** On the same circle, the inscribed hexagon has half-perimeter exactly 3, so $3 < \pi$ is a theorem about the first member of the trinity (Theorem 18.5); and the hexagon spun about a long diagonal sweeps a solid of volume exactly π (Theorem 18.10). Neither is numerology: both are the value $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$ in disguise. The nine-gon, by contrast, is a trisection and is not on Archimedes’ ladder at all (Remark 18.7).
- **True but weaker than advertised.** The trinity is redundant: $\langle 3 \rangle = \langle 3, 6, 9 \rangle$ and even $\langle 6, 9 \rangle = 3\mathbb{Z}$.
- **The actual mechanism.** Everything above is the arithmetic of $\mathbb{Z}/9$ and its subgroup of index 3, transported by casting out nines. There is no additional structure to be found in 3, 6 and 9 beyond this. Concretely, the whole “vortex” picture is the orbit decomposition $9 = 1 + 2 + 6$ of $\mathbb{Z}/9$ under doubling (Theorem 10.1).
- **Nothing is special about ten.** Every trinity statement holds verbatim in base b for every divisor d of $b - 1$ (Theorem 9.3); ten is distinguished only by being the least base whose three-element trinity detects an odd prime (Theorem 9.5).
- **True, and for a reason.** The Fibonacci digital roots really do cycle with minimal period 24, and the trinity really does appear at every fourth index — because $3 = F_4$ and $\gcd(F_m, F_n) = F_{\gcd(m, n)}$ (Theorems 11.1 and 11.2). The pattern is real; the explanation is Fibonacci divisibility, not numerology.
- **The respectable analogue.** The genuine “small primes suffice” statements are the density $2/3$ of the $\{2, 3\}$ -sieve and the determination of 3-smooth Hecke eigenvalues by a_2, a_3 ; in their general forms, the Legendre density $1 - \prod_{p \in S} (1 - 1/p)$ (Theorem 5.3) and multiplicative closure over an arbitrary set of primes (Theorem 7.1).

- **The board, and why only some cells are lit.** On the mod- p Pascal board a cell is lit exactly when the base- p digits of k are dominated by those of n (Lucas, Theorem 27.2), equivalently exactly when adding k and $n - k$ in base p carries nowhere (Kummer, Theorem 27.4); for $p = 2$ this is the bitwise condition $k \& n = k$ and the picture is the Sierpiński gasket (Theorem 27.3). The row counts are the digit product $\prod (d_i + 1)$ (Theorem 27.5), so on the mod-3 board row 24 has exactly nine lit cells, at exactly the multiples of three (Proposition 27.7).
- **The group layer, and its obstruction.** A group transitive on k -subsets of n points has k -set stabilizers of index $\binom{n}{k}$ (Theorem 27.9), so $\binom{n}{k}$ must divide $|G|$ (Theorem 27.10). For the five Mathieu orders that divisibility holds only in a narrow band at the two ends of the row — for $|M_{24}|$ exactly at $k \leq 5$ and $k \geq 19$ — which is why the middles of those rows are dark, and dark for every group of that order, sporadic or not (Corollary 27.11). Conversely every cell is fillable: S_n is k -homogeneous for every k (Theorem 27.12).
- **The synthetic layer, and where it stops.** Keeping the geometry and dropping the group, a Steiner system $S(t, k, n)$ satisfies $|\mathcal{B}| \binom{k}{t} = \binom{n}{t}$ (Theorem 27.14); hence an $S(5, 8, 24)$ has exactly 759 octads, a block stabilizer has index 759 rather than $\binom{24}{8}$ (Theorem 27.16), and there is no $S(5, 8, 25)$ at all (Corollary 27.15). **Not proved here:** the existence of the Mathieu groups, of the Golay code or of $S(5, 8, 24)$, and the maximality of any subgroup.
- **Drawn, and drawn honestly.** The concrete eigenform Δ is exhibited with its coefficients computed in the kernel (Proposition 12.1), the five words checked on them (Proposition 12.2), and the function itself plotted (Figures 2–4). All three of the picture’s visible symmetries are theorems here: period one and decay at the cusp for the plotted truncation (Proposition 12.6), and $\Delta(-1/z) = z^{12}\Delta(z)$ for the true infinite product (Theorem 14.4).
- **Why it stands up straight.** The vertical alignment of the portrait over the cusps and half-periods is a theorem, and needs only two hypotheses — real coefficients and period one (Section 13). Those give a mirror in every line $x = k/2$ (Proposition 13.1), which are exactly the lines where the nome is real (Proposition 13.2); there the function is real, so its phase is 0 or π and can change only at a zero (Theorem 13.3), the horizontal derivative of the magnitude vanishes, and the contours cross at a right angle (Theorem 13.4). For Δ high up the two colours are identified: 0 on $x = 0, \pm 1$ and π on $x = \pm \frac{1}{2}$ (Proposition 13.5). This says nothing about the arcs $|z| = 1$; those need a different argument, and get one in the next item.
- **And why it curls.** The reflection across the arcs $|z| = 1$ is genuine modularity and is proved for the honest infinite product $\Delta = \eta^{24}$, not for a truncation: the logarithmic derivative of Δ is $2\pi i E_2$, the defect in the S -transformation of E_2 is exactly the logarithmic derivative of z^{12} , and the two sides therefore differ by a constant, which the fixed point $z = i$ pins to 1 (Theorem 14.4). With period one this gives invariance under all of $SL_2(\mathbb{Z})$ (Theorem 14.7), the mirror $|\Delta(1/\bar{z})| = |z|^{12}|\Delta(z)|$ (Theorem 14.9), and on the arc itself a phase lock $\overline{\Delta(z)} = z^{12}\Delta(z)$ (Corollary 14.10). Two ingredients are imported rather than proved: Mathlib’s Dedekind eta function with $L\eta = \frac{\pi i}{12}E_2$, and its S -transformation of G_2 .
- **One graph, five heights.** The image of Δ under each of the five words is, on the range checked, Δ multiplied by a constant (Proposition 15.1, Corollary 15.2); a Hecke word there-

fore shifts a logarithmic profile and, if its multiplier is negative, turns the phase portrait by half a turn, and does nothing else (Proposition 15.3). All five multipliers are divisible by three, so every plotted coefficient of every word lies in the trinity (Proposition 15.4). Beyond the checked range the plates display Mordell's theorem, which is not proved here.

- **Comparing two words.** Term by term, $T_2\Delta$ and $T_3\Delta$ are proportional, so their product, difference, quotient, power and remainder are all determined by $\tau(2)$, $\tau(3)$ and τ (Proposition 16.1); the product recovers the multiplier $\tau(6)$ of the word T_2T_3 , and the quotient is a constant carrying no further information. Three divides the product, the difference and the remainder, so those stay in the trinity, and the integer quotient does not — division is exactly the operation that throws the common factor away (Proposition 16.3).
- **Six comparisons, six pictures.** Each panel of the comparison becomes a function on the upper half plane (Definition 17.1), and the six pictures are of three kinds (Propositions 17.2 and 17.3): the two sequences and their difference are Δ rescaled; the product is the Hadamard square of Δ , quadratic and not a rescaling; the quotient and the remainder read the sign of τ , which no rescaling can, and the quotient keeps nothing else. The power has no picture at all, only a logarithm.
- **The polygons, turning on the form.** Setting the inscribed polygons spinning on the portrait is not decoration only: the ring is rigid (Proposition 19.1), its loop is exactly $2\pi/n$ (Theorem 19.2), the modular involution $z \mapsto -1/z$ reverses the frame and so fixes the ring at exactly two instants of each loop (Theorem 19.3), and at every vertex in the upper half plane the phase lock of the arc applies, with the single factor e^{12it} for the twelve-gon (Theorem 19.4). What is *not* claimed is anything about the raster behind them: the colours come from a truncated product, and no theorem here reads them.
- **Cut up, and read.** The dissection of the portrait is exact where it is claimed to be exact: the ternary cells partition the plane, each carries 9^m organised parts m levels down (Theorem 20.3), and the readings of Δ inside them inherit period one and the mirror in $x = 0$ (Proposition 20.5), which is why the character normalised from them is a palindrome (Theorem 20.6). Both Frobeniuses then act: the mirror is a Frobenius isometry fixing the character (Theorem 20.7), the three-shade quantisation is fixed pointwise by $x \mapsto x^3$ on $\mathbb{Z}/3$ (Theorem 20.8), and the trace $\tau(p)$ is $p^2 + p^9$ modulo 27, hence $p(p + 1)$ modulo 3, on the certified range (Theorem 20.9). What is *not* claimed is anything about the raster: the OpenCV contours locate shapes in a picture of a truncation, and no theorem reads a pixel of it.
- **Every frame, cut into triangles.** The two pictures combine, and the combination is exact where it is claimed to be exact. Euclid's cut divides a triangle into nine similar parts that cover it, each of a third of the size, and d cuts give 9^d parts (Theorem 21.3); applied to the fan of the spinning n -gon this cuts each frame into $n \cdot 9^d$ triangles. The frames are one frame turned: rotating by $2\pi/n$ carries the parts of the frame at time t onto the parts of the frame at time $t + 2\pi/n$ (Theorem 21.5), so the multiset of readings, and hence the sorted palette, comes back exactly after one loop (Theorem 21.7); the mirror $z \mapsto -\bar{z}$ leaves every reading unchanged. That the polygons approximate the curve is also a theorem and not a picture: a chord is at most its arc, so every point of the circle is within π/n of a vertex of the frame, and $\pi/n \rightarrow 0$ (Proposition 21.8), alongside the perimeters

rising to 2π (Theorem 18.4). What is *not* claimed is the colouring: parts whose centroid leaves the upper half plane are not read at all, and the colour scale is a clipped logarithm chosen for legibility, with no theorem behind it.

- **The whole plane, and its character table.** The fan repeated on the ladder of circles of radii 3^j and cut Euclid's way to depth d really does fill the plane: the fan of the unit n -gon covers the disc of its inradius $\cos \frac{\pi}{n} \geq \frac{1}{2}$, so ring j covers the disc of radius $3^j/2$, and every point lies in a tile of every depth (Theorem 22.3). Summing $|\Delta|$ over the 9^d parts of a sector gives a matrix of palette traces, rings by sectors, and expanding each ring in the character table of the sector group C_n loses nothing (inversion), distorts no length (Plancherel, up to the factor $\sqrt{n}$ of Frobenius' orthogonality relations, Theorems 22.8 and 22.11), turns the spin into a diagonal matrix $\text{diag}(1, \zeta_n, \dots, \zeta_n^{n-1})$ (Theorem 22.12), and pins each coefficient of the still filling to a line through the origin (Theorem 22.13). The trinity reappears as the sub-table: rows 3, 6, 9 of the table of C_9 are the character table of C_3 (Proposition 22.9). What is *not* claimed is the numerical content of the plate: the coefficients quoted in Remark 22.14 are a computation with a truncated Δ , and no theorem here reads them.
- **Not proved here.** Anything analytic. The sieve results are exact identities for a *fixed* finite set of primes and say nothing when S grows with the window; the Hecke results are the algebraic content of the recursions and give no effective bound of Sturm type (Remark 7.6).
- **Zeta, and moonshine.** In ζ the trinity is exact but thin: the trinity indices carry $3^{-s}\zeta(s)$ and their complement carries ζ with the Euler factor at three removed (Theorem 23.2), and that single factor is all there is, because the Dirichlet coefficients are constant. In j the trinity is loud: every third coefficient is divisible by 3^5 , and j shares its digital roots exactly with the McKay–Thompson series of class 3A and not at all with that of class 2A (Theorem 23.6) — which is the statement that the digital root is a three-adic instrument, nothing more. Those coefficient statements are kernel computations on the range computed, not proofs for all n (Remark 23.7).
- **Ten plates, one tiling.** The same ring ladder of Euclid triangles carries Δ , j , the three McKay–Thompson hauptmoduls, ζ , ξ and three Dirichlet L -functions, each in its own standalone plate and animation (Figure 35). What is proved there is the geometry — the sizes, the covering, the mirror, the loop of the spin and the orthogonality of the characters; the values are numerics, checked against the certified expansions and the classical special values before anything is drawn, and no colour is a theorem (Remark 23.8).
- **Why the plates coincide.** That the plates of different functions look the same is not an artefact and not a coincidence: above an explicit height $Y(C, k, \delta)$ the leading term of a q -series with polynomially bounded coefficients dominates, so any two normalised series paint a height-separated sample with exactly the same ranks, and any two hauptmodul-shaped series paint it with exactly the reverse ranks (Theorem 24.3, Corollary 24.4). The gallery of 109 LMFDB newforms (Section 24) is the experiment that prompted the theorem, and it also shows its limits: low down, and after dividing the universal factor out, the plates separate again and each form is itself. What is proved is the statement about series; the polynomial bound for these particular newforms is Deligne's theorem, imported, and the coefficients are database values, checked but not derived (Remark 24.5).

- **Where to look, and the tree that comes out.** The universality theorem also says that a view high up shows nothing, so the natural question is which window shows most. A search over 92 windows finds, for every newform of the cache, the window in which its palette is furthest from the universal one; at the common best view the palette distances give a table on which the single-linkage chain distance is the largest ultrametric below it, its groups at all thresholds are nested or disjoint — a tree — and any invariant no short step can change is constant on every group (Theorems 25.6, 25.7, 25.8). Measured on the sample, forms of different weight are never closer than 39 tiles and forms differing in complex multiplication never closer than 44, so the lower ranks of the tree are weight- and CM-pure: the taxonomy of Section 25.
- **The same sample, read arithmetically.** Sorting the 109 newforms by the digital roots of their eigenvalues instead of by their pictures gives a second tree, whose three levels are the residues mod 3, the occupied circuit vertices, and the residues mod 9 (Section 26). That it is a tree is proved (Theorem 26.5), and so is the reason its branches are thin: the Eisenstein congruence $a_p \equiv 1 + p^{k-1}$ forbids the residue 1 mod 3, and mod 9 it confines the spectrum to $\{0, 2\}$ or to $\{0, 2, 3, 5, 6, 8\}$ according to the weight mod 6 (Theorems 26.6, 26.7). What the sample supplies — the two layer-3 classes, the 7 layer-9 classes and their sizes, the 11 forms of the thin branch, the trinity carrying at least a third of every form’s mass, the mere 51 distinct spectra — is checked by the Lean kernel on the cached residues, but only for the primes $p \leq 997$ of the sample (Remark 26.9).

Tesla, then, was right in the only sense in which the claim can be made precise: knowing 3, 6 and 9 is knowing divisibility by three, which is knowing the residue mod 9 up to the unit orbit. That is a key — to one ninth of the universe.

A Formalisation

All statements are proved in Lean 4 against Mathlib, in the directory `RequestProject/`:

- `Tesla369.lean` — digital roots and the trinity (Sections 2 and 3);
- `Sieve23.lean` — the sieve at 2 and 3 (Section 4);
- `SieveGeneral.lean` — the general finite sieve (Section 5);
- `Hecke23.lean` — Hecke systems and the 3-smooth part (Section 6);
- `HeckeGeneral.lean` — multiplicative closure (Section 7);
- `HeckeWords.lean` — the five Hecke words (Section 8);
- `BaseGeneral.lean` — the trinity of an arbitrary base and the vortex orbits (Sections 9 and 10);
- `FibVortex.lean` — the Fibonacci digital roots (Section 11);
- `DeltaGraph.lean` — the discriminant form, its coefficients and the drawn function (Section 12);

- **DeltaWordGraphs.lean** — the Hecke image of Δ under each of the five words, and what a multiplier does to a graph (Section 15);
- **T2vsT3.lean** — the pointwise comparison of $T_2\Delta$ with $T_3\Delta$ (Section 16);
- **T2vsT3Delta.lean** — the delta diagram of each of those six comparisons (Section 17);
- **VerticalWalls.lean** — the mirror symmetries of the portrait and the vertical walls over $x = 0, \pm\frac{1}{2}, \pm 1$ (Section 13);
- **DeltaModular.lean** — $\Delta = \eta^{24}$, the circular symmetry $\Delta(-1/z) = z^{12}\Delta(z)$, full modularity, and the reflection in the unit circle (Section 14);
- **Archimedes369.lean** — the inscribed and circumscribed polygons, the sandwich, the doubling ladder and the first rungs (Section 18);
- **ArchimedesNinetySix.lean** — interval propagation along the ladder, giving $\frac{223}{71} < \pi < \frac{22}{7}$ (Theorem 18.6);
- **ArchimedesSpin.lean** — the same figures rotating: the polar profile, the swept annulus, the solids of revolution and the vertices on the modular arc (Section 18);
- **ArchimedesDelta.lean** — the polygons spinning on the portrait of Δ : the loop of the animation, the nested rings, the rigid edge, the S -symmetric instants and the phase lock along the ring (Section 19);
- **DeltaDissection.lean** — Euclid’s ternary dissection of the plane, the readings of Δ inside the parts, and the character they normalise to (Section 20);
- **DeltaFrobenius.lean** — the Frobenius norm on characters, the Frobenius endomorphism on the three-shade glyph, and the trace of Frobenius $\tau(p)$ (Section 20);
- **DeltaTriangles.lean** — the fan of triangles of each frame, Euclid’s ternary cut of a triangle into nine, the parts d levels down, the reading of Δ on them, the sorted palette of a frame, and how the polygons approximate the arc (Section 21);
- **EuclidTiling.lean** — the ring ladder, the tiles of the filling and the proof that Euclid’s triangles fill the plane (Section 22);
- **Zeta369.lean** — the trinity indices inside the Riemann zeta function, the zeta values and the constants of the Eisenstein series (Section 23);
- **Moonshine369.lean** — the coefficients of Klein’s j and of the McKay–Thompson series of classes 1A, 2A and 3A, and their digital roots (Section 23);
- **PaletteCharacterTable.lean** — the reading of Δ on a tile anywhere in the plane, the palette traces of the sectors, the character table of C_n with Frobenius’ orthogonality relations, and the palette read in it (Section 22);
- **PaletteUniversality.lean** — the leading-term estimate for a q -series, the reading by height, and the universality of the palette ranks for normalised and for hauptmodul-shaped series (Section 24);

- `UltrametricTaxonomy.lean` — the chain distance of a table of distances on a finite set: the subdominant ultrametric, the laminar family of its groups (the tree), its root and leaves, and the refinement of an invariant (Section 25);
- `PaletteTaxonomy.lean` — views, the palette distance, the separation score and the existence of a best view, the blindness of a deep view, and the taxonomy tree of a family of forms at a view (Section 25);
- `Spectrum369.lean` — the enneagram of the digital roots: the trinity and the doubling circuit, the nine vertices and their vanishing sums, spectra and their three layers, the Eisenstein congruences that force the thin branches, and the tree the layers carry (Section 26);
- `LmfdbSpectrum369Data.lean` and `LmfdbSpectrum369.lean` — the residues of the 109 surveyed newforms, and the kernel-checked claims of the survey: the layer-3 dichotomy and its congruence, the seven layer-9 classes and their sizes, the thin branch and its labels, the mass on the trinity and the 51 distinct spectra (Section 26).

The spectral survey of Section 26 is computed from the same cache by `paper/spectrum369_survey.py`, which also writes the tree (`paper/lmfdb_spectrum369_tree.txt`), the figures and the Lean data file; it refuses to write anything unless its own checks pass — that the counts add up to the sample size, that the triangle and circuit masses partition it, that layer 9 determines layer 3, and that the layer-3 dichotomy agrees form by form with the Eisenstein congruence mod 3.

The best views and the taxonomy of Section 25 are computed by `paper/palette_taxonomy.py` and drawn by `paper/make_taxonomy_figs.py`; the sample they run over is chosen from the database by `paper/select_lmfdb.py` and cached by `paper/fetch_lmfdb.py`.

The plates of the ten functions are generated by `paper/make_modular_palettes.py` from the evaluators of `paper/modular_functions.py`; both are pure standard-library Python, and both refuse to run unless their recomputations agree with the Lean-certified tables. The LMFDB gallery of Section 24 is generated the same way: `paper/fetch_lmfdb.py` pulls the newform data through the public LMFDB API into the cache `paper/lmfdb_newforms.json`, `paper/lmfdb_forms.py` rebuilds the coefficients from the Hecke eigenvalues and evaluates the forms with a truncation controlled by Deligne’s bound, and `paper/make_lmfdb_gallery.py` draws the plates, the traces and the tables. Each of the three runs its own self-checks — normalisation, the Hecke recursions, Deligne’s bound, the τ table certified in Lean, the eta product for 11.2.a.a, the truncation error, and the verbatim instance of Corollary 24.4 on a separated sample — and stops rather than draw if one fails.

Verification status. The claim of the abstract is a checked one, not a target. The project builds with `lake build` against the pinned Mathlib revision (Lean toolchain v4.28.0); the sources contain no occurrence of `sorry`; and the axiom check (`#print axioms`) on the theorems of the table reports nothing beyond `propext`, `Classical.choice` and `Quot.sound`. Reproducing this needs only `lake build` in the project root.

Statement	Lean name
Definition 2.1	<code>Tesla369.digitalRoot</code>
Lemma 2.2	<code>Tesla369.digitsSum_lt_self</code>
Theorem 2.3	<code>Tesla369.digitalRoot_digitsSum,</code> <code>Tesla369.repeatedDigitSum_eq_digitalRoot</code>
Theorem 3.1	<code>Tesla369.digitalRoot_mem_trinity_iff</code>
Closure/absorption	<code>Tesla369.trinity_add, Tesla369.trinity_mul</code>
Theorem 3.3	<code>Tesla369.trinity_generates_nat, Tesla369.trinity_generates</code>
Theorem 3.4	<code>Tesla369.three_generates, Tesla369.six_nine_generates</code>
Powers of three	<code>Tesla369.digitalRoot_three_pow</code>
Theorem 3.7	<code>Tesla369.digitalRoot_two_pow_period_six,</code> <code>Tesla369.digitalRoot_two_pow_values,</code> <code>Tesla369.digitalRoot_two_pow_notMem_trinity</code>
Remark 3.8	<code>Tesla369.digitalRoot_two_pow_period_six_minimal</code>
Subgroup of $\mathbb{Z}/9$	<code>Tesla369.trinity_zmod, Tesla369.trinity_residues</code>
Theorem 4.1	<code>Sieve23.card_sieved_range</code>
Primes mod 6	<code>Sieve23.prime_mod_six</code>
Primes avoid the trinity	<code>Sieve23.prime_digitalRoot</code>
Lemma 5.1	<code>SieveGeneral.card_filter_range_of_periodic</code>
Lemma 5.2	<code>SieveGeneral.not_sieved_iff_coprime,</code> <code>SieveGeneral.card_survivors_period, SieveGeneral.totient_modulus</code>
Theorem 5.3	<code>SieveGeneral.card_survivors_range,</code> <code>SieveGeneral.card_sieved_range, SieveGeneral.density_sieved</code>
Specialisation to $\{2, 3\}$	<code>SieveGeneral.two_three</code>
Hecke system	<code>Hecke23.HeckeSystem</code>
Non-vacuity	<code>Hecke23.heckeSystem_cardDivisors</code>
Theorem 6.3	<code>Hecke23.eq_on_smooth, Hecke23.eq_on_Smooth3</code>
$a(4), a(6), a(9)$	<code>Hecke23.a_four, Hecke23.a_six, Hecke23.a_nine</code>
S -smooth numbers	<code>Hecke23.Smooth, Hecke23.Smooth3_eq</code>
Theorem 7.1	<code>Hecke23.eq_of_primeFactors_subset,</code> <code>Hecke23.eq_on_Smooth</code>
Corollary 7.2	<code>Hecke23.eq_of_eq_on_primes</code>
$a(0)$ unconstrained	<code>Hecke23.a_zero_undetermined</code>
Corollary 7.4	<code>Hecke23.eq_on_le_of_eq_on_primes_le</code>
Theorem 7.5	<code>Hecke23.exists_separating_bound</code>

Section 8 lives in the namespace `HeckeWords`, whose prefix is omitted in the following table.

Statement	Lean name
Word multiplier	<code>wordEigenvalue</code>
Structure constants (1)	<code>basisMul</code> , <code>hmul</code>
Proposition 8.1	<code>hmul_two_three</code> , <code>eigen_word_two_three</code>
Proposition 8.2	<code>hmul_two_three_two</code> , <code>eigen_word_two_three_two</code> , <code>a_twelve</code>
Proposition 8.3	<code>hmul_three_three</code> , <code>hmul_three_three_three</code> , <code>eigen_word_three_three_three</code> , <code>a_twentyseven</code>
Consistency of the two sides	<code>evalT_word_two_three</code> , <code>evalT_word_two_three_two</code> , <code>evalT_word_three_three_three</code>
Corollary 8.4	<code>eigen_polynomial_two_three</code> , <code>eigen_polynomial_two_three_two</code> , <code>eigen_polynomial_three_three_three</code>
Example 8.5	<code>delta_values</code>

Statement	Lean name
Definition 9.1	<code>Tesla369.digitalRootBase</code> , <code>Tesla369.trinityBase</code>
Theorem 9.2	<code>Tesla369.digitalRootBase_digitsSum</code>
Theorem 9.3	<code>Tesla369.dvd_digitalRootBase_iff</code> , <code>Tesla369.mem_trinityBase_iff</code>
Proposition 9.4	<code>Tesla369.card_trinityBase</code> , <code>Tesla369.exists_trinity_card_three_iff</code>
Theorem 9.5	<code>Tesla369.base_ten_least_odd_prime_trinity</code>
Proposition 9.7	<code>Tesla369.digitalRootBase_add</code> , <code>Tesla369.digitalRootBase_mul</code> , <code>Tesla369.digitalRoot_add</code> , <code>Tesla369.digitalRoot_mul</code> , <code>Tesla369.digitalRoot_cast</code>
Theorem 9.6	<code>Tesla369.digitalRootBase_pow_period</code> , <code>Tesla369.digitalRootBase_pow_notMem_trinityBase</code>
Theorem 10.1	<code>Tesla369.vortex_orbits_partition</code> , <code>Tesla369.vortex_orbit_cards</code> , <code>Tesla369.vortexOrbit_one_eq_units</code>
Theorem 11.1	<code>Tesla369.fib_add_24_modEq</code> , <code>Tesla369.digitalRoot_fib_period_24</code> , <code>Tesla369.digitalRoot_fib_cycle</code> , <code>Tesla369.digitalRoot_fib_period_24_minimal</code>
Theorem 11.2	<code>Tesla369.three_dvd_fib_iff</code> , <code>Tesla369.digitalRoot_fib_mem_trinity_iff</code>

Section 12 lives in the namespace `DeltaGraph`, whose prefix is omitted in the following table.

Statement	Lean name
The form (2)	<code>deltaCoeffs</code> , <code>tau</code>
Proposition 12.1	<code>tau_table</code>
Proposition 12.2	<code>tau_four</code> , <code>tau_six</code> , <code>tau_nine</code> , <code>tau_twelve</code> , <code>tau_twentyseven</code> , <code>tau_word_values</code>
Proposition 12.4	<code>tau_ramanujan_bound</code>
Proposition 12.5	<code>tau_congr_mod_27</code> , <code>three_dvd_tau_iff_three_dvd_sigma</code> , <code>digitalRoot_natAbs_mem_trinity_iff</code> , <code>digitalRoot_tau_table</code>
Proposition 12.6	<code>qParam_add_one</code> , <code>deltaApprox_periodic</code> , <code>abs_qParam</code> , <code>abs_qParam_lt_one</code>

Section 15 lives in the namespace `DeltaWordGraphs`, whose prefix is omitted in the following table.

Statement	Lean name
The certified table	<code>tauTable</code> , <code>tauTable_eq_deltaCoeffs</code> , <code>tauT</code> , <code>tauT_table</code> , <code>tauT_eq_tau</code>
Hecke action (4)	<code>heckeCoeff</code> , <code>wordT2</code> , <code>wordT3</code> , <code>wordT2T3</code> , <code>wordT2T3T2</code> , <code>wordT3T3T3</code>
Proposition 15.1	<code>wordT2_eq</code> , <code>wordT3_eq</code> , <code>wordT2T3_eq</code> , <code>wordT2T3T2_eq</code> , <code>wordT3T3T3_eq</code> , <code>word_multipliers</code>
Corollary 15.2	<code>heckeApprox</code> , <code>heckeApprox_eq_smul</code> , <code>heckeApprox_wordT2</code> , <code>heckeApprox_wordT3</code> , <code>heckeApprox_wordT2T3</code> , <code>heckeApprox_wordT2T3T2</code> , <code>heckeApprox_wordT3T3T3</code>
Proposition 15.3	<code>heckeApprox_periodic</code> , <code>heckeApprox_norm</code> , <code>log_norm_heckeApprox</code> , <code>heckeApprox_neg_phase</code> , <code>word_abs_multipliers_strictMono</code>
Proposition 15.4	<code>word_coeffs_three_dvd</code> , <code>coeff_mem_trinity</code> , <code>word_coeffs_mem_trinity</code>

Section 16 lives in the namespace `T2vsT3`, whose prefix is omitted in the following table.

Statement	Lean name
The two sequences	<code>t2_eq</code> , <code>t3_eq</code> , <code>tauT_ne_zero</code>
The five comparisons	<code>prodT</code> , <code>diffT</code> , <code>ratioT</code> , <code>quotT</code> , <code>powT</code> , <code>modT</code>
Proposition 16.1(1)	<code>prodT_eq</code> , <code>prodT_eq_tau_six</code> , <code>prodT_neg</code>
Proposition 16.1(2)	<code>diffT_eq</code> , <code>diffT_ne_zero</code>
Proposition 16.1(3)	<code>ratioT_eq</code> , <code>abs_ratio_21_2</code> , <code>quotT_eq</code>
Proposition 16.1(4)	<code>powT_pos</code> , <code>three_pow_dvd_powT</code> , <code>powT_one</code>
Proposition 16.1(5)	<code>modT_eq</code> , <code>modT_nonneg</code> , <code>modT_lt</code> , <code>modT_ne_zero</code>
Proposition 16.3	<code>three_dvd_prodT</code> , <code>three_dvd_diffT</code> , <code>three_dvd_modT</code> , <code>prodT_mem_trinity</code> , <code>diffT_mem_trinity</code> , <code>modT_mem_trinity</code> , <code>quotT_notMem_trinity</code>
The tabulated values	<code>comparison_table</code>

Section 17 lives in the namespace `T2vsT3Delta`, whose prefix is omitted in the following table.

Statement	Lean name
Definition 17.1	<code>cApprox</code> , <code>cApprox_int</code> , <code>cApprox_tau</code> , <code>cApprox_periodic</code> , <code>cApprox_congr</code> , <code>cApprox_smul</code> , <code>cApprox_add</code>
Proposition 17.2(1)	<code>panelA_T2</code> , <code>panelA_T3</code> , <code>panelA_antiparallel</code>
Proposition 17.2(2)	<code>tauSqT</code> , <code>panelB_eq</code> , <code>panelB_coeffs_neg</code>
Proposition 17.2(3)	<code>diffT_smul_tau</code> , <code>panelC_eq</code> , <code>panelC_sub</code>
Proposition 17.2(4)	<code>posMaskT</code> , <code>panelD_rat</code> , <code>panelD_int</code>
Proposition 17.2(5)	<code>powLogCoeff</code> , <code>log_powT</code> , <code>powLogCoeff_eq</code> , <code>powLogCoeff_pos</code>
Proposition 17.2(6)	<code>tauPostT</code> , <code>tauNegT</code> , <code>tauPostT_add_tauNegT</code> , <code>modT_split</code> , <code>panelF_halves</code> , <code>panelF_eq</code>
Proposition 17.3	<code>prodT_not_smul_tau</code> , <code>quotT_not_smul_tau</code> , <code>modT_not_smul_tau</code> , <code>panels_smul_or_not</code>
Equation (3)	<code>VerticalWalls.F</code> , <code>VerticalWalls.Fr</code> , <code>VerticalWalls.wallPt</code> , <code>VerticalWalls.deltaApprox_eq_F</code>
Proposition 13.1	<code>qParam_neg_conj</code> , <code>F_neg_conj</code> , <code>F_int_add</code> , <code>F_mirror</code> , <code>mirror_fixed_iff</code>
Proposition 13.2	<code>qParam_im_eq_zero_iff</code> , <code>qParam_wall</code>
Theorem 13.3	<code>F_wall_ofReal</code> , <code>F_wall_im</code> , <code>arg_F_wall</code> , <code>arg_F_wall_eq_zero_iff</code> , <code>arg_F_wall_eq_pi_iff</code> , <code>continuous_F_wall_re</code> , <code>phase_locked_of_ne_zero</code>
Theorem 13.4	<code>F_mirror_slice</code> , <code>norm_F_mirror_slice</code> , <code>sqAbs_mirror</code> , <code>deriv_eq_zero_of_mirror</code> , <code>deriv_sqAbs_wall</code> , <code>deriv_norm_wall</code> , <code>fderiv_sqAbs2_wall_horizontal</code> , <code>grad_sqAbs2_wall_vertical</code> , <code>contour_tangent_horizontal</code> , <code>contour_perp_wall</code>
Proposition 13.5	<code>abs_Fr_sub_lead_le</code> , <code>abs_tau_le</code> , <code>exp_le_of_five_le</code> , <code>deltaApprox_wall_pos</code> , <code>deltaApprox_wall_neg</code> , <code>arg_deltaApprox_wall_even</code> , <code>arg_deltaApprox_wall_odd</code>

Section 14 lives in the namespace `DeltaModular`, whose prefix is omitted in the following table.

Statement	Lean name
Definition 14.1	<code>Delta</code> , <code>Delta_ne_zero</code> , <code>differentiableAt_Delta</code> , <code>differentiableOn_Delta</code>
The upper half plane	<code>neg_inv_mem</code> , <code>ne_zero_of_mem</code> , <code>isPreconnected_upperHalfPlaneSet</code> , <code>coe_S_smul</code>
Proposition 14.2	<code>logDeriv_Delta</code>
Proposition 14.3	<code>G2_S</code> , <code>E2_eq_G2</code> , <code>E2_S_transform</code>
The two logarithmic derivatives	<code>logDeriv_Delta_comp</code> , <code>logDeriv_weight</code>
Theorem 14.4	<code>Delta_neg_inv</code> , <code>Delta_neg_inv'</code>
Proposition 14.6	<code>qParam_one_add_one</code> , <code>eta_add_one</code> , <code>Delta_add_one</code>
Theorem 14.7	<code>DeltaH</code> , <code>DeltaH_S</code> , <code>DeltaH_T</code> , <code>DeltaH_slash</code>
Proposition 14.8	<code>multipliable_eta</code> , <code>qParam_conj</code> , <code>eta_neg_conj</code> , <code>Delta_neg_conj</code>
Theorem 14.9	<code>Delta_inv_conj</code> , <code>norm_Delta_inv_conj</code>
Corollary 14.10	<code>Delta_arc</code> , <code>Delta_arc_phase</code>

Section 18 lives in the namespace `Archimedes369`, whose prefix is omitted in the following table.

Statement	Lean name
Definition 18.1	<code>lower</code> , <code>upper</code> , <code>inArea</code>
Theorem 18.2	<code>lower_lt_pi</code> , <code>pi_lt_upper</code> , <code>inArea_eq</code> , <code>inArea_lt_pi</code>
Theorem 18.3	<code>upper_two_mul</code> , <code>lower_two_mul</code> , <code>lower_lt_lower_two_mul</code> , <code>upper_two_mul_lt_upper</code>
Theorem 18.4	<code>tendsto_lower</code> , <code>tendsto_upper</code> , <code>tendsto_inArea</code>
Theorem 18.5	<code>lower_three</code> , <code>upper_three</code> , <code>lower_four</code> , <code>upper_four</code> , <code>lower_six</code> , <code>upper_six</code> , <code>lower_twelve</code> , <code>upper_twelve</code> , <code>hexagon_bounds</code> , <code>twelve_gon_bounds</code>
Theorem 18.6	<code>harm_mono_lower</code> , <code>harm_mono_upper</code> , <code>bounds_two_mul</code> , <code>bounds_six</code> , <code>bounds_twelve</code> , <code>bounds_twentyfour</code> , <code>bounds_fortyeight</code> , <code>bounds_ninetysix</code> , <code>archimedes_bounds</code>
Remark 18.7	<code>sin_pi_div_nine_cubic</code> , <code>sin_pi_div_nine_bounds</code> , <code>lower_nine_bounds</code>
The sawtooth	<code>wrap</code> , <code>radius</code> , <code>abs_wrap_le</code> , <code>wrap_rot</code>
Theorem 18.8	<code>radius_rot</code> , <code>radius_mul_cos_wrap</code> , <code>le_radius</code> , <code>radius_le_one</code> , <code>radius_edge_center</code> , <code>radius_vertex</code> , <code>isGLB_radius</code> , <code>isLUB_radius</code> , <code>sweptArea_eq</code> , <code>tendsto_sweptArea</code>
Lemma 18.9	<code>frustum_volume</code>
Theorem 18.10	<code>revVolume</code> , <code>revVolume_sum</code> , <code>revVolume_eq</code> , <code>revVolume_two</code> , <code>revVolume_three</code> , <code>revVolume_six</code> , <code>tendsto_revVolume</code>
Theorem 18.11	<code>neg_inv_circle</code> , <code>vertex</code> , <code>vertex_zero</code> , <code>norm_vertex</code> , <code>S_vertex</code> , <code>vertex_twelve_one</code> , <code>vertex_twelve_neg_one</code>
The spinning ring	<code>spinAngle</code> , <code>spinVertex</code> , <code>spinVertexSet</code> , <code>spinVertex_zero</code>
Proposition 19.1	<code>norm_spinVertex</code> , <code>spinVertexSet_subset_of_dvd</code> , <code>norm_exp_sub_one</code> , <code>chord_eq</code> , <code>perimeter_eq</code> , <code>hexagon_chord</code>
Theorem 19.2	<code>spinVertex_eq_iff</code> , <code>spinVertex_shift</code> , <code>spinVertexSet_rot</code> , <code>spinVertexSet_eq_iff</code>
Theorem 19.3	<code>S_spinVertex</code> , <code>S_image_spinVertexSet</code> , <code>S_stable_iff</code>
Theorem 19.4	<code>Delta_spinVertex_phase</code> , <code>spinVertex_twelve_pow</code> , <code>Delta_spinVertex_twelve_phase</code> , <code>norm_Delta_S_pair</code>

Section 20 lives in the namespaces `DeltaDissection` and `DeltaFrobenius`, whose prefixes are omitted in the following table.

Statement	Lean name
Definition 20.1	cellIndex, cell, mem_cell_iff
Proposition 20.2	exists_unique_cell, cell_disjoint, iUnion_cell, mem_cell_succ_iff, dist_le_of_mem_cell, tendsto_cell_diam
Theorem 20.3	floor_three_mul, cell_subdivide, children, card_children, cell_eq_iUnion_children
Definition 20.4	center, center_mem_cell, center_im_pos, mass
Proposition 20.5	mass_pos, mass_period, mass_mirror
The character	raw, rowPeak, glyph, level, pix
Theorem 20.6	glyph_nonneg, glyph_le_one, glyph_row_eq_one, mirrorCol, mirrorCol_involutive, glyph_mirror, pix_mirror
The Frobenius metric	frobInner, frobNorm, frobNorm_sq, mirrorMat
Theorem 20.7	frobInner_mirror, frobNorm_mirror, mirrorMat_glyph, frobInner_glyph_mirror, sqrt_rows_le_frobNorm_glyph, one_le_frobNorm_glyph, frobNorm_glyph_le
Theorem 20.8	pix_frobenius, frobenius_pix
Theorem 20.9	frobTrace, frobTrace_mod_27, frobTrace_zmod_27, frobTrace_zmod_3, three_dvd_tau_iff_not_one_mod_three, primes30, normTrace, traceRow, tau_sq_le, normTrace_sq_le, frobNorm_traceRow_le

Section 21 lives in the namespace `DeltaTriangles`, whose prefix is omitted in the following table.

Statement	Lean name
Definition 21.1	<code>Tri, stdTri, para, shape, size, centroidT, centroid_mem_shape</code>
Definition 21.2	<code>aff, affTri, para_aff, childP, childMap, child, desc</code>
Theorem 21.3(1)	<code>childMap_mapsTo, stdTri_subset_iUnion_childMap, shape_eq_iUnion_child</code>
Theorem 21.3(2)	<code>size_child, size_desc</code>
Theorem 21.3(3)	<code>size_nonneg, dist_le_size</code>
Theorem 21.3(4)	<code>shape_eq_iUnion_desc, card_words</code>
Definition 21.4	<code>fanTri, size_fanTri, size_desc_fanTri</code>
Theorem 21.5	<code>mapTri, para_mapTri, child_mapTri, desc_mapTri, shape_mapTri, fanTri_rot, desc_fanTri_rot, conjTri, para_conjTri, child_conjTri, desc_conjTri</code>
Definition 21.6	<code>reading, frameReadings, readings, sortedReadings</code>
Theorem 21.7	<code>card_frameReadings, card_readings, spinVertex_add_nat, fanTri_shift, sum_range_shift, readings_rot, sortedReadings_rot, centroidT_conjTri, reading_conjTri</code>
Proposition 21.8	<code>norm_exp_sub_exp_le, dist_circle_spinVertexSet_le, tendsto_dist_circle</code>

Section 22 lives in the namespaces `EuclidTiling` and `PaletteCharacterTable`, whose prefixes are omitted in the following table.

Statement	Lean name
Definition 22.1	ringTri, tile, ringTri_zero_level
Proposition 22.2	size_mapTri, size_ringTri, size_tile, size_tile_diag, card_tiles
Theorem 22.3(1)	mem_shape_fanTri, mem_shape_ringTri
Theorem 22.3(2)	half_le_cos_pi_div, closedBall_subset_ring, iUnion_ring_eq_univ
Theorem 22.3(3)	exists_tile, iUnion_tiles_eq_univ
Definition 22.4	reflUp, reflUp_of_pos, im_reflUp_pos, reflUp_conj, readingRefl, readingRefl_eq_reading
Definition 22.5	paletteTrace, traceMatrix, readingRefl_nonneg, paletteTrace_nonneg
Proposition 22.6(1,2)	ringTri_periodic, paletteTrace_periodic, paletteTrace_rot
Proposition 22.6(3)	swapTri, sigma9, sigma9_involutive, child_swapTri, desc_swapTri, swapTri_mapTri, neg_conj_spinVertex, conjTri_fanTri, conjTri_mapTri_real, ringTri_mirror, readingRefl_conjTri, readingRefl_swapTri, wordPerm, paletteTrace_mirror
Definition 22.7	zeta, isPrimitiveRoot_zeta, chi, norm_chi, chi_symm, chi_add_right, conj_chi, chi_periodic, charTable, norm_charTable, charTable_zero_left, charTable_symm
Theorem 22.8	sum_chi, sum_chi_left, dvd_iff_eq, charTable_row_orthogonality, charTable_col_orthogonality
Proposition 22.9	chi_scale, chi_nine_three
Definition 22.10	fourierCoeff, paletteCoeff, paletteChar
Theorem 22.11(1)	fourier_inversion, paletteTrace_eq_sum
Theorem 22.11(2)	sum_normSq_fourierCoeff, sum_normSq_paletteCoeff, frobNormC, frobNormC_paletteChar
Theorem 22.11(3)	paletteCoeff_zero
Theorem 22.12	sum_range_shift_int, fourierCoeff_shift, paletteCoeff_rot, norm_paletteCoeff_rot
Theorem 22.13	sum_range_reflect_int, fourierCoeff_mirror, paletteCoeff_mirror

Section 23 lives in the namespaces Zeta369 and Moonshine369, whose prefixes are omitted in the following table.

Statement	Lean name
Definition 23.1	trinityIdx, nonTrinityIdx, mem_trinityIdx_iff, mem_nonTrinityIdx_iff, trinityEquiv
Theorem 23.2	tsum_trinity, tsum_nontrinity, tsum_multiples, multiplesEquiv, trinityIdx_eq, tsum_trinity_two, trinity_density
Zeta values	riemannZeta_six, riemannZeta_neg_one, riemannZeta_neg_three, riemannZeta_neg_five, riemannZeta_neg_six, digitalRoot_zeta_even_denominators
Theorem 23.3	eisenstein_constants_from_zeta, digitalRoot_eisenstein_constants, zeta_neg_five_eq_neg_inv_tau_three
Definition 23.4	jCoeff, T1, T2, T3
Proposition 23.5	jCoeff_table, jCoeffNegOne_eq, T1_table, T2_table, T3_table, T2_zero, T3_zero
Theorem 23.6(1)	three_pow_five_dvd_jCoeff_three_mul, two_pow_eleven_dvd_jCoeff_two_mul, digitalRoot_jCoeff_three_mul, digitalRoot_T2_three_mul, digitalRoot_T3_three_mul, digitalRoot_tau_three_mul
Theorem 23.6(2)	jCoeff_congr_T3_mod_729, digitalRoot_jCoeff_eq_T3, dr_eq_of_mod_nine, digitalRoot_jCoeff_table, digitalRoot_T3_table
Theorem 23.6(3)	jCoeff_congr_T2_mod_4096, digitalRoot_T2_ne_jCoeff, digitalRoot_T2_table
Definition 24.1	PaletteUniversality.qSeries, PaletteUniversality.hauptSeries, PaletteUniversality.PolyBound, PaletteUniversality.Normalized, PaletteUniversality.paletteRank
Lemma 24.2	PaletteUniversality.norm_qSeries_sub_qq, PaletteUniversality.norm_hauptSeries_sub_inv
Theorem 24.3	PaletteUniversality.qHeight, PaletteUniversality.qSeries_norm_lt_of_height, PaletteUniversality.hauptSeries_norm_lt_of_height
Corollary 24.4	PaletteUniversality.paletteRank_eq_of_normalized, PaletteUniversality.paletteRank_eq_of_haupt, PaletteUniversality.haupt_order_reverse
Definition 26.1	Spectrum369.Spec, LmfdbSpectrum369.counts
Theorem 26.2	Spectrum369.trinity_union_circuit, Spectrum369.mem_circuit_iff_isUnit, Spectrum369.circuit_eq_orbit, Spectrum369.trinity_absorbing, Spectrum369.dbl_iterate_six
Theorem 26.3	Spectrum369.vertex_sum_univ, Spectrum369.trinity_vertices, Spectrum369.vertex_sum_trinity, Spectrum369.vertex_sum_circuit
Definition 26.4	Spectrum369.sup9, Spectrum369.hexad, Spectrum369.sup3
Theorem 26.5	Spectrum369.spectral_17_tree_laminar, Spectrum369.hexad_determined, Spectrum369.spectral_cluster_laminar
Theorem 26.6	Spectrum369.eisenstein_three_avoids_one, LmfdbSpectrum369.layer_three_dichotomy,

Section 27 lives in the namespaces `PascalBoard`, `MathieuBoard` and `SteinerSystem`, whose prefixes are omitted in the following table.

Statement	Lean name
Definition 27.1	<code>digit</code> , <code>Lit</code> , <code>Dominates</code> , <code>digit_succ</code> , <code>digit_eq_zero_of_lt</code>
Theorem 27.2	<code>lit_iff_dominates</code> , <code>not_dvd_choose_of_lt</code> , <code>le_of_dominates</code> , <code>dominates_succ_iff</code>
Theorem 27.3	<code>lit_two_iff_testBit</code> , <code>lit_two_iff_and</code>
Theorem 27.4	<code>dark_iff_carry</code> , <code>dominates_iff_no_carry</code>
Theorem 27.5	<code>litCount</code> , <code>litCount_zero</code> , <code>litCount_rec</code> , <code>litCount_eq_prod</code>
Corollary 27.6	<code>row_all_lit_pow_sub_one</code>
Proposition 27.7	<code>row3_mod3</code> , <code>row6_mod3</code> , <code>row9_mod3</code> , <code>row12_mod3</code> , <code>row24_mod3</code> , <code>litCount_three_369</code> , <code>litCount_three_12</code> , <code>litCount_three_24</code>
Definition 27.8	<code>IsSetTransitive</code>
Theorem 27.9	<code>index_stabilizer_eq_choose</code> , <code>ncard_card_eq</code>
Theorem 27.10	<code>choose_dvd_card_of_isSetTransitive</code> , <code>not_isSetTransitive_of_not_dvd</code>
Corollary 27.11	<code>choose_24_dvd_iff</code> , <code>choose_12_dvd_iff</code> , <code>not_isSetTransitive_middle_24</code> , <code>not_isSetTransitive_six_12</code>
Theorem 27.12	<code>exists_perm_smul_finset</code> , <code>perm_isSetTransitive</code> , <code>index_stabilizer_perm_eq_choose</code>
Definition 27.13	<code>SteinerSystem</code> , <code>SteinerSystem.trivial</code>
Theorem 27.14	<code>biUnion_powersetCard</code> , <code>card_blocks_mul</code> , <code>choose_dvd</code>
Corollary 27.15	<code>card_blocks_5_8_24</code> , <code>no_5_8_25</code>
Theorem 27.16	<code>index_stabilizer_block_mul</code> , <code>index_stabilizer_octad</code>